Q-CRAFT Explorer Companion Guide

A course in climate-aware fiscal projection

Teal Insights & NatureFinance

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Preface

What Q-CRAFT is

Q-CRAFT is the IMF Fiscal Affairs Department's tool for projecting public finances under climate warming. The full name is the Quantitative Climate Risk Assessment Fiscal Tool. It takes a country's macroeconomic and fiscal data, adds assumptions about demography, productivity, prices and fiscal policy, and projects the debt-to-GDP ratio to 2099 under a baseline and six warming scenarios.

The Fiscal Affairs Department built it for its own climate technical assistance. The September 2023 mission to Uganda is one example: a five-day workshop with ministry staff, written up as *Uganda: PFM Climate Assessment* (IMF, 2024). Ministries then carry the results into their own documents. Uganda's Fiscal Risk Statement for FY 2024/25 has a chapter on climate fiscal risks whose figures are sourced to "QCRAFT (2023)".

This course teaches you to run Q-CRAFT both ways: in the official IMF Excel workbook, which the Fund publishes on its website, and in **Q-CRAFT Explorer**, an open-source Python reimplementation of that workbook. Teal Insights and NatureFinance develop the Explorer and the code is MIT licensed. The economics is the IMF's on either surface. The web interface, the automatic data loading and the record of what you assumed are ours.

> **ℹ This is not an IMF product**
>
> Q-CRAFT Explorer and this course are an independent project by Teal Insights and NatureFinance. Neither is an official IMF product and neither carries IMF endorsement. Both are meant to be complementary to the IMF's own training materials and to the Q-CRAFT User Guide (Tim and Rahman, 2024), which remain the authoritative references. Where this guide explains a concept it cites the User Guide section behind it, so you can always check the teaching against the source.

Two surfaces, one model

The workbook is where Q-CRAFT lives officially, and it is built in Excel deliberately. Every finance ministry has Excel, staff already know it, and a Fund tool has to run in nearly two hundred countries without assuming data or infrastructure that may not be there. Q-CRAFT is designed simple so that it runs pretty much anywhere.

The Explorer runs that same economics on the web, and it exists for four reasons.

1. **Data currency.** A posted workbook ships with the WEO vintage it was built on, and that vintage ages between releases. The Explorer's bundled data is kept current.
2. **Guidance where you need it.** The help for choosing a parameter sits beside the parameter, at the moment you are choosing it. In a week with a deadline in it, that is the moment the help gets read.
3. **Faster to the analysis.** Fewer manual steps sit between the question and the chart, so more of the hour goes on the policy thinking.
4. **The export packet.** Your assumptions, your written remarks on them and presentation-ready output come out together, so less of the write-up gets copied across by hand.

Chapter 2 sets the four out at greater length. After that the course mostly stops arguing them and shows them, one at a time, in the flow of the work.

The questions it answers

Q-CRAFT answers a narrow set of questions, and it answers each one by comparing two runs of the same model. Four of them, in the register they usually arrive in:

- If warming follows the high scenario, where is our debt ratio in 2050? And in 2099?
- How much of that path is climate damage, as against where the debt ratio was heading anyway?
- Does climate damage on its own take us through the ceiling in our fiscal rule, and in which year?
- Which assumption moves the answer most: the demographic outlook, the interest rate, or how far spending can adjust when growth disappoints?

None of those is a forecast. Each one is a difference between two projections built on assumptions you chose, which is why the assumptions are worth arguing about and worth writing down.

Who this is for

This course is for anyone who has to run, interact with, or otherwise understand this class of fiscal projection tool. Ministry of finance economists come first, because they run the tool and then defend the numbers to the officials who sign them off. IFI and technical assistance staff come second, because they build capacity around tools like this one and a teaching sequence is easier to hand over than a workshop. Researchers and analysts come third, because they read the output and have to decide what it licenses them to say.

The tool is broadly applicable and deliberately conservative

This guide teaches what the tool does, why it exists, and what it leaves out. The honest way to hold the third part is as a pairing with the first two.

The strengths. Q-CRAFT runs for most of the world, on data that is already published. It needs no bespoke country dataset, no new survey and no waiting for a mission, so a team can have a first pass by the end of the week. And the arithmetic is open to inspection. This reimplementation is verified against the IMF's original workbook, with baseline parity exact for 147 of 147 tested countries and climate-scenario parity confirmed for ratio metrics. Chapter 2 sets out what that claim covers and what it does not.

The limitations. The scenarios capture one channel, the slow effect of temperature on productivity. They exclude tipping points, nonlinearities, natural disasters, sea-level rise, the public spending that adaptation takes, and whatever channel is specific to your own country (User Guide, pp. 5-6). Every one of those exclusions is a cost left out, so the results read conservative: a lower bound on fiscal impact rather than a central estimate of the total. Chapter 6 sets this out with the sources.

The two halves are one design choice seen from two sides. A tool that runs anywhere on published data is a tool that cannot carry the channel only your country has. Knowing that is part of knowing the model. This is the first in a series of guides to the models behind sovereign climate-fiscal analysis, and each one takes the same three questions: what the model is for, what it computes, and where it stops.

What you will be able to do

Three objectives, weighted equally:

1. **Run the tool, on both surfaces.** Select a country, set the parameters, read the charts and export the results, in the IMF workbook and in the Explorer.
2. **Understand what you are doing when you run it.** One equation, three inputs, and a clear view of which of your assumptions is moving the answer.
3. **Interpret the output appropriately.** Say what a result supports and what it does not, with its strengths and its limits attached.

The first objective takes an afternoon. The second and third are the ones that hold up in a meeting.

How the course is organised

Seven modules, each built around something you can do at the end of it.

Module	What it gets you
0. How to use this course	The capstone stated up front, a self-assessment, and the path that fits your background
1. What Q-CRAFT does and why it exists	A full projection in ten minutes, and the map: three numbers, one equation
2. The model, in three steps	An equation, a growth engine, a climate overlay: the whole tool, traceable end to end
3. Choosing the parameters	Every parameter set and defended, with the rationale written down
4. A worked example, end to end	One country from assumptions to a fiscal risk paragraph, then two more with less help
5. What the tool can and cannot tell you	The strengths, the documented exclusions, and which questions belong to a different tool
6. The capstone	The capstone brief and rubric, and a look back at where you started

The tool covers most of the world, so the examples move country by country and the mechanism picks the country. Demography reads clearest where the workforce is already shrinking, set against one where it is still growing fast. Productivity convergence reads clearest in an economy catching up to the frontier. The climate channel reads clearest where exposure is high. Module 4 runs one country the whole way through, and that country is Uganda, because it is the tool's golden-master verification country and every figure in the walkthrough can be checked against published documents. The exercises that follow it run on other countries, and the capstone runs on yours.

Start at Module 0. It takes about ten minutes and it will tell you which parts of the later modules you can read quickly. Module 2 is the spine: it assembles the whole tool in three steps, and everything after it either chooses the inputs, runs the thing or reads the output.

The Glossary defines key terms and the References section gives annotated pointers to the primary sources underlying the methodology.

There is also an appendix on co-design and the SovTech vision. It sits outside the learning path on purpose: it is about the project behind the tool rather than about using it, and nothing in the capstone depends on it. Read it if you want to shape what V2 does.

What this course defers to

This is an educational companion to the IMF's User Guide (Tim and Rahman, 2024), which remains the authoritative methodology reference, and to the IMF's own training materials, which remain the authoritative teaching materials. Where this guide explains a concept, it cites the relevant User Guide section so you can go deeper. Where the two ever differ, the User Guide is right.

> ℹ This is an initial version
>
> Q-CRAFT Explorer is a starting point rather than a finished product. We want to co-design the next version with you. See the co-design appendix for how to get involved.

> Try the App
>
> **Open Q-CRAFT Explorer** to follow along as you read. The source code is on GitHub.

Colophon

This course is open source under the MIT license, and it builds completely from what is in its repository. It sets in Inter, IBM Plex Serif and IBM Plex Mono, three faces under the SIL Open Font License, bundled with the book rather than fetched from a font service. Clone the repository, run `quarto render docs/companion-guide`, and you get this book in the same three faces on a machine with no network access. Opening a page fetches no font from anyone's server, which matters in a ministry network that blocks outside hosts.

Teal Insights also publishes a house edition of the same content, set in licensed Klim Type Foundry faces where the license permits. Those font files are not in the repository, because the license that covers them covers tealinsights.com. Where they are absent, every font stack falls through to the open faces and the book still sets. No word, number or figure in the course depends on which edition you are reading.

The figures are built from the repository too. `scripts/build_course_map.py` draws the course map that opens every module, and `scripts/build_parameter_context.py` draws the source-data figures in Chapter 4 from the same Parquet files the Explorer runs on.

Chapter 1

How to use this course

In this module
You will see the deliverable the whole course builds toward, so you can judge every later module against it. You will calibrate your own starting point, in about ten minutes. Then you will pick the path that fits, so nobody sits through material they do not need.

Fast path

Short on time? Answer the three questions in Section 1.5 and read the path table in Section 1.6. That is about ten minutes, and it tells you which of the later modules you can skip.

1.1 A ministry has already published this deliverable

The document this course teaches you to write is an ordinary ministry-of-finance product. Fiscal risk statements, budget framework papers, medium-term debt reports: most finance ministries publish something in this genre every year, and the climate section is the part that has been arriving recently. At least one published fiscal risk statement already carries a climate section built with the tool this course teaches, with the figures sourced to it by name.

Chapter 5 opens that document and works through it, page by page, against numbers you can check. Until then the useful thing to know is that the target is a real published section rather than an exercise.

You will produce two things by the end of the course:

1. **An export packet.** The parameters you chose, a written rationale for each judgment call, and the polished charts and CSV that Q-CRAFT Explorer exports.
2. **A two-paragraph draft** in the register of your ministry's fiscal-risk documentation. In your case that is whatever your ministry publishes. If it publishes nothing of the kind yet, the published section in Chapter 5 is the model.

That packet is the capstone, and it is what you defend to the senior officials who sign the document off. Chapter 7 contains the brief and the rubric it will be marked against. Every module in between either feeds that packet or tells you when not to trust it.

> **i** This course defers to the User Guide
>
> The IMF's User Guide (Tim and Rahman, 2024) remains the authoritative methodology reference. This course explains the concepts behind the tool and cites the User Guide section for each one, so you can go deeper where you need to.

1.2 By the end of this module you can

- **Locate** your own starting knowledge against three anchored descriptions of practice
- **Diagnose** which of three concept-inventory items you answered from intuition rather than from the debt dynamics equation
- **Select** the path (A, B or C) that matches your background, and justify the choice
- **Describe** the capstone deliverable in one sentence to a colleague who has not taken the course

1.3 Where you are in the course

The map below opens every module, and this is the first time you meet it.

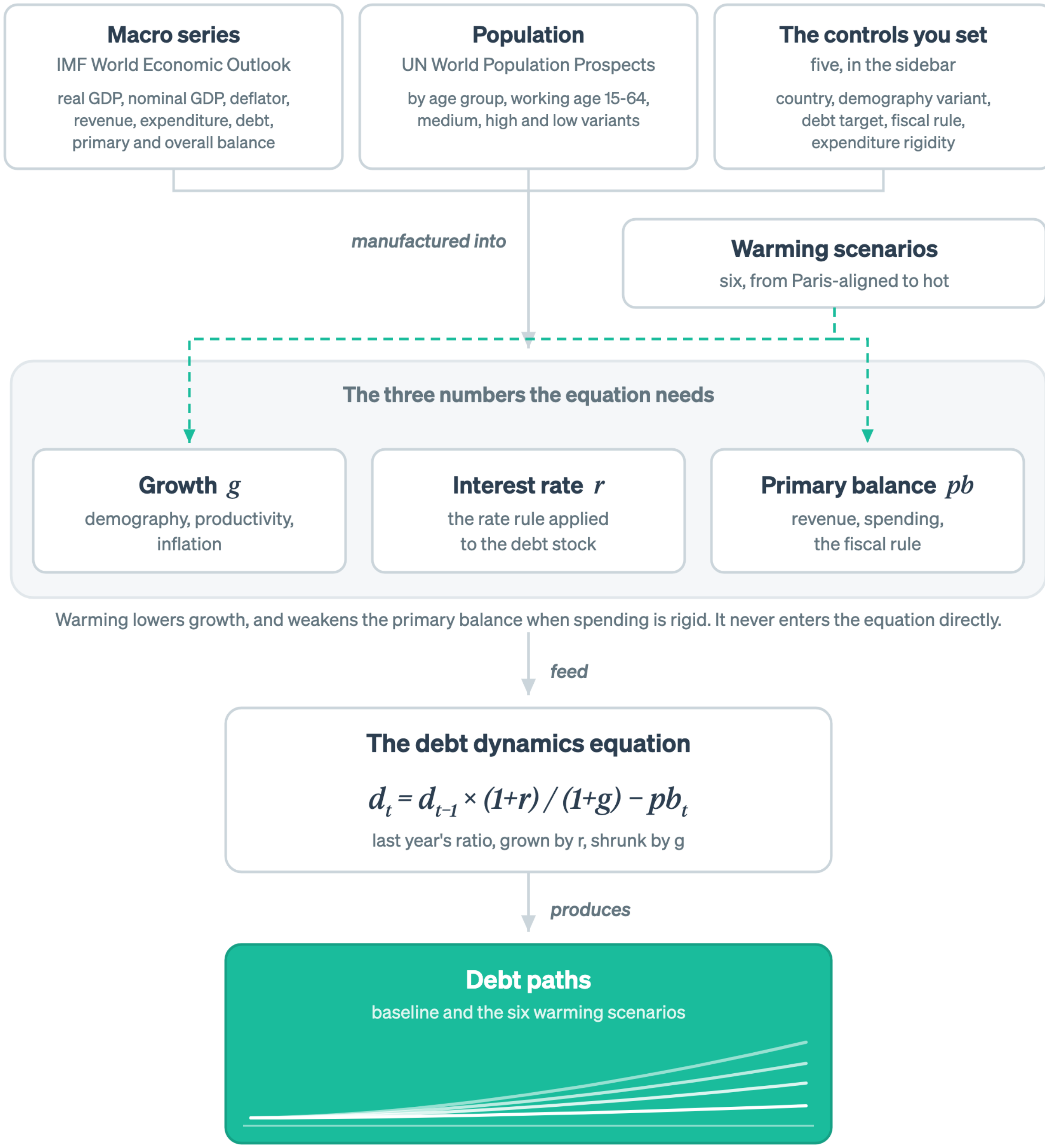


Figure 1.1: The course map. Two published data sources and the controls you set are manufactured into the three numbers the debt dynamics equation needs. This module fixes the destination on the right.

Only the destination is lit, because that is the part to hold on to now. The same map opens every module, with that module's own nodes lit.

1.4 Self-assessment: which of these describes your Monday?

Three rows, one answer each: long-term fiscal projection, climate-fiscal analysis, and the tool itself. Pick the description closest to true. There is no scoring and no one sees it. The point is to route you, and to give you something to compare against in Chapter 7.

On long-term fiscal projection:

	Description
A1	I have not worked with a debt sustainability analysis. I know what the debt-to-GDP ratio is and roughly why it matters.
A2	I have read DSAs and can follow the tables. I could not build the projection myself without help.
A3	I have produced or reviewed a DSA for a country. I could set up a baseline projection this week if asked.

On climate-fiscal analysis:

	Description
B1	I work on climate policy or climate finance. I have not connected it to a debt projection.
B2	I have seen climate scenarios used in fiscal analysis and can describe what they are for.
B3	I have run or interpreted a climate-scenario fiscal projection for a country.

On the tool itself:

	Description
C1	I have never opened Q-CRAFT, in Excel or on the web.
C2	I have opened it and clicked around. I could not defend the parameter settings I used.
C3	I have run Q-CRAFT for a country and explained the output to someone else.

Write your three answers down. You will be asked for them again at the end.

1.5 Three questions before you start

Three questions follow, and they check which intuitions you are carrying in rather than what you have memorised. Answer each one, commit to it in writing, then open the answer.

! DRAFT FOR TEAL: concept-inventory question 1 (interest-growth differential)

Question. A country's government runs a primary balance of exactly zero: revenue equals non-interest spending, to the last unit of local currency. The nominal interest rate on its debt is 9 percent. Nominal GDP growth is 6 percent. Over the next year, the debt-to-GDP ratio will:

(a) Fall, because the government is not adding to its debt.
(b) Stay flat, because a zero primary balance is by definition debt-stabilizing.
(c) Rise, even though the government borrowed nothing new for programmes.

Answer

(c) Rise. Debt grows at the interest rate. The economy it is measured against grows more slowly. So the ratio climbs by roughly (9 - 6) / 1.06 in the first year, which is about 2.8 percent of the starting ratio. A country starting at 50 percent of GDP ends the year near 51.4 percent, having changed no policy at all.

If you picked (b), that is the most expensive misconception in this material: that a zero primary balance stabilizes debt. It stabilizes debt only when the interest rate equals growth. Every other time, the gap between the two is doing work someone has to pay for. Step 1 of Chapter 3 is built around that gap.

If you picked (a), you may be thinking of the deficit rather than the ratio. The stock of debt is flat in local currency. The ratio is not, because the denominator moved.

! DRAFT FOR TEAL: concept-inventory question 2 (expenditure rigidity)

Question. In Q-CRAFT Explorer you set expenditure rigidity to 1.0 and run a climate scenario. What have you assumed?

(a) Spending falls in step with GDP, holding the expenditure-to-GDP ratio at its baseline level.
(b) Spending stays at its baseline level in local currency even as GDP falls, so the whole revenue loss lands on the deficit.
(c) Spending is capped by the fiscal rule, whatever happens to GDP.

Answer

(b). Rigidity 1.0 means spending is sticky. The wage bill, pensions and existing commitments do not move when growth disappoints, so the whole climate-driven revenue shortfall becomes borrowing. That is the worst case, and the tool makes it the default on purpose.

If you picked (a), you have the scale inverted. That is rigidity 0.0. The inversion matters, because it flips the sign of your headline risk number. It is the easiest way to publish a wrong figure from this tool.

If you picked (c), you have merged two separate controls. The fiscal rule and expenditure rigidity are different parameters and they act on different scenarios. Chapter 4 separates them.

! DRAFT FOR TEAL: concept-inventory question 3 (what the output is)

Question. A Q-CRAFT run shows debt-to-GDP at 66 percent in 2099 under the Hot scenario against 47.5 percent in the baseline. A colleague asks what that means. Which answer is defensible?

(a) The country's debt will be about 66 percent of GDP in 2099 if the world warms on that path.
(b) Under a set of stated assumptions, climate damage of this magnitude would add roughly 19 percentage points of GDP to the debt ratio by 2099 relative to the same projection without climate damage.
(c) Climate change will cost the country 19 percent of GDP.

Answer

(b). The number that survives scrutiny is the *difference between two runs of the same model*, stated with its assumptions attached. Q-CRAFT compares scenarios. It does not forecast.

Answer (a) treats a 2099 projection as a prediction. Nobody forecasts 2099, and the tool's own User Guide says the results are not forecasts (pp. 5-6).

Answer (c) converts a debt-ratio gap into a cost figure, which it is not. The gap is the extra borrowing implied by lower growth under stated assumptions about how spending responds, and it leaves out disasters, sea-level rise and adaptation spending entirely. Chapter 6 is about that class of error.

The 47.5 and 66 percent figures are real. They come from a five-day IMF workshop with the staff of one finance ministry in September 2023, reported in the C-PIMA high-level summary (IMF, 2024), and Chapter 5 runs the case they came from.

1.6 Pick your path

The three paths cover the same tool. They differ in how much of the economics is rebuilt for you along the way.

Path	Choose it if
A. Guided	You answered mostly A1/B1/C1. Common if you came to this from the climate side and not from the debt arithmetic.
B. Standard	You answered mostly A2/B2/C1 or C2. Common if you know debt dynamics already and the tool is the new part.
C. Fast	You answered A3 and C2 or C3. You have built projections and want the tool's specifics and the judgment calls.

The figure below lays the three routes across the seven modules, so you can see what each one actually abridges.

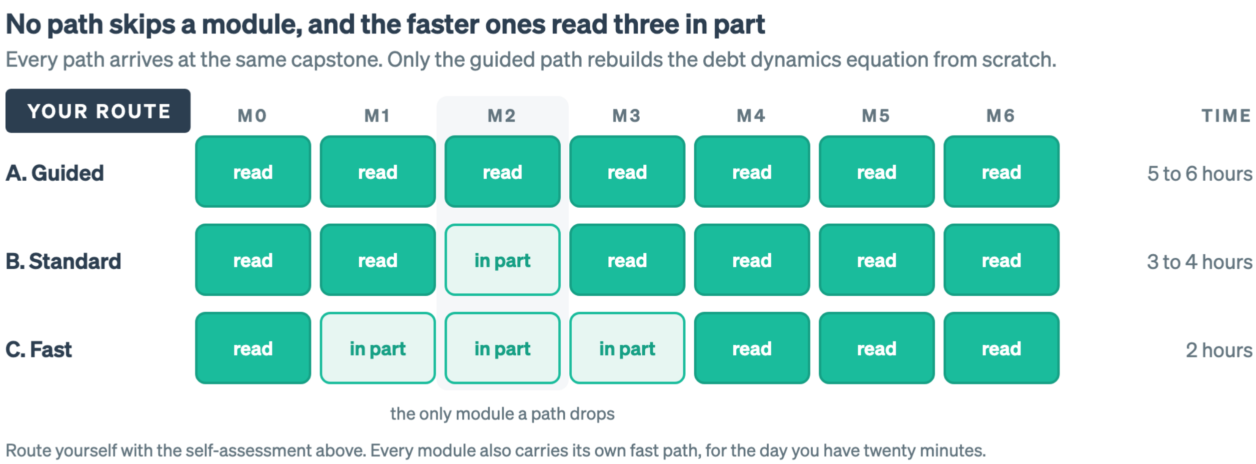


Figure 1.2: What the faster paths abridge is always the same thing: material that rebuilds economics you may already have. Nothing about running the tool, reading its output or defending an assumption is dropped on any route.

No route drops a module. What the faster paths abridge is the material that rebuilds economics you may already own, so choosing a path is answering one question: how much of it do you want rebuilt for you on the way through.

Chapter 3 is the one module nobody should skip, because it is the only place the whole tool is assembled in one sitting. If you already own the debt dynamics equation, its Step 1 will be twenty minutes of revision and its collapsible depth layer is there if you want it. Steps 2 and 3 are new material for almost everyone.

Every module carries a fast-path marker near the top, telling you the shortest useful route through it.

> **A note on time**
>
> Budget three to four times whatever you think this will take. That ratio is the standard finding on expert time estimates for other people's learning, and it holds here. The ten-minute run in Chapter 2 does take ten minutes. The rest does not.

1.7 Wrapper: what you should have now

- The capstone in one sentence: an export packet with documented rationale, plus a two-paragraph fiscal risk draft.
- Three self-assessment answers written down for comparison in Chapter 7.
- Three concept-inventory answers, and a note on which ones you got from intuition rather than from the identity.
- A path.

On your desk this week: find the most recent fiscal risk documentation your own ministry publishes and read the section that would carry this analysis. If there is none, read the published section Chapter 5 works through. Either way, that is the genre you are writing into.

Chapter 2

What Q-CRAFT does and why it exists

In this module
You will run a full projection in ten minutes, before anything is explained. Then the model is built in front of you in two stages. First the base machine: a long-term fiscal projection model with no climate in it, resting on one equation and the three numbers it needs, which are growth, the interest rate and the primary balance. Then the warming block bolts onto that finished machine at two points and the same equation becomes climate-aware. The module closes with the parity evidence, so you can decide now whether to trust the numbers: baseline parity is exact for 147 of 147 tested countries.

Fast path

Short on time? Do Section 2.4 (the ten-minute run), Section 2.5.2 (the three-number table) and the map at the top of Section 2.6 (where warming attaches). Skip the rest and come back to it.

2.1 By the end of this module you can

- **Run** a full projection in Q-CRAFT Explorer for a country of your choice and export the results
- **Name** the three numbers the debt dynamics equation needs and **identify** what builds each one
- **Distinguish** what the base projection model does from what the warming block adds to it
- **Locate** any parameter you are asked about against the number it moves, without searching
- **Explain** to a colleague why a Python reimplementation of an Excel tool needs a parity test, and what "147 of 147" does and does not cover

2.2 Where you are in the course

The map below is the same one from Chapter 1, with a different part of it lit.

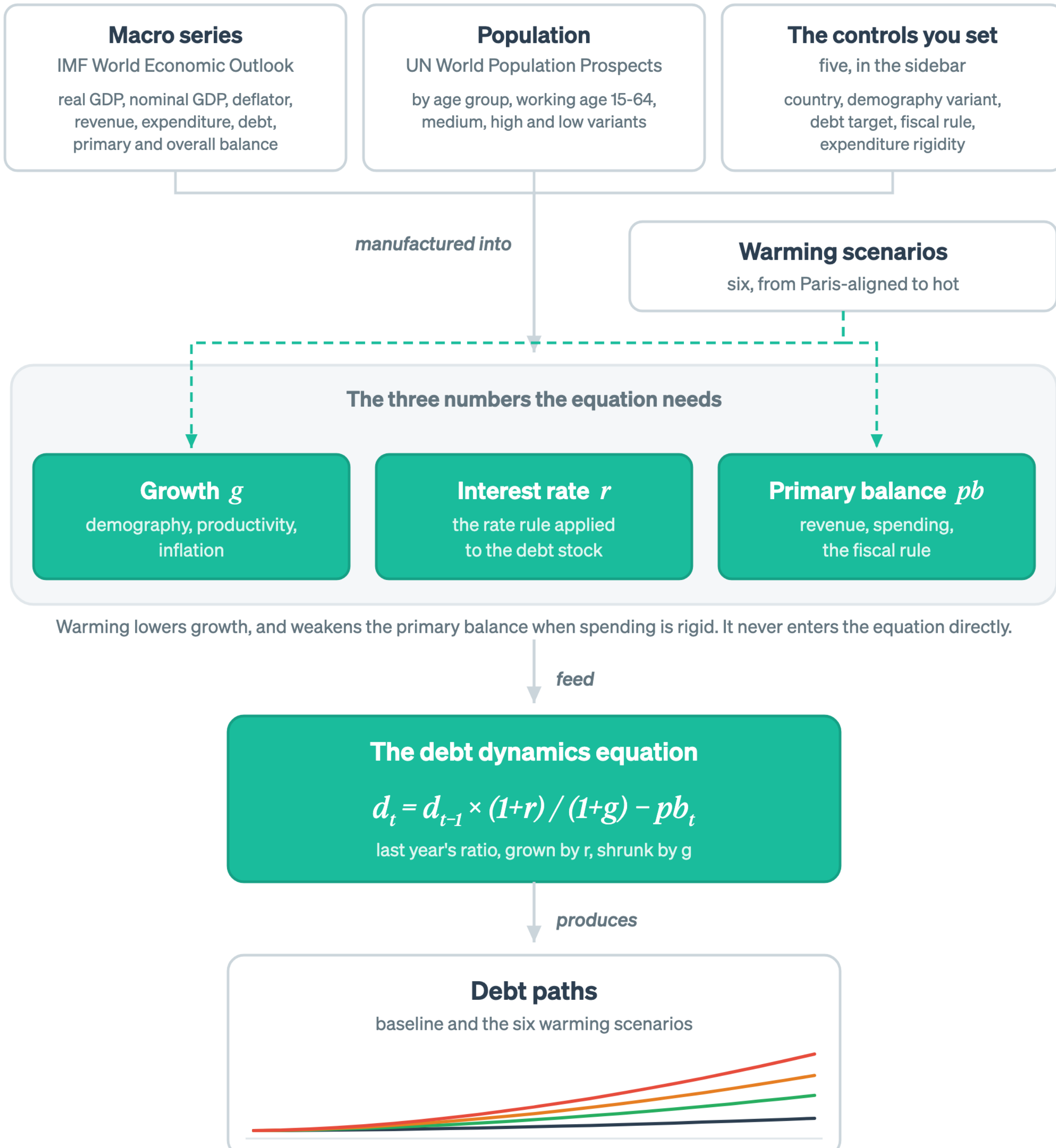


Figure 2.1: This module covers the middle of the chain: the three numbers, and the debt dynamics equation they feed.

Four nodes are lit: growth, the interest rate, the primary balance, and the equation that turns them into a debt path. That group is the base machine this module builds first. The warming block at the bottom of the map arrives second, and the same picture comes back with only that block lit when it does.

2.3 Q-CRAFT runs in a published Excel workbook, and in this Explorer

The IMF publishes Q-CRAFT as an Excel workbook, downloadable from its website. Building it that way was a deliberate decision and a sound one. Every finance ministry has Excel, staff already know it, and a Fund tool has to run in nearly two hundred countries without assuming data or infrastructure that may not be there. Q-CRAFT is designed simple so that it runs pretty much anywhere. The workbook is well structured, clearly documented, and covers 197 economies.

Q-CRAFT Explorer runs that same economics on the web. Four things are different, and they are the reasons it exists.

1. **Data currency.** A posted workbook ships with the WEO vintage it was built on, and that vintage ages between releases. The Explorer carries current bundled data, October 2024 as this is written, and refreshing it is a release rather than a download and a rebuild.
2. **Guidance where you need it.** The help for choosing a parameter sits next to the parameter. That is the difference between reading about expenditure rigidity and choosing a value for it while looking at what your own country's spending has actually done, which is what Chapter 4 has you do.
3. **Faster to the analysis.** Fewer manual steps sit between the question and the chart, so more of the hour goes on the judgment calls that are the actual work.
4. **The export packet.** Your assumptions, your written remarks on them, and presentation-ready output come out together, so the record of what you decided travels with the numbers and less of the write-up gets copied across by hand. Chapter 4 is where you build one, Chapter 5 is where you export it, and Chapter 7 is where you hand it over.

The engine underneath the Explorer is a standalone Python package with extensive automated testing, checked against the workbook cell by cell (Section 2.7). The code is open source under the MIT license, so anyone can examine, verify, and extend it.

⚠ TODO: the Excel half of the both-ways promise

The preface promises Q-CRAFT both ways, and the material for the workbook half does not exist yet: where to download it, how its sheets map onto the three numbers below, and how to run the worked case in Chapter 5 there as well as here. Everything the course currently teaches is on the Explorer. Scope decision for review.

2.4 Run a country in ten minutes

Run the tool before you read how it works. The explanation lands better on something you have already seen move. Six steps follow, and the whole sequence takes about ten minutes.

The figure below is the shape of those ten minutes, one panel per place you will be looking.

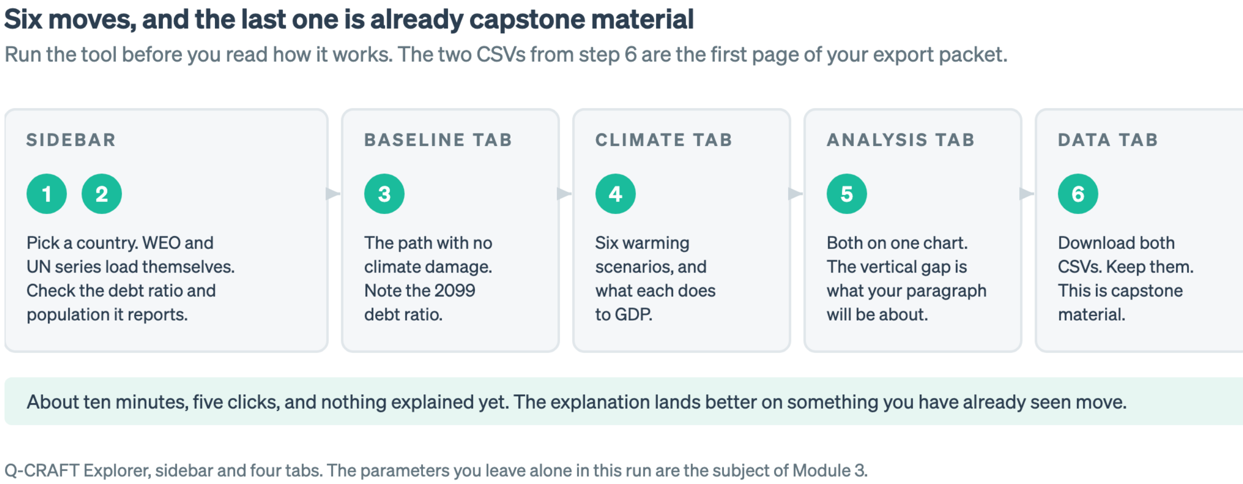


Figure 2.2: Nothing here is explained, on purpose. Both prediction and explanation land better on a chart you have already watched move, which is why the reading starts after the clicking.

Two moves happen in the sidebar and one in each of the four tabs, and the last of them produces a file you keep. Now do it.

> Zero to a projection
>
> Open **Q-CRAFT Explorer** and follow these steps:
>
> 1. **Select a country** from the sidebar dropdown. Pick the one you work on. If you want a run you can check against a published result, pick Uganda, which is the tool's verification country and the worked case in Chapter 5. Either way, WEO macroeconomic data and UN population projections load automatically. Nothing to download, nothing to paste.
> 2. **Read the sidebar context.** It shows the latest WEO debt-to-GDP ratio and total population. Check them against what you know before going further.
> 3. **Open the Baseline tab.** This is the country's fiscal trajectory with no climate damage. Note the debt-to-GDP number at 2099.
> 4. **Open the Climate tab.** Six warming scenarios, their effect on GDP. Note where the scenarios separate from each other, and in which year.
> 5. **Open the Analysis tab.** Baseline and all climate scenarios on one debt chart. The vertical gap between the baseline line and the Hot Unadapted line is the number your fiscal risk paragraph will be about.
> 6. **Open the Data tab and download both CSVs.** That is the first piece of your capstone export packet. Keep it.

If you picked Uganda, you have now done with five clicks the analysis that ministry staff produced in a five-day workshop with a visiting IMF team in September 2023. The workshop's finding, published in the C-PIMA high-level summary: GDP loss by end of century could pass 4 percentage points, with debt rising to 66 percent of GDP against 47.5 percent in the baseline (IMF, 2024).

Your numbers will not match theirs exactly, even on the same country. That workshop used the 2023 WEO vintage and its own parameter choices, and the Explorer ships with WEO October 2024. Which vintage and which parameters were used is precisely the kind of thing that gets lost, and precisely what the export packet is for.

> ⚠ SCREENSHOT-TODO
>
> Annotated screenshot of the Analysis tab for Uganda, the worked case, with the baseline-to-Hot-Unadapted gap called out on the chart. Do not ship this module without it: the sentence "the vertical gap is the number your paragraph is about" is doing visual work with no visual.

2.5 The base machine, before any climate

Before Q-CRAFT is a climate tool it is a long-term fiscal projection model. Give it a country and it projects the debt-to-GDP ratio to 2099 with no warming in the picture at all. That machine is what this section teaches. It is worth learning on its own, because it is the same machine underneath every debt sustainability analysis you have read.

Nothing below mentions climate. Warming arrives in the next section, and it arrives as one block that bolts onto what you are about to read.

2.5.1 One equation produces the whole debt path

Next year's debt ratio is this year's, grown by the interest rate, shrunk by economic growth, less whatever the government paid down.

That sentence is the whole model. The compact form is the sentence in shorthand:

$$d_t = d_{t-1} \times \frac{1 + r_t}{1 + g_t} - pb_t$$

Symbol	Meaning
d_t	Debt-to-GDP ratio in year t
r_t	Effective nominal interest rate on government debt
g_t	Nominal GDP growth rate
pb_t	Primary balance as a share of GDP (revenue minus non-interest spending)

Three of those four are inputs. The fourth, d, is the answer, and it is also last year's input, which is why the tool walks the projection forward one year at a time.

The sign of r minus g decides whether the ratio climbs on its own. When interest rates exceed growth, the debt ratio rises automatically, and the government must run a primary surplus even to hold debt still. When growth exceeds interest rates, the ratio declines even with modest deficits. Step 1 of Chapter 3 works this through with numbers, and Steps 2 and 3 build the rest of the tool on top of it.

Prefer it in numbers? One year of the debt dynamics equation

Start a country at a debt ratio of 60 percent of GDP. Put the effective interest rate at 8 percent, nominal growth at 6 percent, and the primary balance at a deficit of 1 percent of GDP.
The amplifier goes first. Divide 1.08 by 1.06 and you get 1.019, so last year's ratio is carried forward at a little over one for one:

$$60 \times \frac{1.08}{1.06} = 61.1$$

Then the primary balance. A deficit of 1 percent of GDP is $pb = -1$, and subtracting a negative adds:

$$61.1 - (-1.0) = 62.1$$

Next year's ratio is 62.1 percent of GDP. The government borrowed one point of GDP and the ratio rose by 2.1, because the interest-growth gap supplied the other 1.1 with nobody deciding anything.
Sensitivity. Raise growth by one point, to 7 percent, and next year's ratio lands at 61.6 instead of 62.1. Half a point in one year looks like nothing. Hold the same two settings for ten years and the paths land at 83.2 and 76.3 percent of GDP. That is the whole climate story in this course in advance: warming takes a fraction of a point off g, and the equation compounds it for seventy years.

2.5.2 Where the three numbers come from

Growth has three suppliers, the interest rate has one rule, and the primary balance has two sides. The table below is the thing to remember. The sections after it are for the number you happen to be arguing about.

The number	What builds it	What you control in V1
Growth, g	Working-age population, output per worker, and the price path, combined into nominal GDP growth	The UN population variant. Productivity and inflation sit at the Excel tool's defaults.
Interest rate, r	One of three rate rules, applied to the debt stock	Nothing. V1 uses the Excel default.
Primary balance, pb	Revenue, which tracks nominal GDP, minus primary expenditure, which grows with productivity, inflation and total population, plus any fiscal rule adjustment	The debt target, and the fiscal rule on or off

Read the last column carefully. V1 exposes five controls in the sidebar, counting the country selection, and the other assumptions sit at the Excel tool's defaults. That is a deliberate scoping choice rather than an oversight, and Chapter 6 says what it costs you.

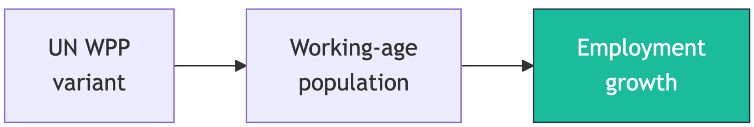
WIDGET-TODO: three-numbers-to-one-equation

Embed the intuition widget here (lane 2, run 3: `apps/qcraft-web/widgets/*`). The widget was specified as "seven-modules-to-one-equation" before the teaching frame changed. Rebuild it on the chain above, data and assumptions producing g, r and pb, with "seven modules" demoted to an architecture aside. Iframe under Quarto for now, native React once the site moves to Astro. The prose above must carry the full idea on its own, because the default state of any interactive has to work for readers who never touch it.

2.5.2.1 Growth, g, is three things multiplied together

Growth is the number with the most suppliers, and the one your assumptions move most.

Demography sets how fast the workforce grows.



The workforce sets how fast the economy can grow. Q-CRAFT takes population by age group (children, working age, elderly) from UN World Population Prospects, in three variants: medium, high, and low. Growth in the working-age population, ages 15 to 64, drives employment growth after the WEO forecast horizon, and employment growth drives potential GDP. The contrast is sharp across the tool's country set. Ethiopia's working-age population is projected to keep growing for the rest of the century, which pushes potential growth up and debt ratios down. Thailand's has already turned and is falling, which does the opposite, and it does it before any climate damage is applied.

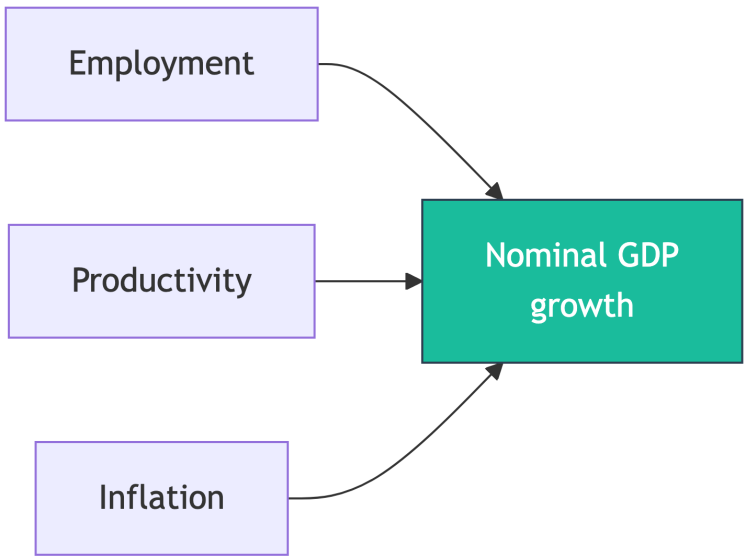
Productivity sets output per worker, and it moves gradually rather than in a step.

The measure is GDP per employed person, from World Bank data. You give the model a starting growth rate and a long-run rate, and it slides between them along an S-shaped path, which economists call a logistic curve. Vietnam is the case that makes the shape concrete: a catch-up economy raising output per worker quickly now, with the rate expected to ease as the level approaches the frontier. The resulting productivity level relative to the OECD is a useful realism check, because assumptions that imply a low-income country overtaking OECD productivity by 2099 deserve another look.

Inflation turns real growth into the nominal growth the debt ratio actually responds to.

Inflation follows the same gradual S-shaped path as productivity. The starting rate should reflect the WEO projection for the final forecast year, and the ending rate should reflect the long-run target, typically 2 to 3 percent for advanced economies and 3 to 6 percent for emerging and developing economies. Where the central bank publishes an explicit target, that number is the reference point.

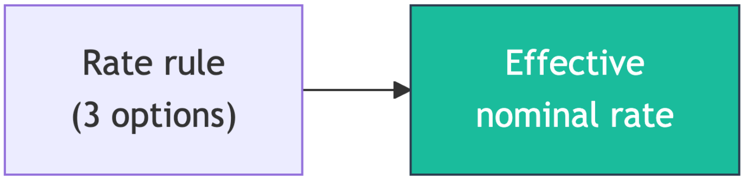
The three combine into nominal GDP growth.



Nominal GDP growth decomposes into employment growth, productivity growth, and inflation, and the three multiply rather than add. Through the WEO forecast period, which currently runs to 2029, Q-CRAFT uses the IMF's own projections. Beyond that horizon the model takes over, using the assumptions set in the dashboard.

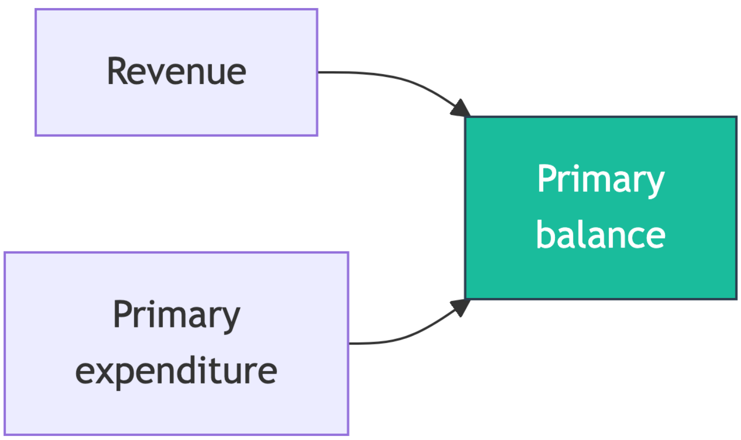
> **ℹ Prefer it in numbers? The growth decomposition**
>
> Put employment growth at 2 percent, productivity growth at 3 percent, and inflation at 5 percent.
> Real growth is the first two multiplied together:
>
> $$1.02 \times 1.03 = 1.0506$$
>
> which is 5.06 percent. Nominal growth puts the price path on top of that:
>
> $$1.0506 \times 1.05 = 1.1031$$
>
> which is 10.31 percent. That is the g the debt dynamics equation uses. Adding the three rates instead of multiplying them gives 10.00 percent, close enough to be tempting and wrong by three tenths of a point every year.
> **Sensitivity.** Take one point off productivity growth, down to 2 percent, and nominal growth lands at 9.24 percent instead of 10.31. The loss is a little larger than the point you removed, because the terms multiply.

2.5.2.2 The interest rate, r, is one rule applied to the debt stock



This number is what carrying the debt costs. Q-CRAFT offers three rules: a constant nominal rate, a constant gap between the interest rate and growth, or a constant real rate. Country expertise matters here, because the choice should reflect concessional lending terms, sovereign risk premia, and the maturity structure of domestic financial markets. For a country moving from concessional to commercial borrowing, this assumption does more work than any other. Zambia's shift into Eurobond issuance through the 2010s is the standard example, and the debt-service consequences arrived faster than any demographic or productivity trend could have.

2.5.2.3 The primary balance, pb, is revenue minus non-interest spending



The two sides grow on different rules. Revenue grows with nominal GDP, holding the revenue-to-GDP ratio constant. Primary expenditure, meaning spending excluding interest payments, grows multiplicatively with productivity, inflation, and total population: (1 + productivity growth) × (1 + inflation) × (1 + population growth), plus any fiscal rule adjustment. When the fiscal rule is enabled, it adjusts spending to steer debt toward a target ratio.

> **i** Prefer it in numbers? Why the expenditure formula multiplies
>
> Put productivity growth at 3 percent, inflation at 5 percent, and population growth at 2 percent. Primary expenditure of 100 units next year is
>
> $$100 \times 1.03 \times 1.05 \times 1.02 = 110.3$$
>
> Adding the three rates instead would give 110.0. Three tenths of a unit in the first year, which is nothing.
> **Sensitivity.** The two versions keep separating, because each year's gap compounds on the last. Run both for seventy years and the multiplicative path ends about 22 percent above

the additive one, which is a different debt projection rather than a rounding difference. The multiplicative form is the one the engine implements.

2.5.3 That is the base machine

You now have a complete long-term fiscal projection model: two published data sources, five controls, three manufactured numbers, one equation, one debt path. It works for any country in the tool, and nothing in it knows that the climate exists.

! DRAFT FOR TEAL: prefer it as a kitchen? The same chain, in an analogy

Some readers hold the chain more easily as a kitchen.
The pantry holds ingredients somebody else bought. The WEO macroeconomic series and the UN population projections. You did not choose them, you cannot change them, and every cook in every country works from the same shelves.
The seasoning is yours. The controls in the sidebar are the salt: the demography variant, the debt target, the fiscal rule, expenditure rigidity. Small quantities, large effect, and the reason two cooks working from one pantry produce different dishes.
Three components are prepared before anything else happens. Growth, the interest rate, the primary balance. Everything the tool does before the final step is preparing those three.
There is one recipe and it never changes. The debt dynamics equation. Same method for every country, every scenario, every year.
Where the analogy breaks. A cook tastes and adjusts, and this model does not. Nothing downstream feeds back into the ingredients, so cutting spending inside the model never slows growth, which is the partial-equilibrium limit stated below. And a recipe is a claim about what will come out of the oven. A projection to 2099 is not. It is a comparison between two dishes made from deliberately different seasoning.

i Two properties of the base machine worth knowing

It is partial equilibrium. The tool does not let fiscal policy feed back into GDP growth or interest rates: cut spending inside the model and growth does not respond. Economists call that a partial-equilibrium model (User Guide, p. 5). It is the reason a Q-CRAFT result is a comparison rather than a forecast.
It is seven functions. Under the hood the engine is seven pure functions, one per box in the sections above, composed into a single call. That is a fact about the code rather than about the economics, and it lives in the repository.

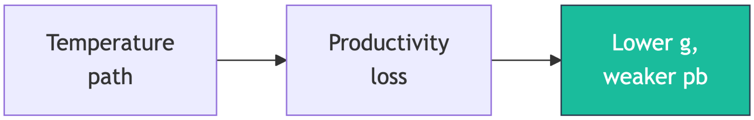
2.6 The warming block snaps on

Everything above runs without climate. Now add one block.

Figure 2.3: The same map, with the base chain greyed out and the warming block highlighted. The block docks onto growth and the primary balance. Nothing else in the chain changes.

The warming scenarios do not add a term to the equation and do not change how any of the three numbers is built. They reach into the finished machine at two nodes. Growth falls, and where spending is rigid the primary balance weakens with it. The equation then does what it already did.

2.6.1 Warming lowers growth, and weakens the primary balance when spending is rigid



Lowering growth is the only way climate enters the equation. The damage estimates are empirical, drawn from the FADCP Climate Dataset (Centorrino, Massetti, and Tagklis, 2024), building on the Kahn et al. (2021) methodology for long-term macroeconomic effects of climate change. Six scenarios run from Paris-aligned, below 2°C warming, to hot-unadapted, high warming with a slow policy response. Impacts begin in 2030, and before that year all scenarios match the baseline (User Guide, p. 19). The channel is productivity: reduced GDP growth means lower revenue, higher expenditure-to-GDP ratios, and accelerating debt.

The second arrow is the one people miss. Revenue tracks nominal GDP, so a smaller economy collects less. Spending does not automatically shrink to match, and how much of it does is the expenditure rigidity control in Chapter 4. At full rigidity the whole revenue shortfall becomes borrowing. That is why the same warming scenario produces a much wider debt fan in one country than another.

The number	Under warming	What you control
Growth, g	Damage applied from 2030, rising with the temperature path	The scenario you read against baseline
Primary balance, pb	Revenue falls with GDP; spending falls only as far as rigidity allows	Expenditure rigidity, 0.0 to 1.0
Interest rate, r	Untouched by the climate block	Nothing

The channel is temperature, so hot low-latitude economies take the largest modelled hit. Bangladesh shows both halves of that sentence at once. It is exactly the kind of economy the modelled channel bites hardest, and it is also the economy that shows what the channel leaves out, because sea-level rise is not in this model at all (Chapter 6).

The six climate scenarios

Scenario	Warming (wrt present)	Based on	Description
Paris-Aligned (1.5°C)	+0.7°C	SSP1-2.6	International commitments met, with significant emission cuts

Moderate (2°C)	+1.6°C	SSP2-4.5	Current trends continue, with no new aggressive mitigation
High (4°C+)	+2.5°C	SSP3-7.0	Countries scale back mitigation, response fragmented
Hot (3°C)	+3.5°C	SSP3-7.0 (90th pctile)	Same emissions as High, using the 90th percentile of temperature increases
Hot + Adapted	+3.5°C	SSP3-7.0 (90th pctile)	Same temperatures as Hot, with faster adaptation
Hot + Unadapted	+3.5°C	SSP3-7.0 (90th pctile)	Same temperatures as Hot, with very slow adaptation

Warming values are IPCC best estimates for 2081-2100 relative to present (User Guide Table 1). Adding about 1.1°C for present-to-pre-industrial warming gives the more familiar above-pre-industrial framing: Paris below 2°C, Moderate about 2.7°C, High about 3.6°C, Hot about 4.6°C.

One naming difference is worth knowing before a meeting. A published fiscal risk statement may report a subset of these under its own labels. The worked case in Chapter 5 is one: it carries five scenarios and calls the last one "Vulnerable" rather than "Hot Unadapted". Same scenario, different label, and this course uses the Explorer's naming.

These scenarios are conservative in ways that matter for how you write up the results. Chapter 6 sets out what they leave out.

Self-check: where does it live?

A colleague sends you five questions in one email. For each, name which of the three numbers it moves, and where the tool builds that number. Commit to your five answers before opening the panel.

1. "Our fiscal rule ceiling is 50 percent of GDP. Does the model know that?"
2. "The projection assumes we keep growing at 6 percent forever. Does it?"
3. "Why does the debt path only start diverging across scenarios in 2030?"
4. "We are shifting from IDA credits to Eurobonds. Where does that show up?"
5. "Spending per person keeps rising in the projection. Who decided that?"

   DRAFT FOR TEAL: self-check answers

   Four of the five land on two of the three numbers, which is the point of the exercise: most questions that arrive as five separate problems are two.

   1. **The primary balance.** The debt target and the fiscal rule on/off switch both sit on the spending side, and both are yours to set.
   2. **Growth.** It is built from employment, productivity and inflation along a

gradual path, so it does not sit at the WEO rate forever.

3. **Growth, and the primary balance behind it.** The climate scenarios move both, and impacts start in 2030 by construction, with every scenario equal to the baseline before that (User Guide, p. 19).
4. **The interest rate.** The rate rule is where concessional-to-commercial transitions live. In V1 it sits at the Excel default, which is a limitation worth stating in your write-up.
5. **The primary balance.** Primary expenditure grows with productivity, inflation and total population, multiplicatively. Nobody decided it per person. It falls out of that formula.

If you missed 3, reread the warming section above. If you missed 5, the expenditure growth formula is the thing to hold: $(1 + a) \times (1 + b) \times (1 + c)$, never $a + b + c$.

2.7 The Python engine is checked against the Excel workbook, cell by cell

A reimplementation is worth nothing if it quietly disagrees with the original. This is the one place in the course where the verification story appears, and it is here because you should decide now whether to trust the numbers.

The figure below is the check itself on top, and underneath it the two claims that come out of the check, drawn to scale against each other.

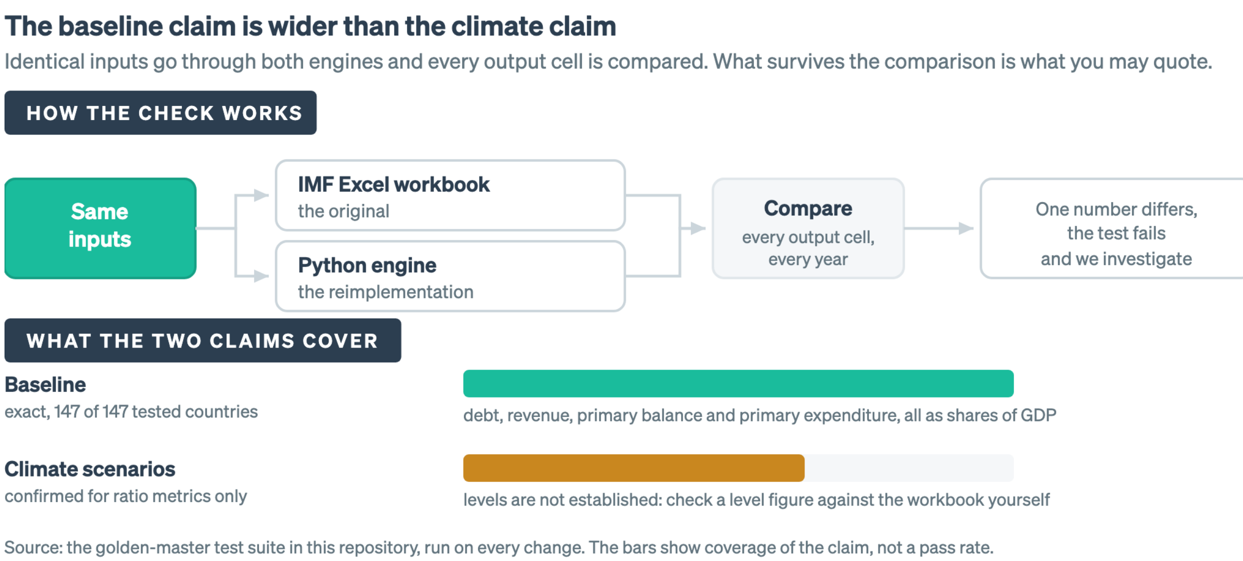


Figure 2.4: The bars are drawn to the reach of each claim rather than to a pass rate. Both claims passed. One of them was simply asked a narrower question.

Identical inputs go through both the original Excel workbook and the Python engine, and every output cell for every projection year is compared, which is why the two bars are different lengths: the baseline claim covers four series exactly, and the climate claim stops at the ratios.

Think of the comparison as an auditor reconciling two copies of the same ledger. If a single number differs, the test fails and we investigate.

Where that stands. Baseline parity is exact for 147 of 147 tested countries: zero percentage-point deviation on debt-to-GDP, revenue, primary balance and primary expenditure as shares of GDP. An additional 25 parameter sensitivity combinations were tested and passed. Climate-scenario parity is confirmed for ratio metrics, meaning debt-to-GDP, revenue and expenditure as percent of GDP, across the tested countries and scenarios.

The second claim is narrower than the first. Climate-scenario agreement has been established for the ratios you will quote in a Fiscal Risk Statement. It has not been established across every level series in the workbook. If your analysis turns on a climate-scenario figure in levels rather than as a share of GDP, check it against the Excel tool yourself.

The test suite is public and runs on every change. You do not have to take the claim on trust, because you can run it.

2.8 Wrapper: what you can now do

- You have run a full projection and exported the data. That file is capstone material.
- You can name the three numbers the equation needs and say what builds each one.
- You can separate the base projection model from the warming block, and name the two nodes the block attaches to.
- You can say what the parity claim covers (baseline, 147 of 147, exact) and what it does not (climate levels).

Common error at this stage: treating the tool as seven things to learn. It is one equation with three inputs, the inputs have suppliers, and warming is one block on top. If you can answer the "where does it live" self-check, you have the map.

On your desk this week: when someone hands you a fiscal projection, ask which of r, g and pb their headline number moves. It works on projections that have nothing to do with climate.

Chapter 3

The model, in three steps

> **In this module**
> Q-CRAFT is three things stacked on each other. An equation that turns three numbers into a debt path. A projection model that manufactures those three numbers year after year. A climate overlay that changes one of them. Read the three in order and every result the tool produces becomes traceable.

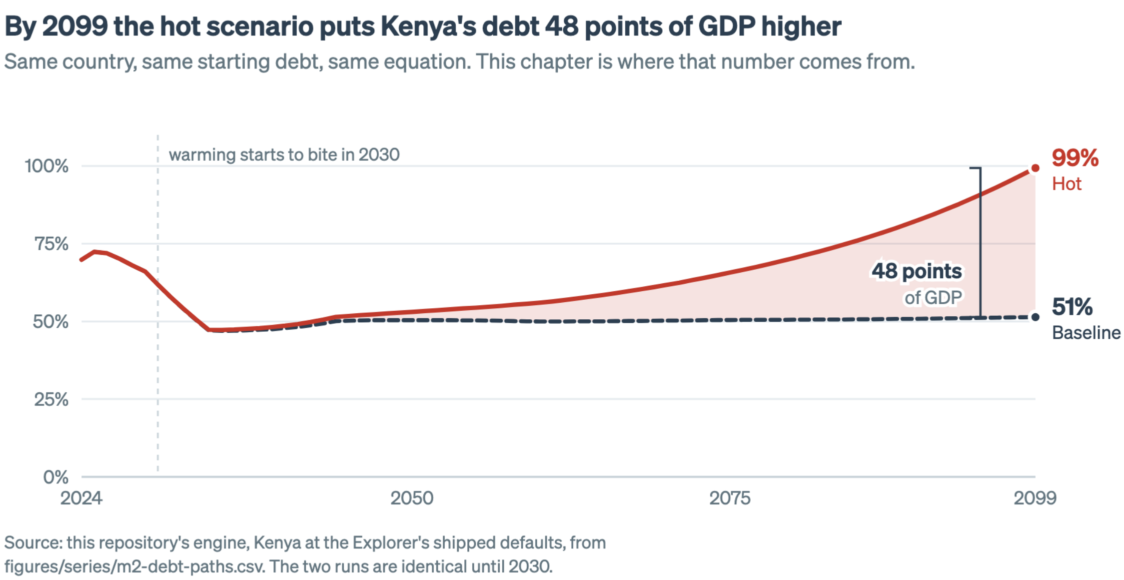


Figure 3.1: Both runs load the same country data and start from the same debt stock. The gap is what a single assumption about warming does once it has seventy years to compound, and every step of this chapter is a piece of it.

Forty-eight points of GDP is a large number to take on trust, and nothing about it is obvious. Both runs load the same Kenyan data. Both start from the same debt stock. Both use the same equation. One of them assumes the world warms along a hot path, and seventy years later the debt ratio is nearly double.

This module takes that number apart. It has three steps, and they build:

1. **An equation.** Three numbers a year, and it gives you a debt path.
2. **A growth engine.** Where those three numbers come from, which is most of what Q-CRAFT is.
3. **A climate overlay.** One block that changes one of the three, and the fan opens.

By the end you can point at any part of the chart above and say which step put it there.

3.1 By the end of this module you can

- **Compute** next year's debt ratio from this year's, given r, g and the primary balance
- **Trace** each of the three numbers back to the data source or the assumption that produced it
- **Explain** how a temperature path becomes a productivity growth rate, and why the estimate is not country-specific
- **Distinguish** the part of a climate result that comes from the damage estimate from the part that comes from your own spending assumption

---

3.2 Step 1. The equation

> **You are here.** Step 1 of 3: the equation. It needs three numbers a year and nothing has supplied them yet.

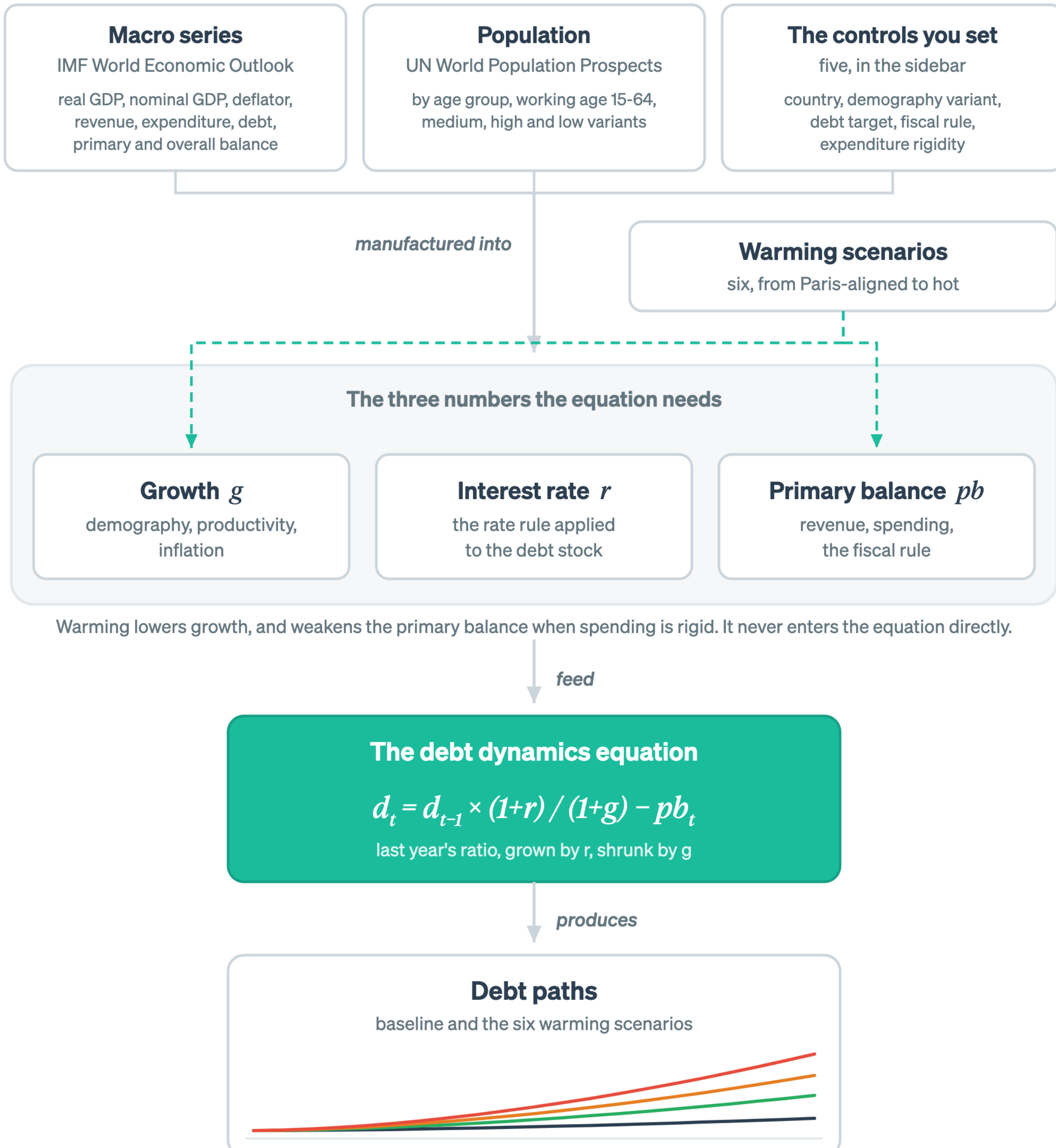


Figure 3.2: Step 1 is the equation node on its own. It needs three numbers a year, and nothing on this map has supplied them yet.

3.2.1 The identity is a stock and two flows

A government's debt is a stock. Next year it is this year's stock, plus interest owed on it, plus whatever it has to borrow to cover the gap between non-interest spending and revenue.

$$\text{Debt next year} = \text{debt this year} + \text{interest on it} + (\text{non-interest spending} - \text{revenue})$$

That last bracket is the primary deficit. Flip its sign and it is the primary balance. Nothing controversial has happened yet.

Now divide by GDP, because a stock of currency means nothing until it is set against the economy carrying it. The numerator grows at the interest rate. The denominator grows at the rate of nominal GDP. Two things are racing:

$$d_t = d_{t-1} \times \frac{1 + r_t}{1 + g_t} - pb_t$$

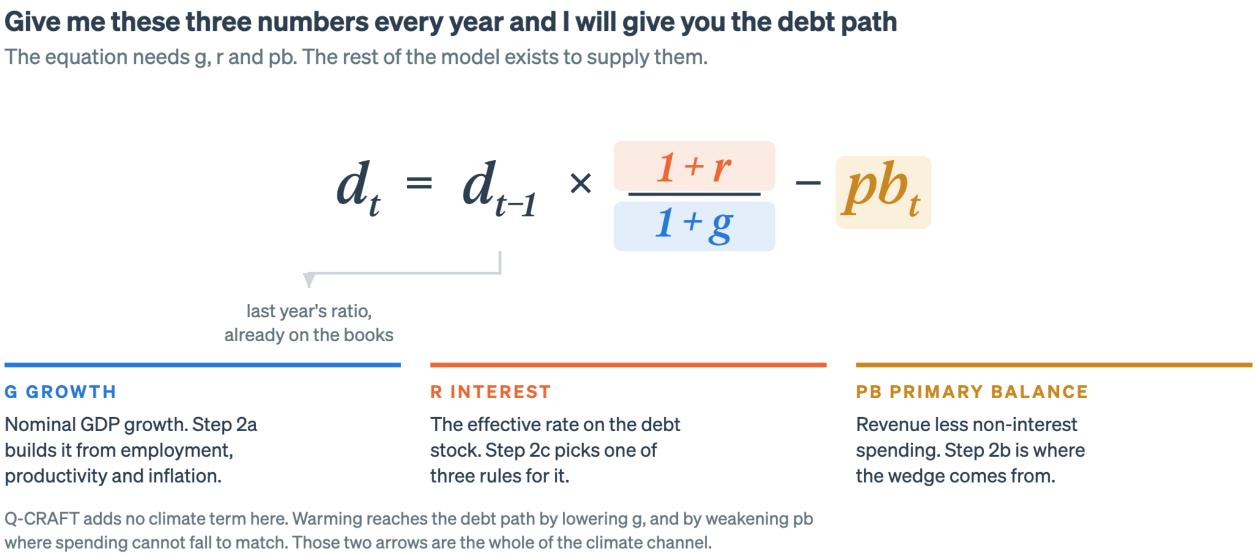


Figure 3.3: The first term is not one of the three. Last year's ratio is a stock you inherit, so the only things a projection has to manufacture are the growth rate, the interest rate and the primary balance.

The first term is not a decision. Last year's ratio is a stock the government inherits. The other three are the numbers a projection has to manufacture, and Step 2 is the manufacturing.

3.2.2 One year forward, and only half the movement is borrowing

Start a country at 60 percent of GDP. Give it an effective interest rate of 8 percent, nominal growth of 6 percent, and a primary deficit of 1 percent of GDP.

Multiply by 1.08 and divide by 1.06. That is a factor of 1.0189, so the ratio carries forward at 61.1. Then add the point of GDP the government borrowed. The answer is **62.1 percent**.

Two things happened. One point came from borrowing, which everyone expects. The other 1.1 points came from the amplifier, which nobody voted for.

Run it three years and the second effect grows, because it applies to a bigger stock each time.

End of year	Ratio carried forward	Plus borrowing	Debt ratio
1	61.1	+1.0	**62.1**

End of year	Ratio carried forward	Plus borrowing	Debt ratio
2	63.3	+1.0	**64.3**
3	65.5	+1.0	**66.5**

Borrowing contributes exactly 1 point a year, every year. The amplifier contributes 1.1 points in year one and 2.2 by year three.

One sensitivity. Run the identical three years with growth at 7 percent instead of 6 and the ratio lands at 64.7 rather than 66.5. One percentage point of growth, held for three years, is worth 1.8 points of debt ratio. Held for seventy, it is worth a different fiscal policy. That sentence is the entire reason a climate model that only touches growth can move a debt path this far.

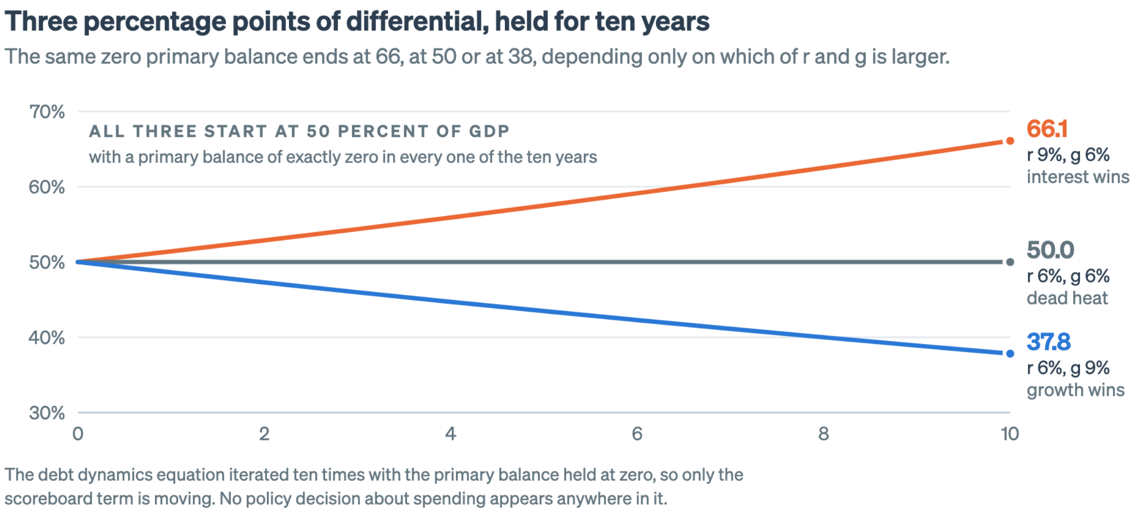


Figure 3.4: The middle line is the only place on this chart where a zero primary balance holds the ratio still, and it needs r and g to be exactly equal. Every other position moves the ratio with nobody borrowing for a new programme.

> ⚠ WIDGET-TODO: the debt sandbox
>
> Embed the debt sandbox widget here (lane 2, run 3: `apps/qcraft-web/widgets/*`). Iframe under Quarto for now, native React once the site moves to Astro. The widget lets a reader set r, g and pb and watch ten years of the equation redraw, which is the same exercise as the table above with the arithmetic done for them. This callout is the anchor point. The prose above carries the full idea on its own, because the default state of any interactive has to work for readers who never touch it.

> ℹ Going deeper: the derivation, and what the interest-growth gap does
>
> **Building the ratio from the stock.** Write the stock recursion in local currency, with D for debt, PB for the primary balance and Y for nominal GDP:
>
> $$D_t = D_{t-1}(1 + r_t) - PB_t$$

Divide both sides by Y_t, and use $Y_t = Y_{t-1}(1 + g_t)$:

$$\frac{D_t}{Y_t} = \frac{D_{t-1}(1 + r_t)}{Y_{t-1}(1 + g_t)} - \frac{PB_t}{Y_t}$$

which is the equation above with $d = D/Y$ and $pb = PB/Y$. The only substantive step is dividing by a denominator that is itself growing, and that step is where the whole interest-growth story comes from.

The amplifier has a name. $\frac{1+r}{1+g}$ is usually reported as a single number, the interest-growth differential $(r - g)/(1 + g)$. Its sign is the most consequential fact about a country's debt position, and in the short run no minister chooses it.

Take a debt ratio of 50 percent of GDP and a primary balance of exactly zero, so the second term drops out and only the amplifier is left.

r	g	After 1 year	After 10 years	What is happening
9%	6%	51.4	66.1	Interest wins. The ratio climbs with no new programme borrowing.
6%	6%	50.0	50.0	Dead heat. A zero primary balance stabilizes debt only here.
6%	9%	48.6	37.8	Growth wins. The ratio falls while the primary account is balanced.

Three percentage points of differential, held for a decade, is the difference between 66.1 and 37.8 percent of GDP. Nothing in that table is a decision about spending.

The balance that holds the ratio still. Set next year's ratio equal to this year's and solve for the primary balance:

$$pb^* = d \times \frac{r - g}{1 + g}$$

That is the debt-stabilizing primary balance, a standard quantity in any debt sustainability analysis. It says that the worse your interest-growth gap and the higher your existing debt, the larger the surplus you need to stand still.

Run it for the country in the table. Debt at 50 percent of GDP, r at 9 percent, g at 6 percent:

$$pb^* = 0.50 \times \frac{0.09 - 0.06}{1.06} = 0.0142$$

The arithmetic asks for a primary surplus of about 1.4 percent of GDP, every year, to keep the ratio at 50. A government running a primary deficit of 1 percent in that position is 2.4 points of GDP away from standing still.

Take one point off growth, to 5 percent, and the required surplus rises from 1.4 to 1.9 percent of GDP. The bill for standing still went up by half a point because a macro variable moved. Hold that thought until Step 3, where a climate scenario does exactly this and nothing else.

Three claims a colleague might send you. For each, decide whether you agree and what you would check.

1. "Our debt ratio fell last year, so the fiscal position improved." Not necessarily. A ratio can fall on its denominator. Uganda's own Fiscal Risk Statement makes this point about FY2022/23: the ratio fell to 46.9 percent from 48.4, and the statement attributes part of the fall to nominal GDP rising on high inflation, warning the effect would subside as inflation fell. Check whether the stock of debt fell or the denominator moved.
2. "We should compare our primary deficit to the regional average." Wrong benchmark. The sustainable primary deficit depends on your own r, g and starting debt. The comparison that means something is your actual primary balance against your own pb^*. Two countries with identical deficits can be in opposite positions.
3. "Inflation is helping us, because it inflates away the debt." Half right, and the wrong half is expensive. Unexpected inflation raises nominal g and deflates the ratio once. It then feeds into the effective rate on new and floating-rate borrowing, so r follows g up with a lag. In Q-CRAFT specifically, revenue and primary expenditure both grow with inflation, so a permanently higher inflation assumption largely cancels out of the ratio.

Step 1 lands here. Give me r, g and pb every year and I will give you the debt path. The rest of the model exists to supply those three.

3.3 Step 2. From an equation to a projection model

> **You are here.** Step 2 of 3: the growth engine. Three numbers a year, seventy years, from published data and five controls.

An equation that needs r, g and pb is useless until something produces them for 2031, and 2032, and every year to 2099. That is the bulk of Q-CRAFT, and it is three separate small machines. This step takes them in the order the equation reads: growth, then the primary balance, then the interest rate.

3.3.1 Step 2a. Growth is built from three published series

Q-CRAFT does not estimate a production function. There is no capital stock in it, no factor shares, no total factor productivity residual. What it uses is an accounting identity, which is a much weaker and much more portable claim:

$$\text{output} = \text{people working} \times \text{output per person working}$$

That is true by the definition of output per worker. Turn it into growth rates and add prices, and you have nominal GDP growth:

$$1 + g = (1 + \text{employment growth}) \times (1 + \text{productivity growth}) \times (1 + \text{inflation})$$

Three inputs, each from somewhere you can check.

- **Employment growth** comes from the UN's World Population Prospects, working-age population aged 15 to 64, medium variant by default. Nobody sets it. It arrives with the country.
- **Productivity growth** is GDP per employed person. You set where it starts and where it ends, and the tool slides it between the two along a logistic path.
- **Inflation** is the GDP deflator, on the same start-and-end treatment.

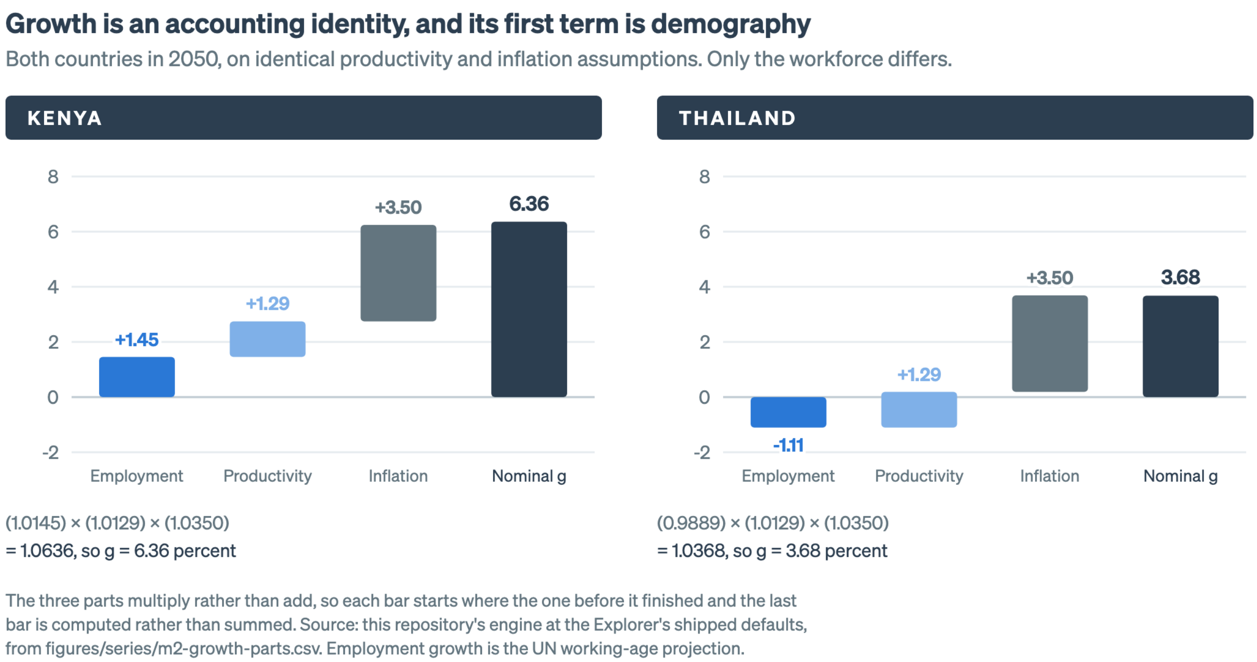


Figure 3.5: The productivity and inflation assumptions are the tool's defaults and are identical across the two panels, so the whole of the difference in the final bar is the working-age population. Demography is not a detail here; it is the first term.

The toy numbers, from the chart. Kenya in 2050 has employment growing at 1.45 percent, productivity at 1.29 and inflation at 3.50. Multiply: $(1.0145)(1.0129)(1.0350) = 1.0636$, so g is 6.36 percent.

Thailand in the same year has the same productivity and the same inflation, because those are the tool's defaults and neither has been changed. Its working-age population is falling at 1.11 percent a year. Multiply: $(0.9889)(1.0129)(1.0350) = 1.0368$, so g is 3.68 percent.

Two and a half points of nominal growth separate them, and the entire difference is demography. Put that back into Step 1's amplifier and it is the difference between a debt ratio that drifts down and one that climbs.

3.3.1.1 The World Economic Outlook hands over in 2030

The three parts above do not drive the projection from the first year. For the first stretch the tool uses the IMF's own numbers, and it only starts doing its own arithmetic once those run out.

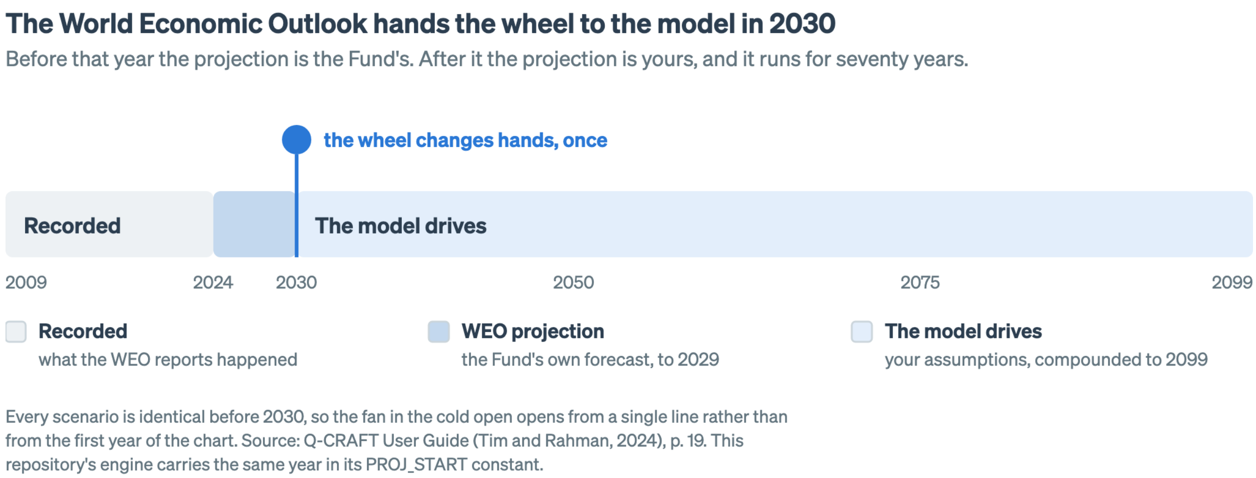


Figure 3.6: The handover is why a Q-CRAFT result is not a forecast. For the first twenty years you are reading the Fund's projection, and for the seventy after it you are reading your own assumptions compounded.

This matters for reading results. A Q-CRAFT chart is two different objects glued together: a short WEO forecast, then a long mechanical extrapolation of assumptions you chose. Nobody is forecasting 2087. The tool is answering what happens if these rates hold.

> **Going deeper: why an identity rather than a production function**
>
> **What the choice buys.** A production function needs a capital stock. Capital stock series are estimated rather than observed, they are missing or unreliable for a large share of the world, and building them means assumptions about depreciation and initial conditions that a user cannot check. Q-CRAFT is designed to run for 171 economies on data that is already published, and the identity route needs only population and GDP per worker.
> **What the choice costs.** Because there is no capital, there is no investment channel. A scenario cannot reduce growth by reducing the return to investment, and adaptation spending cannot raise growth by adding to the capital stock. Everything has to arrive through productivity, which is why Step 3's climate damage shows up as a cut to productivity growth and nowhere else.
> **A wording note.** The IMF User Guide describes the baseline as "grounded in a simple production function and standard debt dynamic equation" (p. 5). The arithmetic the tool actually performs is the identity above: employment times productivity, with no capital term and no factor shares. The two descriptions point at the same lines of code. This guide uses "accounting identity" because it is the more precise description of what is being assumed, and because readers who hear "production function" reasonably expect a capital stock that is not there.
> **The productivity path.** Between the start rate you set and the end rate you set, the tool does not interpolate in a straight line. It uses a logistic curve, so growth stays near the start rate for a while, falls quickly through the middle of the century, then flattens onto the end rate. That shape is a claim: convergence toward the frontier happens fastest when a country is furthest from it. Chapter 4 covers how to choose the two endpoints and what evidence you can cite for them.

3.3.2 Step 2b. The primary balance is a wedge between two rules

Revenue and spending in Q-CRAFT are not forecasts of policy. They are rules, and the rules are simple enough to state in two sentences.

Revenue rides nominal GDP. The revenue-to-GDP ratio is held at its last observed value and carried forward. The assumption behind this is unchanged tax policy: no new taxes, no new exemptions, no change in collection efficiency. Whatever nominal GDP does, revenue does.

Spending follows people, productivity and prices. Primary expenditure grows by total population, times productivity, times inflation. The assumption behind this is unchanged service levels: the same real provision per person, priced up each year.

Notice that the two demographic terms are not the same one. Growth uses the working-age population, because that is who works. Spending uses the total population, because that is who is served. In a country whose workforce shrinks faster than its population, those two pull in opposite directions, and that is the fiscal squeeze demography produces on its own with no climate anywhere near it.

The gap between revenue and spending is the primary balance, and it is the only one of the three numbers a government sets directly.

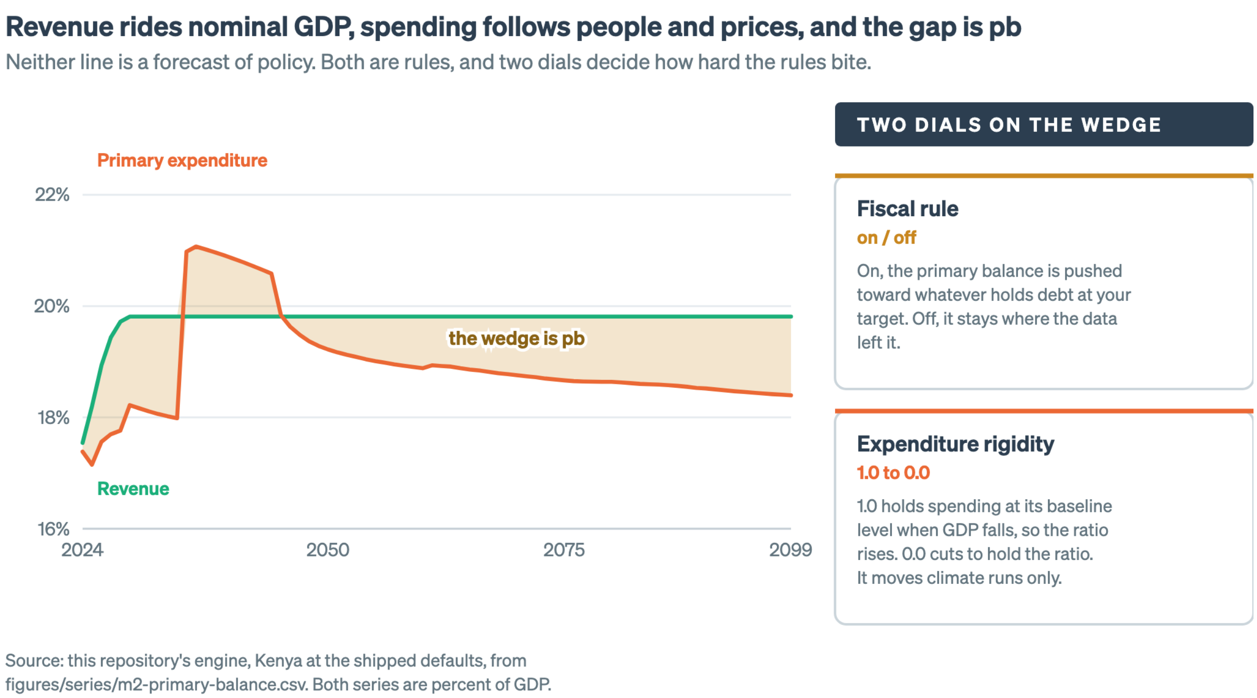


Figure 3.7: Revenue is a fixed share of GDP by assumption, so it cannot rescue a projection on its own. Everything interesting in the wedge happens on the spending line, which is where both dials act.

The toy numbers. Kenya's revenue settles at 19.81 percent of GDP and stays there for seventy years, because that is what the rule says. Primary expenditure in 2050 is 19.22 percent. The wedge is 0.59 percent of GDP, and that is the primary balance the equation gets for that year.

Is 0.59 enough? Step 1 already told you how to check. Kenya's debt-stabilizing primary balance in 2050 is 0.59 percent of GDP. The two are the same number to two decimal places, which is not a coincidence: the fiscal rule is on, and holding debt at the target is exactly what the rule is doing.

Two dials sit on the wedge, and both of them appear again in Step 3.

The **fiscal rule** decides whether the primary balance is left where the spending rule put it or pushed toward whatever holds debt at your target. It works by adjusting next year's primary expenditure by this year's fiscal gap, which is the distance between the actual primary balance and the debt-stabilizing one from Step 1. With the rule on and a 50 percent target, Kenya's baseline converges on 50 and stays. With it off, the ratio goes wherever the spending rule takes it.

Expenditure rigidity decides what happens to spending when GDP disappoints. At 1.0, spending stays at its baseline level in local currency, so a smaller GDP means a higher spending-to-GDP ratio and a worse balance. At 0.0, spending is cut to hold the ratio where the baseline had it. The dial changes nothing in the baseline itself. It only bites when a scenario makes GDP smaller than the baseline, which is to say it only bites in Step 3.

> **i** Going deeper: why the rule matters more than it looks
>
> **The rule fires in both directions.** It is not a one-way consolidation device. It adjusts spending when debt is rising and above target, and it also adjusts when debt is falling and below target, which relaxes rather than tightens. In the years when debt is moving toward the target on its own it does nothing. The adjustment is a level, in local currency, added after the multiplicative growth terms rather than folded into them.
>
> **It is a baseline-only device.** Q-CRAFT applies the rule when building the baseline. It does not apply it inside the climate scenarios, which run on the baseline's spending path with no further consolidation. That asymmetry is deliberate: the climate scenarios ask what happens if warming arrives and policy does not respond, which is the right question for a fiscal risk statement and the wrong one for a policy plan.
>
> It also means a large part of any baseline-to-scenario gap is the rule, not the damage. Step 3 puts a number on that for Kenya, and the number is bigger than most readers expect.
>
> **A run with the rule off looks alarming and often is not.** Turn the fiscal rule off for Kenya and the baseline debt ratio falls to zero, because the spending rule leaves a persistent primary surplus and the amplifier works in the country's favour for most of the projection. The baseline is floored at zero and the climate scenarios are not, so the climate runs go through zero into negative territory and the chart inverts. That floor asymmetry is a real feature of the tool and it is covered in Chapter 6. For now the practical rule is that a baseline sitting on zero is a signal to check your assumptions rather than a finding.
>
> **Revenue cannot rescue anything.** Because the revenue-to-GDP ratio is fixed by assumption, there is no channel in this model through which better collection or a broader base improves the debt path. If your question is about revenue mobilisation, this is the wrong tool and Chapter 6 says which one is right.

3.3.3 Step 2c. The interest rate is a choice between three rules

The effective interest rate is the average rate the government pays on its whole stock of debt. It is not a market yield, and Q-CRAFT does not model one. It offers three rules for projecting it forward, and they are three different views about what stays constant.

Hold the nominal rate flat. Take the last observed effective rate and carry it to 2099. The view is that the government's borrowing costs are what they are and nothing systematic changes them.

Hold the interest-growth differential flat. Take the last observed gap between r and g and preserve it, so the interest rate follows nominal growth down as the economy matures. The view is that rates and growth move together.

Hold the real rate flat. Fix a long-run real rate and let the nominal rate follow inflation. The view is that the real cost of borrowing is the structural quantity.

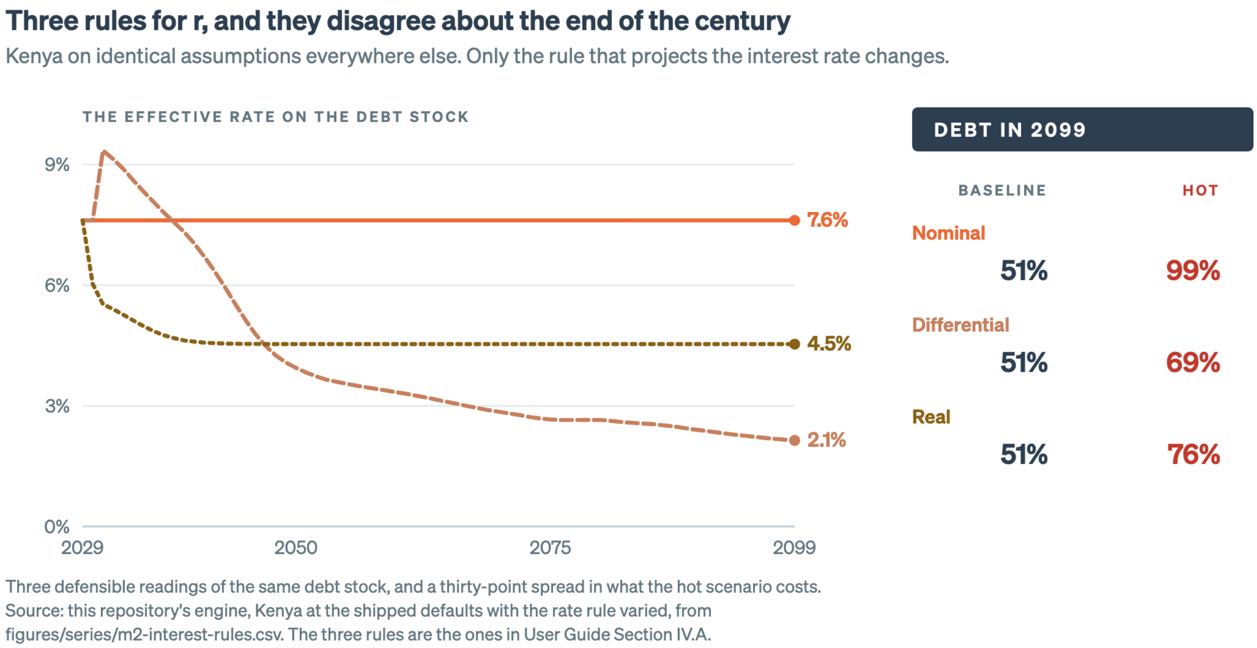


Figure 3.8: The rule is the assumption that moves the climate answer most, and it is the one the Explorer does not yet expose. If you need the other two, that is a reason to run the workbook alongside it.

The toy numbers. For Kenya the three rules give an effective rate in 2099 of 7.6, 2.1 and 4.5 percent. All three baselines end at 51 percent of GDP, because the fiscal rule is doing the work in each of them. Under the hot scenario, where no rule intervenes, they end at 99, 69 and 76.

Thirty points of GDP of disagreement, from one dropdown, with every other assumption identical. This is the single assumption that moves a climate answer most, and the Explorer's V1 does not put it in the sidebar: it runs the first rule. If the answer matters, run the workbook alongside and try the other two.

> **i** Going deeper: why the differential rule is so much gentler
>
> Under the flat-nominal rule Kenya pays 7.6 percent in 2099 while nominal growth is 4.6 percent, so the amplifier is above one and every point of debt compounds. Under the differential rule the interest rate is tied to growth, so when climate damage lowers g it lowers r with it, and the amplifier barely moves.

That is not the interest rate rule being clever. It is the rule assuming away part of the risk, by construction. A country whose borrowing is mostly concessional and long-dated does have an effective rate that responds slowly to anything, which argues for the flat-nominal rule. A country refinancing at market rates every few years has a rate that moves with conditions, which argues for one of the other two. The choice is a claim about your debt portfolio, and it is worth a sentence in your export packet either way. Chapter 4 sets out what to look at before choosing.

3.3.4 Step 2 lands here

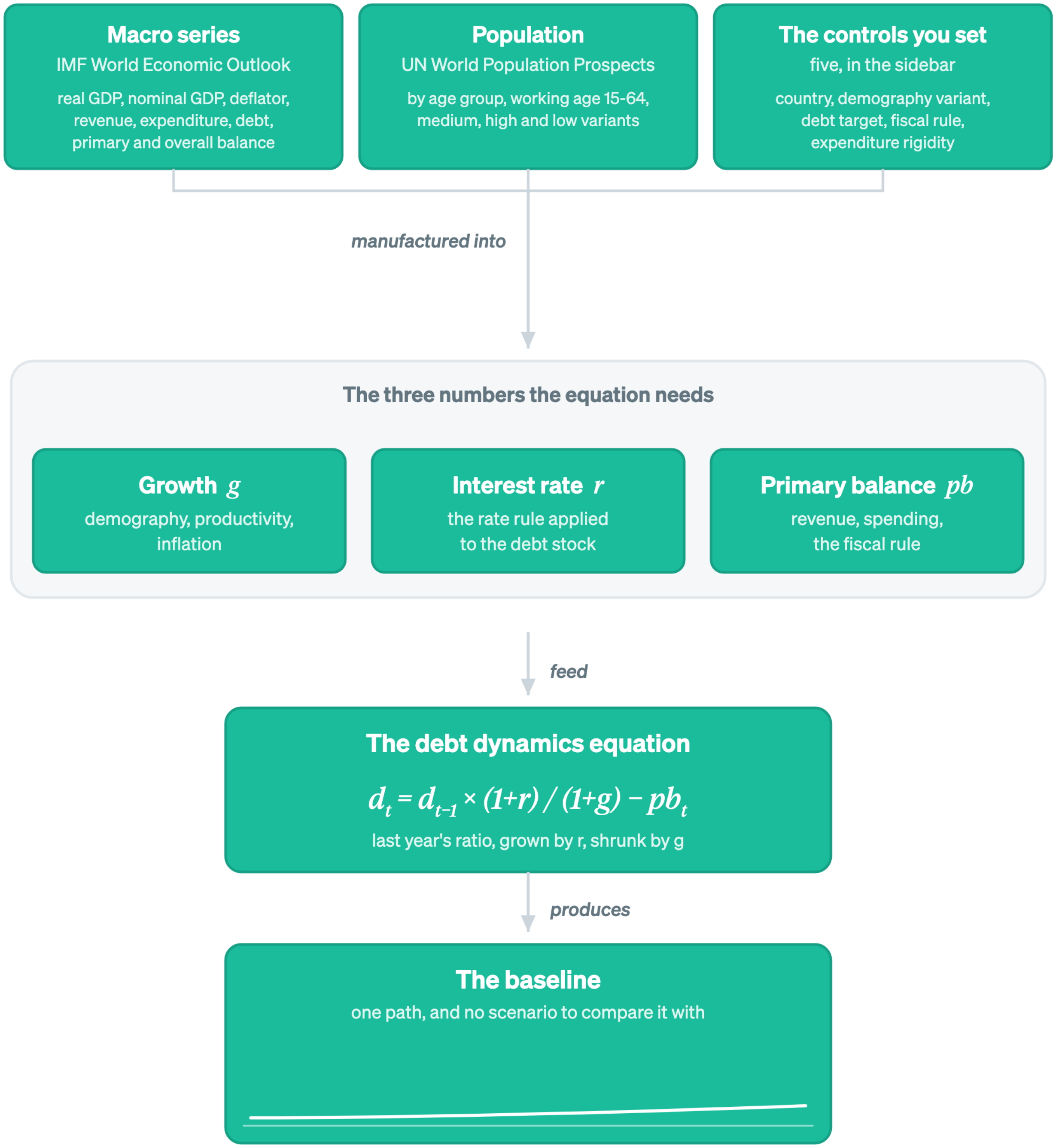


Figure 3.9: The end of Step 2: a complete long-term fiscal projection model. The warming block is not missing from this drawing, it has not been built yet. Run this chain and you get the baseline.

Two published data sources, five controls, three small machines, one equation. Run this chain and you get a baseline: a complete long-term fiscal projection model with no climate in it at all. Everything to this point would work identically in a world with a stable climate.

Predict, observe, explain: run the baseline before the fan opens

Predict. Take a country you know. Write down whether its baseline debt ratio in 2099 is above or below where it starts, and one sentence on why, naming r and g.
Observe. Open the Q-CRAFT Explorer, select that country and read the Baseline tab. Note the debt trajectory, then look at the Fiscal Balances chart and note whether the primary balance is in surplus or deficit.
Explain. Reconcile the two. If the primary balance is in deficit and the ratio is not exploding, the amplifier is working in that country's favour: nominal growth is beating the effective interest rate, which is the normal position for a country borrowing largely on concessional terms. If you expected a deficit to mean rising debt, that is the misconception from Chapter 1, and you have now watched it fail on real data.
Then do it again with a country that borrows commercially, and watch the same chart tell the opposite story. The arithmetic did not change between them. The sign of r minus g did.

SCREENSHOT-TODO

Baseline tab with the debt trajectory and the Fiscal Balances panel side by side, for a country where the primary balance is in deficit and the ratio is stable, so the reconciliation is visible in one image. Kenya works, and using the chapter's own country keeps the numbers checkable against the figures above.

3.4 Step 3. Snapping on the climate block

You are here. Step 3 of 3: the overlay. One block, two arrows, no new equation.

The climate part of Q-CRAFT is smaller than most people expect. It is a table of numbers, one per country per scenario per year, that says how much smaller GDP is than it would otherwise have been. That table comes from somewhere, and this step is where it comes from.

3.4.1 Step 3a. The econometrics measures deviation from a country's own norm

The estimates behind every climate number in Q-CRAFT trace to one paper: Kahn, Mohaddes, Ng, Pesaran, Raissi and Yang (2021), "Long-term macroeconomic effects of climate change", published in *Energy Economics*. Here is what it did, in the order the argument runs.

It used a long global panel. 174 countries, annual data, 1960 to 2014. Not a simulation of a warmer world, and not a small sample of hot countries. Fifty-five years of every country's growth against fifty-five years of every country's weather.

The variable that does the work is deviation, not heat. This is the move that makes the paper different from most of the literature. The regressor is not a country's temperature. It is the gap between this year's temperature and that country's own historical norm, where the norm is a moving average of the past thirty years. A country at 25 degrees that has always been at 25 degrees is at its norm and the model expects no effect. A country at 8 degrees that has always been at 6 is two degrees above its norm and the model expects one.

The reason for building it this way is stated in the paper: temperature is trended in almost every country, so putting the level into a growth regression forces a trend into growth that the data does not show. Deviations have no trend, so they can be used.

A sustained deviation costs output every year it lasts. The estimated effect is on the growth rate rather than the level, so the loss does not arrive once and stop. The headline estimate, for a persistent above-norm rise of 0.01 degrees a year: per-capita real output growth is lower by 0.0543 percentage points a year. Deviations of precipitation showed no statistically significant effect and drop out.

Adaptation is how fast the norm catches up. This is the part worth slowing down for, because it is where the tool's adapted and unadapted scenarios come from.

The norm is a trailing average. If temperature rises steadily, the norm chases it and always lags. How far it lags depends on how many years the average covers. The paper works this out exactly: under a steady warming trend, the average gap between this year's temperature and an m-year trailing norm settles at $\frac{m+1}{2}$ years' worth of warming.

So a thirty-year window leaves a gap of about 15 years' warming. A fifty-year window leaves about 25 years' worth. Same temperature path, a gap two thirds larger, and a proportionally larger drag on growth every year. A short window is a country whose economy treats the new temperature as normal quickly. That is what adaptation means here, and it is the only thing adaptation means in this model.

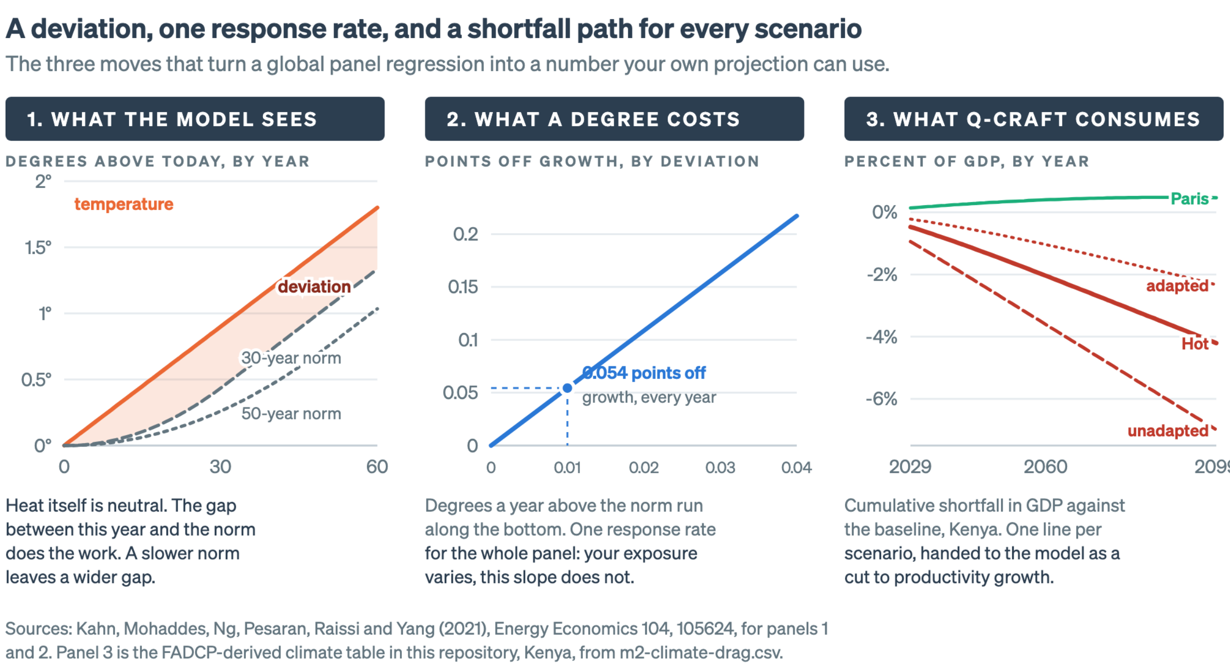


Figure 3.10: Panel two is the one to argue with. A single response rate for every country is what makes the estimate portable, and it is also what makes it a floor rather than a country-specific answer.

The response is pooled across countries. One number, 0.0543 percentage points per 0.01 degrees a year, applies to every country in the panel. The regression has a country fixed effect, so each country gets its own average growth rate, but there is a single long-run coefficient on temperature deviation for all 174 of them. The IMF User Guide states this plainly about the estimates Q-CRAFT uses: "The authors do not estimate country-specific impact of temperature on GDP. Rather, temperature projections vary between countries, leading to different GDP impacts" (Box 4, p. 34).

Keep that sentence. It is the single most important thing to know about how conservative a Q-CRAFT climate result is, and Step 3d comes back to it.

The toy numbers, adapted against unadapted. Kenya's cumulative GDP shortfall under the hot scenario in 2099 is 4.20 percent. The same temperature path with a twenty-year window gives 2.33 percent. With a fifty-year window it gives 6.99 percent.

Check those against the window arithmetic. Twenty years leaves 10.5 years of warming in the gap against thirty years' 15.5, a ratio of 0.68. Fifty years leaves 25.5, a ratio of 1.65. The tool's own numbers give 0.55 and 1.66. The unadapted case lands almost exactly where the arithmetic puts it. The adapted case does slightly better than the window width alone predicts.

Going deeper: what the paper found, and what changed between drafts

The global counterfactuals. The paper runs its estimates forward against IPCC temperature projections. Under an unmitigated path of 0.04 degrees a year, world real GDP per capita is more than 7 percent lower by 2100 than in a reference case where temperature keeps rising along its own 1960 to 2014 trend. Under the Paris path of 0.01 degrees a year, about 1 percent. The published abstract adds that losses reach 13 percent if the variability of climate conditions rises alongside the average.

Note what the comparison is against. The reference case is not a world that stopped warming. It is a world where each country's temperature continues its own historical trend. Q-CRAFT inherits that convention, so a Q-CRAFT baseline is a continued-historical-warming world rather than a no-climate-change world. In a few countries this makes the mitigation scenarios look mildly positive, which is why Kenya's Paris line sits slightly above zero in the third panel above rather than at it.

The window is the adaptation lever, and the losses move a long way with it. The working paper's own counterfactual table gives world losses by 2100 under the unmitigated path of 4.44 percent with a twenty-year window, 7.22 with thirty, and 9.96 with forty. The paper's conclusion is that adaptation reduces the loss and is highly unlikely to eliminate it.

One thing to know if you go to the working paper. The freely available 2019 working papers (Dallas Fed Globalization Institute WP 365, NBER 26167) are an earlier draft of the 2021 journal article, and they differ in two places that matter. The working paper reports point estimates of 7.22 and 1.07 percent where the published abstract says "more than 7 percent" and "about 1 percent". More substantively, the working paper concludes that the findings "apply equally to poor or rich, and hot or cold countries", while the published abstract states that the marginal effects of temperature shocks vary across climates and income groups. Cite the published article for anything load-bearing.

3.4.2 Step 3b. What the IMF does with those estimates

The paper gives a response rate. Turning that into something a fiscal projection can use takes two more steps, and the IMF's Fiscal Affairs Department did both.

It built the temperature side. The FADCP Climate Dataset (Massetti and Tagklis) assembles country-level historical temperature from the CRU observational record and country-level projected temperature from CMIP6 climate models, for each emissions pathway.

It re-ran the counterfactuals. Centorrino, Massetti and Tagklis (2024), "Climate Effects on GDP Growth: Updated Estimates of Kahn et al. (2021)", applies the Kahn method to that dataset and produces, for 171 economies, a per-country per-scenario per-year path of GDP impact. That table is what Q-CRAFT reads. It is also why Q-CRAFT's numbers do not exactly reproduce the numbers in the published paper: the paper used the CMIP5-generation RCP pathways, and the FADCP dataset uses the CMIP6-generation SSP pathways.

So is it country-specific? Yes and no, and the distinction is worth being exact about.

	Country-specific?	Why
Your warming path	**Yes.**	CMIP6 projections for your grid cells, population-weighted
Your historical norm	**Yes.**	Your own observed record, so your deviation is your own
The response to a degree of deviation	**No.**	One pooled coefficient, the same for every country

Exposure is yours. The response rate is the world's. Two countries with the same projected deviation get the same estimated growth drag, whatever their income, their institutions, their sectoral mix or their existing heat.

The six scenarios, and one thing about their names. Paris is SSP1-2.6. Moderate is SSP2-4.5. High is SSP3-7.0. Hot uses the same SSP3-7.0 emissions as High but takes the 90th percentile of the climate models rather than the median, so **Hot is more severe than High**, not less. Hot Adapted and Hot Unadapted are the same Hot temperature path with the norm window set to 20 and 50 years. Kenya's 2099 shortfalls run: Paris +0.47 percent, Moderate −0.06, High −1.94, Hot −4.20, Hot Unadapted −6.99. The dataset deliberately excludes SSP5-8.5, on the ground that its emissions are implausibly high.

3.4.3 Step 3c. The drag enters through g, and leans on pb

Everything above produces one number per year: how much smaller GDP is than the baseline said it would be. Q-CRAFT converts that into a cut to productivity growth and hands it to Step 2a. Impacts start in 2030.

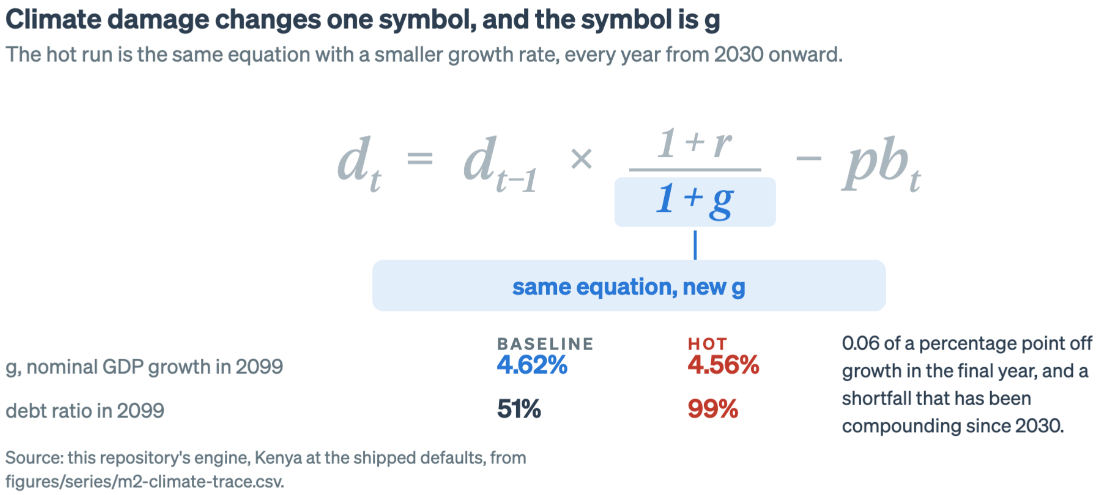


Figure 3.11: Nothing structural changes between a baseline and a climate run. The model recomputes the same line with a smaller growth rate, which is why reading a climate result always means comparing two runs rather than reading one.

That is the whole of the first channel. The equation is not modified. No climate term is added. Employment is unchanged, because warming does not move the UN population projection. Inflation is unchanged. The interest rate is unchanged. One symbol is smaller.

The second channel is the primary balance, and it only exists because of a dial. Revenue is a fixed share of GDP, so when GDP falls revenue falls with it and the revenue ratio does not move. Spending at rigidity 1.0 stays where the baseline put it in local currency. Smaller GDP under the same spending means a higher spending ratio, so the wedge narrows and the primary balance gets worse.

Watch this happen in Kenya's numbers. In 2050 the baseline primary balance is 0.59 percent of GDP and the hot run's is 0.40. By 2099 the baseline is 1.41 and the hot run is 0.72.

Now put that against what Step 1 said each of them needs. The baseline needs 1.47 to hold its ratio still and runs 1.41, so it is roughly stable. The hot run needs 2.83 and runs 0.72, so it is short by more than two points of GDP every year, on a debt stock that is already twice as large. That is what a spiral looks like in this equation, and it is why the fan in the cold open bends upward rather than drifting.

How much of the 48 points is the rigidity assumption? A lot, and you should know the number before you quote the headline.

Expenditure rigidity	Kenya's debt in 2099, hot	Gap over the baseline
1.0, spending fully sticky	99	48 points
0.5	77	26 points
0.0, spending fully flexible	55	4 points

Forty-four of the forty-eight points depend on assuming a government watches GDP fall for seventy years and never adjusts. That assumption is defensible for a fiscal risk statement, whose job is to describe what happens if nothing is done. It is not a forecast, and presenting it as one will not survive the first serious question. Say which dial you used.

> **i** Going deeper: the mechanics, and how to check them
>
> **How the shortfall becomes a growth rate.** The climate table holds a cumulative GDP index shortfall for each year. The engine takes the year-on-year first difference of that index and adds it to the baseline's productivity growth rate for that year. For Kenya in 2099 the baseline productivity growth is 1.200 percent and the hot run's is 1.144, a difference of 0.056 percentage points. That single year's cut is almost invisible. Seventy of them compounded, against an interest rate that does not move, is the entire result.
>
> **Why the numbers get large slowly.** In 2050 Kenya's cumulative GDP shortfall under Hot is 1.5 percent and the debt gap is 2.6 points. In 2099 the shortfall is 4.2 percent and the gap is 48 points. The shortfall roughly tripled and the gap grew about eighteen-fold. Three things drive the divergence: the primary balance falls further behind what it needs every year, the interest-growth differential widens as growth slows, and the baseline has a fiscal rule pulling it back to target while the climate scenarios have none.
>
> **Two things you can check for yourself.** Open the Data tab and export both runs. Revenue as a percent of GDP should be identical in the baseline and every scenario, to the decimal. Employment growth should also be identical. If either differs, something has gone wrong, because neither is supposed to respond to warming at all.

3.4.4 Step 3d. What this structure is good for, and what it cannot see

The design choices in Steps 1 to 3 have consequences, and the honest way to hold them is as a pair rather than as a caveat bolted onto a result.

What the structure buys.

- **It is empirical.** The damage estimate comes from fifty-five years of observed data across 174 countries, not from a calibrated guess about the future.
- **It is disciplined.** Warming touches growth, and the primary balance through a dial you set. There is no free parameter for how bad you feel climate change will be.
- **It runs anywhere.** Because it needs only published temperature records, climate model output, WEO series and UN population, it works for 171 economies without a bespoke country dataset or a mission.

What the structure costs.

- **One response rate for the world.** The pooled coefficient means the estimate cannot know that your agriculture is rain-fed, that your capital sits on a floodplain, or that your labour force works outdoors. Countries with concentrated exposure get the average response.
- **Only the slow productivity channel.** The scenarios contain no natural disasters, no sea-level rise, no tipping points, no non-market damages, and no adaptation spending. Every one of those exclusions is a cost left out.
- **No feedback in either direction.** Damage does not reduce investment, because there is no capital. Spending cuts under a low rigidity setting do not reduce growth, because nothing feeds back into productivity. The tool is a partial-equilibrium set-up and the User Guide says so.

Put together: a Q-CRAFT climate result reads as a lower bound on fiscal impact through the modelled channel, rather than as a central estimate of total impact. Chapter 6 works through what that licenses you to write and what it does not, including the specific cases where the arithmetic can mislead.

3.4.5 Step 3e. Back to the chart at the top

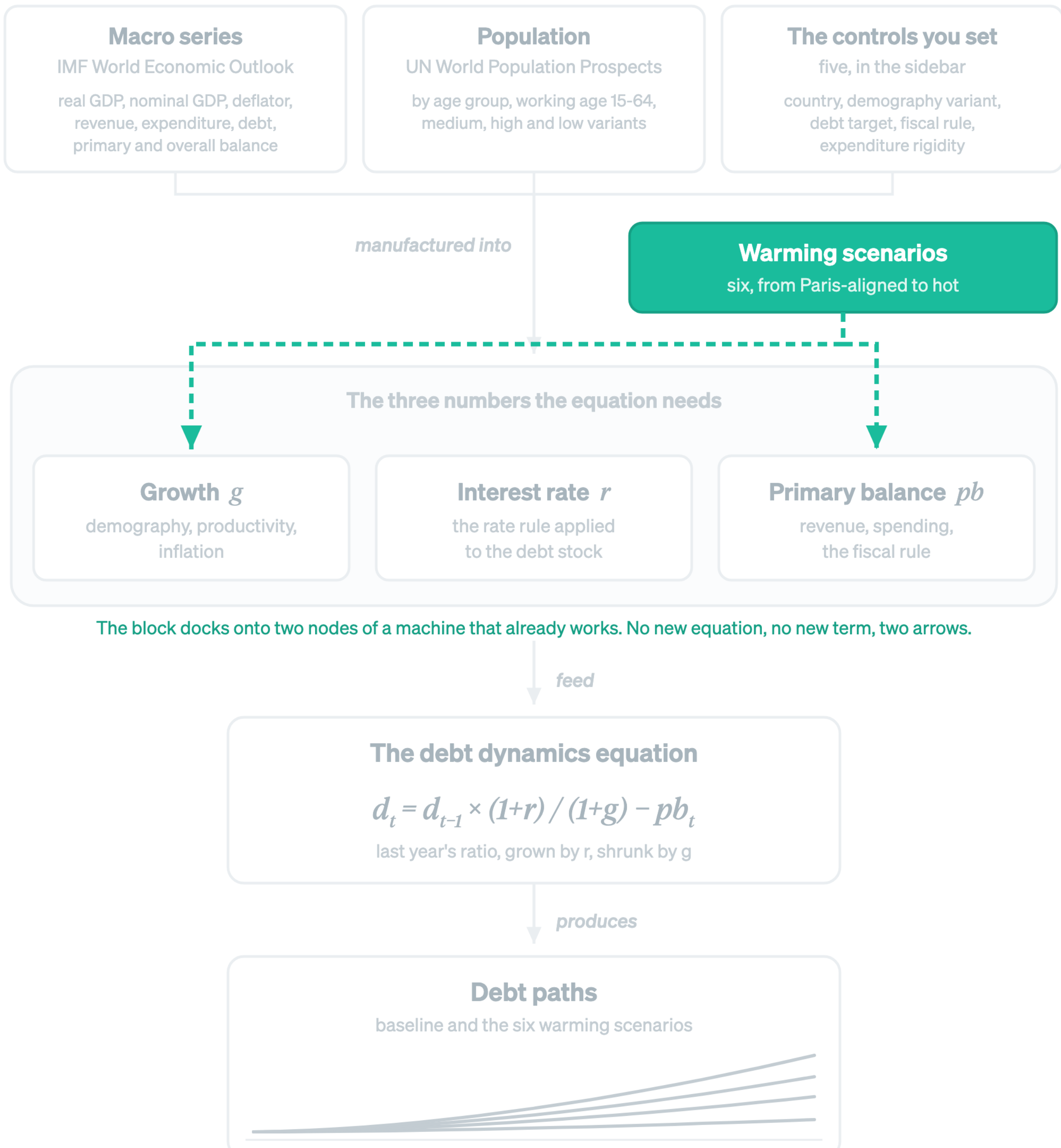


Figure 3.12: Step 3 adds one block and two arrows. The equation is untouched, the sources are untouched, and the interest rate never hears about the weather.

Take Kenya at 2050, halfway along the cold open, and walk the whole model in one paragraph.

The hot scenario's temperature path leaves Kenya further above its thirty-year norm than the baseline's does. The pooled response rate turns that gap into a cut to productivity growth, which by 2050 has accumulated into a GDP level 1.5 percent below the baseline. Employment did not change, and inflation did not change, so g falls from 6.36 to 6.30 percent. Revenue falls with GDP and holds its ratio. Spending does not fall, so the primary balance drops from 0.59 to 0.40 percent of GDP. Feed the smaller g and the smaller pb into Step 1's equation with an unchanged r of 7.6 percent, and the debt ratio comes out at 53.1 against the baseline's 50.4.

Two and a half points, in a year that is only twenty years into the projection. Then run the same arithmetic forty-nine more times, each on a stock that is a little larger than the last, with a primary balance that falls further behind what the equation needs, and against a fiscal rule that holds the baseline down but never touches the scenario.

That is the 48 points. Every part of it is now something you can point at.

3.5 Self-check: one question per step

Three questions, and each one is a step

Answer all three before opening the block below. If you can only do two, the one you missed tells you which step to reread.

1. **Step 1.** A country's debt is 70 percent of GDP, its effective interest rate is 6 percent and nominal growth is 4 percent. It runs a primary surplus of 1 percent of GDP. Is its debt ratio rising or falling next year, and what surplus would hold it still?
2. **Step 2.** Two countries have identical productivity and inflation assumptions. One has nominal growth of 6.4 percent and the other 3.7 percent. Name the input that accounts for the difference, and name the published source it comes from.
3. **Step 3.** Two countries face the same projected temperature path and have the same historical temperature norm. Q-CRAFT gives them different debt outcomes under the hot scenario. Give two reasons that have nothing to do with the climate estimate.

   DRAFT FOR TEAL: self-check answers

   1. **Rising, and it needs 1.35 percent.** The amplifier is $1.06/1.04 = 1.0192$, so the ratio carries forward at 71.3 before any borrowing. Subtract the 1 point surplus and it lands at 70.3, up from 70. The debt-stabilizing balance is $0.70 \times (0.06 - 0.04)/1.04 = 0.0135$, so a surplus of 1.35 percent of GDP holds it still. Running 1 percent against a required 1.35 is the whole story, and it is a third of a point of GDP a year. If you answered "falling, because it is running a surplus", that is the misconception Chapter 1 tested for, and Step 1 is worth another twenty minutes.
   2. **Employment growth, from the UN World Population Prospects.** With productivity and inflation held identical, the only remaining term in $(1 + e)(1 + p)(1 + i)$ is employment, which comes from the working-age population projection and is not something anyone in the tool sets. The two countries in the chapter are Kenya at plus 1.45 percent and Thailand at minus

1.11. Nothing about either country's climate exposure, borrowing or policy enters this answer.

3. **Any two of: their starting debt ratio, their effective interest rate, their revenue and spending ratios, their fiscal rule setting, their expenditure rigidity setting.** The climate estimate produces one number, a cut to productivity growth, and it is the same cut for both. Everything that happens after that cut runs through Steps 1 and 2, where the two countries differ in every input. This is the answer that matters most in a meeting, because it is how you answer "why is our number worse than theirs" without reaching for the climate science.

If the third one gave you trouble, that is the diagnostic the whole chapter was built for: the climate block is small, and most of what a climate result contains was already there in the baseline.

3.6 Wrapper: the three steps in one breath

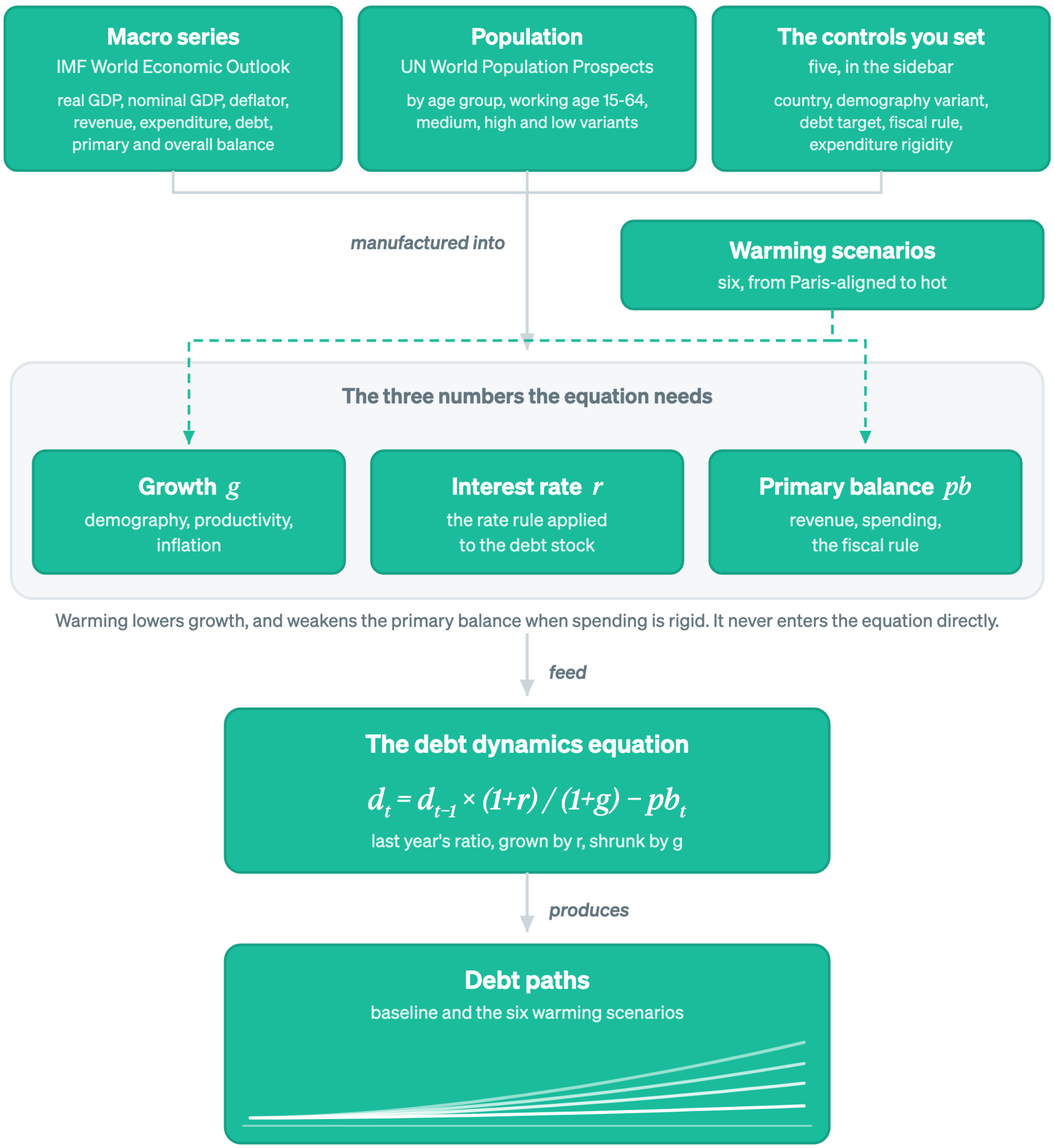


Figure 3.13: The three steps in one picture: an equation, a growth engine that feeds it, and a climate overlay that docks onto two of the three numbers.

An equation carries a debt ratio forward, amplified by interest against growth, less what the government paid down. A growth engine manufactures the three numbers it needs, out of two published data sources and five controls you set. A climate overlay lowers one of the three, and leans on a second through your spending assumption.

What you can now do.

- Compute a debt ratio one year forward from r, g and pb, and say which of the three you could defend to somebody who disagreed.
- Trace any number in a Q-CRAFT output back to the data source, the assumption, or the estimate that produced it.
- Say what "adapted" means in this tool, which is the width of a moving average and nothing else.
- Separate the part of a climate result that comes from the damage estimate from the part that comes from your own rigidity dial.

Common errors on this material. Reading a falling debt ratio as a fiscal improvement. Assuming a balanced primary account stabilizes debt. Drawing climate as a fourth term in the equation. Quoting a scenario gap without saying which rigidity setting produced it. Reading "Hot" as milder than "High".

Where this goes next. Step 2 named five controls and treated them as given. Chapter 4 is where you choose them, one at a time, with the evidence you would cite for each. Chapter 5 runs a country end to end and writes the result up. Chapter 6 is the boundary of what any of it supports.

On your desk this week. Open the Explorer, pick your own country, and read off three numbers: nominal growth in 2050, the effective interest rate, and the primary balance. Then compute the debt-stabilizing primary balance from Step 1 and compare it to the actual. That is one division, and it is the sentence most credible fiscal risk write-ups open with.

Chapter 4

Choosing the parameters

In this module
You will set all five of the Explorer's controls for a country and write down why you set each one where you did. Every parameter gets the same treatment, and the treatment arrives beside the control rather than ahead of it: what it is, why it matters, how to set it, what the underlying source actually shows, a predict-observe-explain run in the app, and one judgment call to defend. The rationale lines you write here become the assumption section of your export packet, and they are what survives you leaving the ministry.

Fast path

Already run the tool? Do the five **Defend your choice** self-checks and the five **Document it** blocks, and skip the What/Why prose. That is about 25 minutes and it is the part the capstone rubric marks.

4.1 Warm-up

From Chapter 2, without looking:

1. Which of the three numbers do the debt target and the fiscal rule move?
2. Expenditure rigidity of 1.0 means spending is what?

(Answers: the primary balance, through the spending side. Sticky, so the full revenue loss lands on the deficit.)

4.2 The assumption nobody wrote down

A year after a projection is built, nobody can say whether 1.2 percent productivity growth was a considered view about the country or a default nobody touched. The analysis is not wrong. It is unusable, because it cannot be defended.

Q-CRAFT Explorer cannot stop you leaving a default in place. What it can do is make the record, so your parameters and your reasons travel with the output. This module is about earning the right to that record.

4.3 By the end of this module you can

- **Set** all five Explorer parameters for a country and **justify** each against country evidence
- **Predict** the direction and rough magnitude of each parameter's effect before you move the slider
- **Defend** an expenditure rigidity choice against a colleague who prefers a different one
- **Produce** the assumption-rationale section of the export packet, one entry per parameter

4.4 Where you are in the course

The map below lights the left-hand end of the chain, where you work in this module.

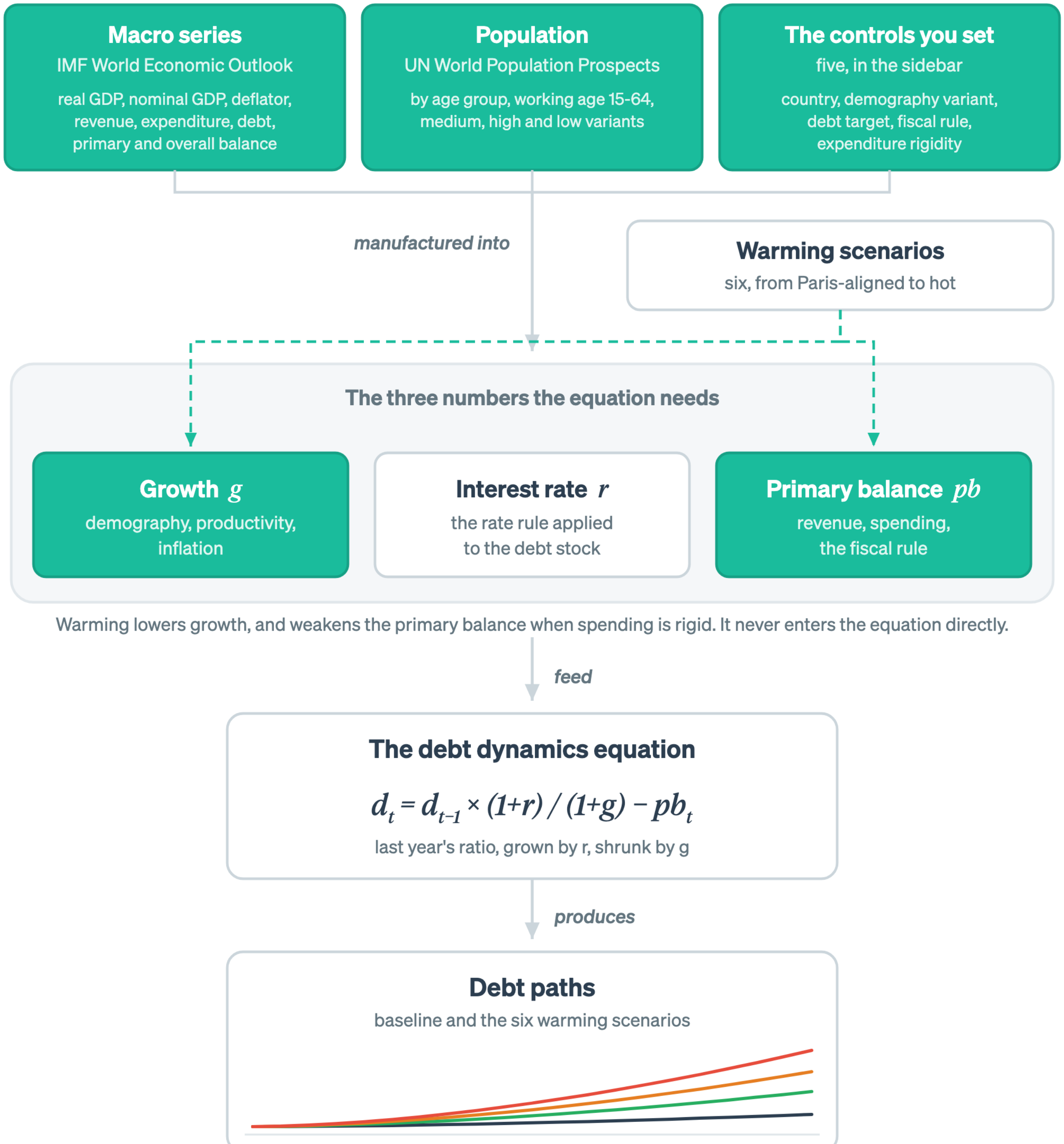


Figure 4.1: This module is the start of the chain: the two sources the country selection loads, the controls you set on top of them, and the two numbers they move.

Three nodes are lit. The data you load and the controls you set on top of it move growth and the primary balance. The interest rate is dark because V1 does not expose it.

4.5 Four tabs, and what each one is for

Four tabs hold everything the tool shows you, and you will be moving between them and the sidebar constantly. The **Baseline** tab is the no-climate-change projection. The **Climate** tab shows what warming does to GDP, scenario by scenario. The **Analysis** tab overlays baseline and climate debt paths on one chart. The **Data** tab gives you the grid and the CSV exports. Reading those charts properly is Chapter 5, and for now you only need to know where to look when a parameter moves.

4.6 Five controls, and four of them shape the projection

Q-CRAFT Explorer has five user-facing parameters. The first, country selection, loads the data. The remaining four shape the projection.

The figure below wires each control to the number it moves, which is the map to have in your head before you touch any of them.

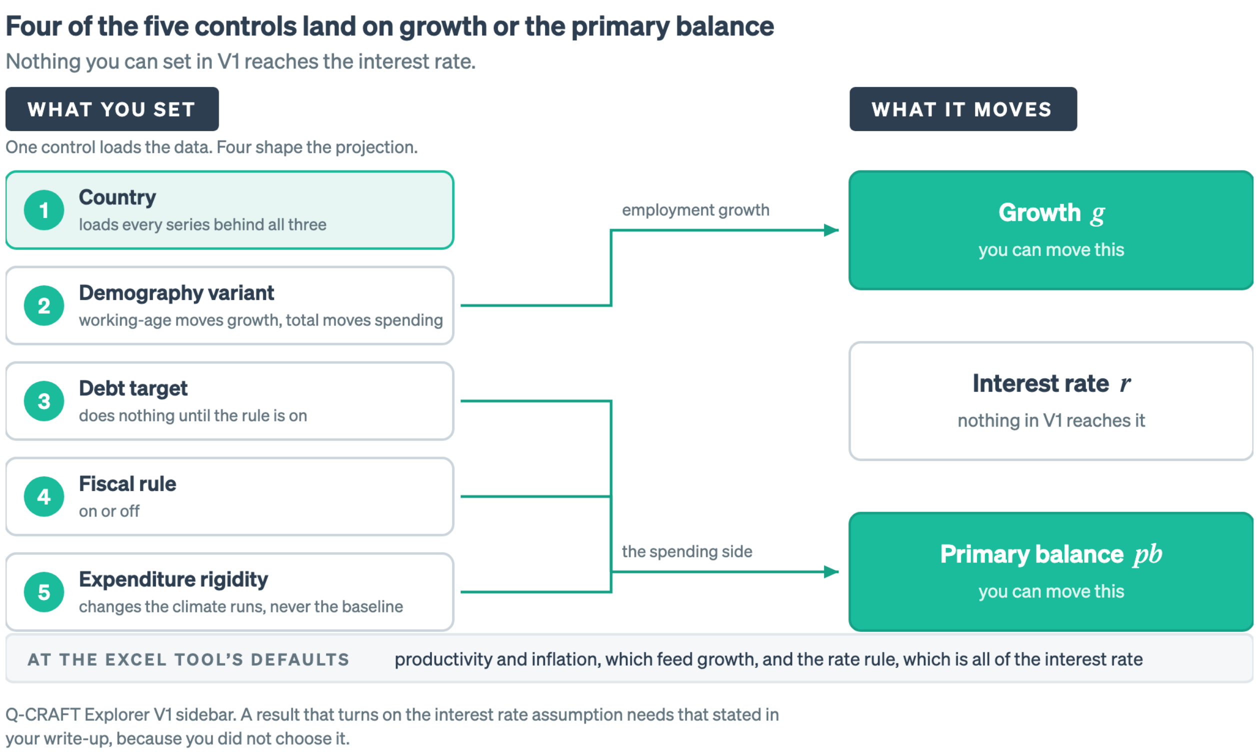


Figure 4.2: Demography is the one control that arrives in two places, because working-age population drives growth while total population drives spending. Productivity, inflation and the whole of the interest rate sit at the workbook's defaults.

Three of the four shaping controls land on the primary balance, one lands on growth, and nothing you can set reaches the interest rate at all.

Productivity, inflation and interest rate assumptions are not exposed in V1, because they sit at defaults matched to the original Excel tool. That is a limitation with consequences, and Chapter 6 says what they are. It also means the four parameters below carry more of your analytical judgment than they would in the full workbook.

Two of those three defaults are worth seeing against the record they are meant to continue. Productivity growth slides from 5.0 percent to 1.2 percent, and inflation from 5.0 percent to 3.5 percent, for every country in the tool.

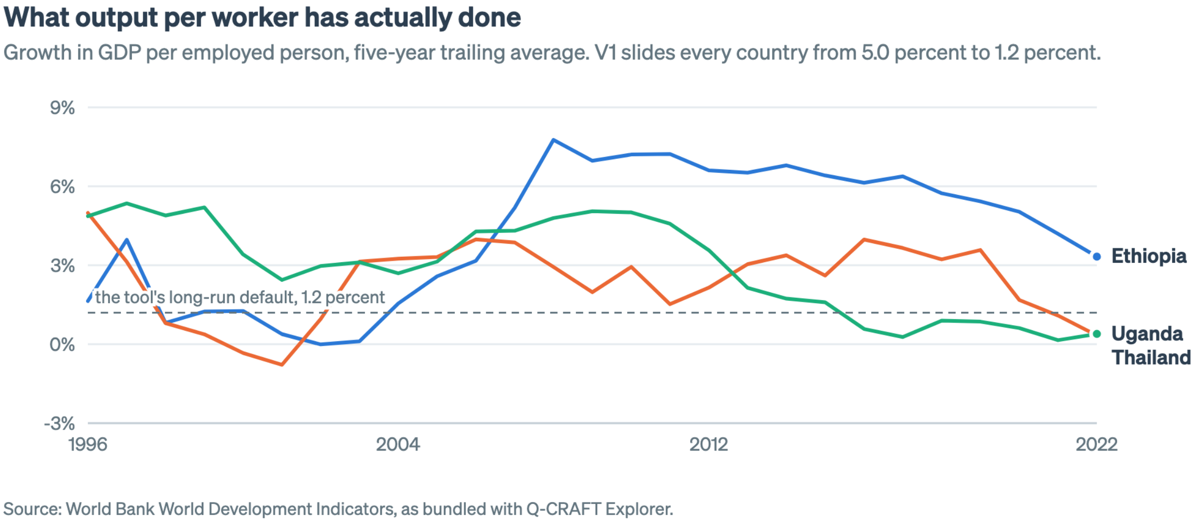


Figure 4.3: V1 slides productivity growth from 5.0 percent to 1.2 percent for every country. Set that path against what output per worker has actually done before you quote a result that depends on it.

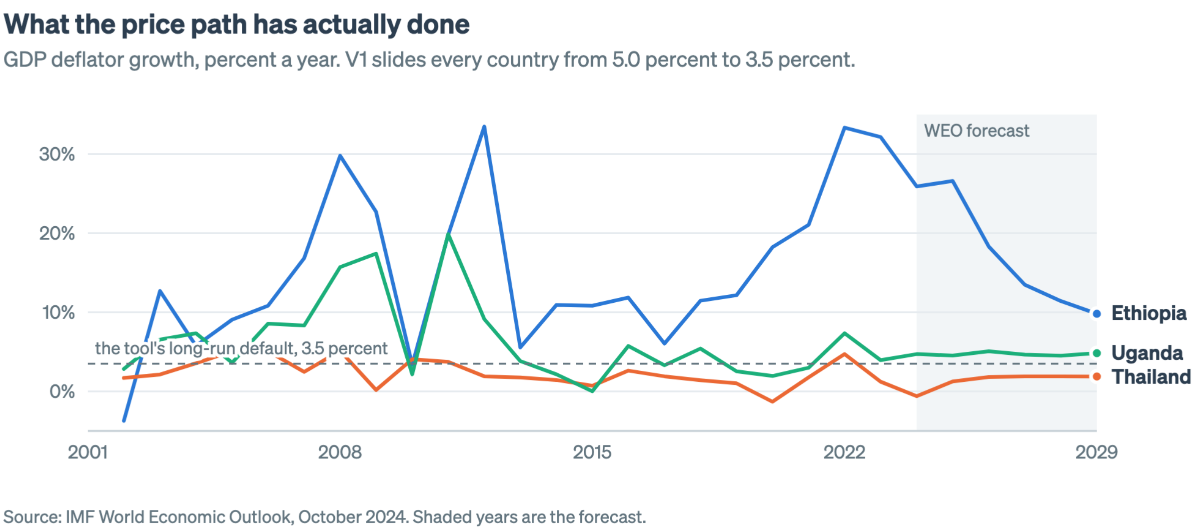


Figure 4.4: V1 slides inflation from 5.0 percent to 3.5 percent for every country. The three records here start in very different places, which is what the single default is averaging over.

Neither figure tells you the default is wrong. Both tell you what it is averaging over, which is the thing to say out loud when a result turns on either assumption. Where the gap between your country's record and the default is large, the sentence for your write-up is that V1 uses the Excel tool's global default and your country sits somewhere else, with the direction named.

> **Guidance where you need it**
>
> This is the second of the four reasons in Chapter 2, in the form you meet it. Every figure in this module is drawn from the same Parquet inputs the Explorer runs on, by `scripts/build_parameter_context.py` in the repository, and each one sits beside the parameter it informs rather than in an annex at the back. The Explorer is gaining interactive context panels that do the same job inside the app, so you can look at the source data for your own country while you set the control rather than afterwards.

4.6.1 Country selection

What it is. The country dropdown lists every country for which Q-CRAFT Explorer has complete data coverage across all required input datasets. Selecting a country loads macroeconomic data from the IMF's World Economic Outlook, currently the October 2024 vintage, and population projections from the UN World Population Prospects, 2022 revision.

Why it matters. The country you select determines all historical data and every WEO forecast-period projection. There is no manual data entry: the tool loads real GDP, nominal GDP, revenue, expenditure, debt, interest rates, and demographic projections automatically.

How to set it. Choose the country you are analyzing. The sidebar then displays two pieces of context: the latest WEO debt-to-GDP ratio and total population. Use them as a sanity check, and investigate before proceeding if either number does not match your understanding.

> Country coverage
>
> Q-CRAFT Explorer currently covers 197 countries, and coverage is expanding as verification progresses. If your country is not listed, it may be added in a future update. See the IMF User Guide (Tim and Rahman, 2024), Section II.B for details on what data is loaded for each country.

What the source shows. The starting debt ratio is not a small detail, because the whole projection compounds off it.

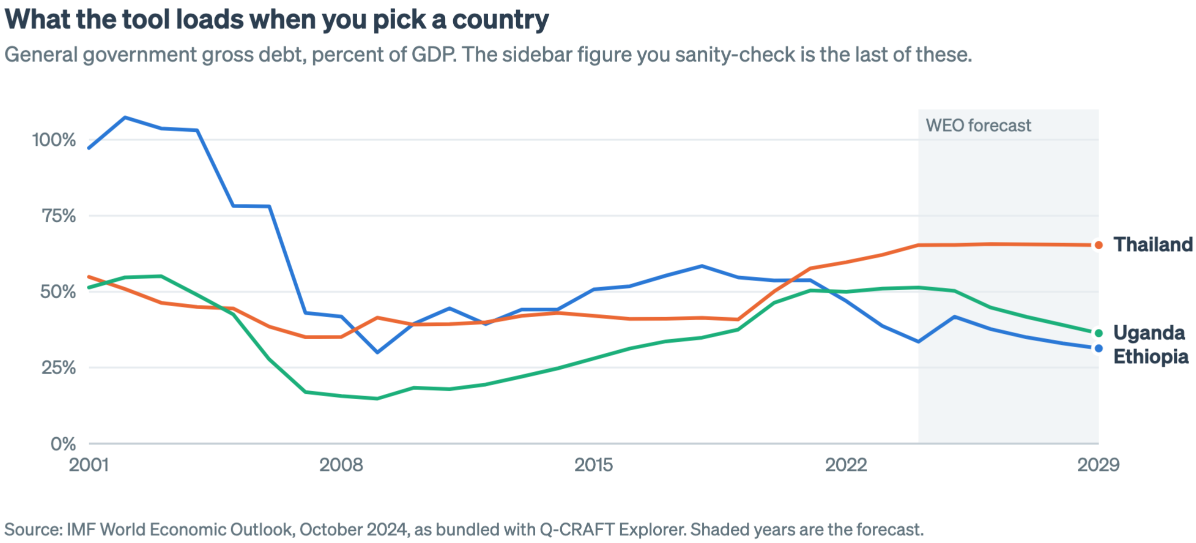


Figure 4.5: Three countries the tool already holds data for. Debt ratios differ by more than a factor of three across them, which is why the first thing you do after selecting a country is check the number the sidebar reports.

Three countries the tool already holds, and their debt ratios differ by more than a factor of three, which is why the first thing you do after selecting a country is check the number the sidebar reports.

Predict, observe, explain

Predict. Before you select your country, write down the debt-to-GDP ratio you expect the sidebar to report, from your own knowledge of its latest position.
Observe. Select the country and read the sidebar figure.
Explain. If your number and the tool's differ by more than a point or two, work out which one is on a different basis before you go any further. Three causes are common: gross versus net debt, fiscal year versus calendar year, and the WEO vintage lagging the ministry's own outturn. Each one is a legitimate difference, and each one will be the first question you are asked about your results.

! Defend your choice: country and vintage

A reviewer asks: "Is this the current debt number?" Take the worked case. You are using WEO October 2024, and Uganda's own FY2024/25 Fiscal Risk Statement reports public debt at 46.9 percent of GDP at June 2023 and projects 49.2 percent at June 2024. What do you say?

DRAFT FOR TEAL: a defensible answer on vintage
Name the vintage, name the basis, and state which direction the difference runs. "This is WEO October 2024 on a calendar-year basis. The ministry's June 2024 figure is 49.2 percent on a fiscal-year basis. The starting point differs by under a point, and the climate result is a difference between two runs from the same starting point, so it is not sensitive to that gap." The last clause is the one that

ends the conversation, and it is only available to you because you understand what the tool computes.

Document it

In your export packet, record: **data vintage** (WEO October 2024, UN WPP 2022), **starting debt ratio as loaded**, and **any known difference from your ministry's own figure, with the reason**.

4.6.2 Demography variant

What it is. The UN publishes population projections in three variants, Medium, High and Low, reflecting different fertility assumptions. This parameter selects which variant Q-CRAFT uses for population and labor force projections through 2100.

Why it matters. Working-age population growth, ages 15 to 64, drives employment growth in Q-CRAFT's production function after the WEO forecast horizon. Higher population growth means a larger labor force, higher potential GDP, and, all else equal, more favourable debt dynamics. Countries facing demographic decline see slower growth and harder fiscal trajectories. Total population growth matters separately, because primary spending grows with it (see Chapter 2 for the expenditure growth formula).

How to set it. Three variants, and one of them is the default choice:

- **Medium** is the standard assumption for most policy analysis. Use it unless you have a specific reason to explore demographic scenarios.
- **High** and **Low** bracket the uncertainty. They earn their place for countries where fertility trends are uncertain or where demographic policy changes are anticipated.
- For countries in rapid demographic transition, with a declining working-age population, compare Medium against Low to see how sensitive the fiscal path is to the demographic outcome.

For detailed methodology, see the IMF User Guide, pp. 10-12 and Section IV.A on demography and employment.

What the source shows. The three variants are the UN's own account of how uncertain its projection is, and how far apart they run depends on where the country is in its demographic transition.

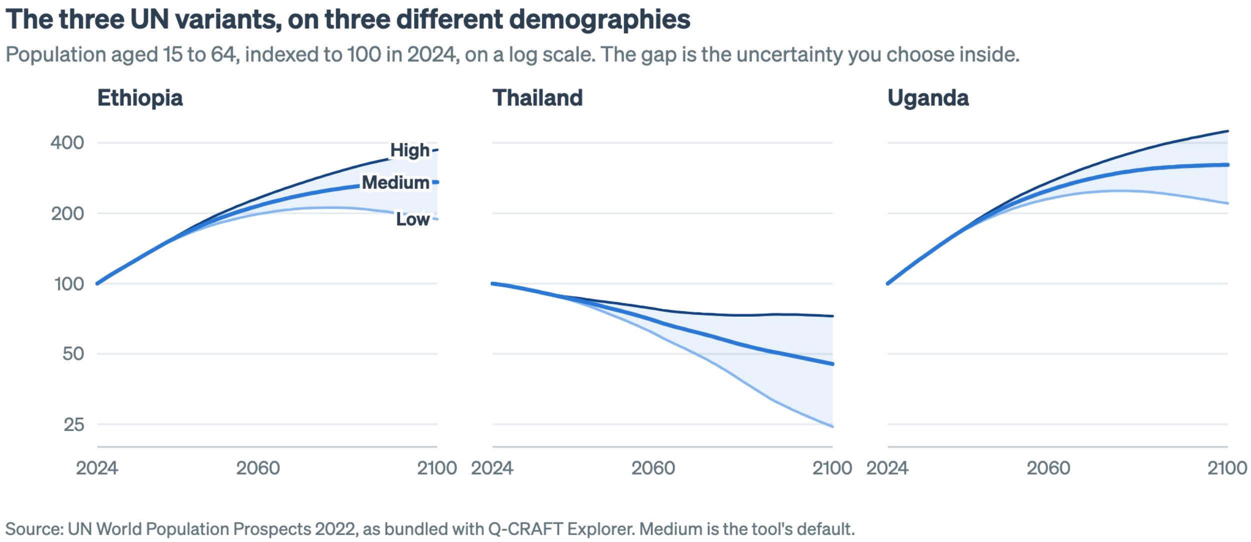


Figure 4.6: The demography control picks one of these three lines. In a country whose working-age population is still climbing the variants separate slowly. In one where it has turned down, the choice between Low and High is the difference between a workforce at a quarter of today's and one at three quarters.

Where the working-age population is still climbing the three variants barely separate, and where it has turned down the choice between Low and High is the difference between a workforce at a quarter of today's and one at three quarters.

Predict, observe, explain

Predict. Ethiopia has one of the youngest populations in the world, and Thailand's working-age population has already started to fall. Before you touch the control, will switching from Medium to High move Ethiopia's 2099 debt ratio a lot, a little, or barely at all? Write down a direction and a rough size in percentage points, then write a second prediction for Thailand.
Observe. Run Ethiopia, switch between Medium and High, and watch the Baseline tab's debt trajectory. Then run Thailand and do the same. Compare the size of the two moves.
Explain. In countries with young, growing populations the variants barely separate over the projection, because the working-age share is already rising in every one of them. In countries facing demographic decline the gap widens sharply: slower working-age growth means lower GDP growth and rising expenditure-to-GDP ratios at the same time. Both effects run through g. The expenditure channel runs through total population and the growth channel runs through working-age population, and those two populations move differently, which is why the net effect is not obvious in advance.

! Defend your choice: demography

A colleague argues you should use the Low variant, because your country's fertility rate has been falling faster than the UN projected. Do you switch?

DRAFT FOR TEAL: a defensible answer on the demography variant

Probably not for the headline run, and certainly worth a sensitivity run. Medium is the reference variant, and using anything else as your headline puts you in the position of defending a demographic forecast inside a fiscal risk document. That is not the argument you want to be having. The colleague's point is testable in two minutes: run Low, record the difference in the 2099 debt ratio, and put the number in the packet as a sensitivity. If the difference is small, you have retired the objection with evidence. If it is large, you have found something worth a paragraph.

Document it

Record: **variant chosen**, **why** (one line), and **the 2099 debt ratio under at least one alternative variant** as a sensitivity.

4.6.3 Debt target (% of GDP)

What it is. The debt target is the debt-to-GDP ratio the fiscal rule steers toward over time. The range runs from 0 to 200 percent of GDP.

Why it matters. The target does nothing until the fiscal rule is enabled, and then it determines how hard spending has to adjust. A target below current debt calls for consolidation, which cuts primary expenditure. A higher target calls for less adjustment.

How to set it. Start from the country's actual fiscal framework. Many countries have explicit fiscal rules with legislated debt targets, and you should use those where they exist. Uganda's FY2024/25 Fiscal Risk Statement reads its climate results against a 50 percent of GDP fiscal rule ceiling, and reports that public debt passes it in the High, Hot and Vulnerable scenarios. Where no formal target exists, the debt-stabilizing primary balance is a useful benchmark, and it is a standard DSA metric (see the IMF User Guide, Section IV.A, and Step 1 of Chapter 3).

Three rough starting points, none of them authoritative IMF guidance:

- **Low-income countries (LICs):** 40 to 50 percent of GDP is a common range in practice, broadly consistent with IMF-World Bank Debt Sustainability Framework thresholds.
- **Emerging markets (EMs):** 50 to 70 percent of GDP, depending on the country's market access, institutional quality, and fiscal track record.
- **Advanced economies (AEs):** can sustain higher ratios, 60 to 100 percent and above, though the appropriate target depends on growth prospects and market confidence.

For detailed methodology, see the IMF User Guide, pp. 15-18 on fiscal rule assumptions and the baseline scenario.

What the source shows. Nothing, in this case. The target is a policy choice rather than a published series, so there is no source figure for it. The nearest thing to evidence is the country's own record, which is the debt-ratio figure at the top of this module, read against whatever anchor your fiscal framework already sets.

Predict, observe, explain

Predict. With the fiscal rule enabled, you are about to move the debt target from 40 percent to 80 percent. Write down what happens to primary expenditure in the early years, and what happens to the 2099 debt ratio.
Observe. Set the target to 40, read the primary expenditure path and the debt path in the Baseline tab. Set it to 80 and read them again.
Explain. A lower target forces faster consolidation, so expenditure is cut harder to bring debt down. A higher target leaves more spending room. The adjustment is gradual rather than immediate, because it works through the fiscal gap over several years. If you predicted an instant jump in expenditure, correct that: the rule is a feedback loop keyed to last year's debt position, not a switch.

Defend your choice: the target

You are asked to run the projection against a 40 percent target rather than the 50 percent ceiling your fiscal risk documentation already uses, "to be prudent." What do you do, and what do you say?

> **DRAFT FOR TEAL: a defensible answer on the debt target**
> Run both, and be explicit about what the target is doing in the model. In Q-CRAFT the debt target is not a judgment about what is sustainable. It is the anchor the simulated fiscal rule steers toward, so changing it changes the assumed policy reaction rather than the country's risk. The honest framing gives both numbers: "At the 50 percent anchor, the Hot scenario breaches the ceiling by X. At a 40 percent anchor, staying inside it requires Y percent of GDP a year of additional primary balance." That answers the request without smuggling a policy preference into a risk assessment.

Document it

Record: **target used**, **its source** (legislated rule, Charter for Fiscal Responsibility, DSF threshold, analyst judgment), and **the alternative target you tested**.

4.6.4 Fiscal rule (Yes / No)

What it is. The fiscal rule, when enabled, adjusts primary expenditure to steer debt toward the target ratio. When it is disabled, spending follows baseline trends whatever the debt level does.

Why it matters. Without a fiscal rule, debt dynamics are purely mechanical: they follow the interest-growth differential and whatever primary balance falls out of revenue and expenditure trends. With the rule on, the model simulates active fiscal policy, so the government cuts spending as debt rises or spends more as fiscal space opens.

How to set it. Run it both ways, in this order:

- **Start with the fiscal rule off** to see the "no policy change" baseline. That reveals the underlying trajectory: is debt stable, rising, or falling under current trends?

- **Turn the fiscal rule on** to see how active policy changes the picture. That answers a different question: how much consolidation would reaching the debt target take?
- Most countries have some form of fiscal anchor in practice, though enforcement varies. The IMF User Guide notes that Q-CRAFT's fiscal rule is an approximation, because it does not model the GDP effects of fiscal consolidation (User Guide, p. 17, footnote 11).

The adjustment lands on the level of primary expenditure rather than on its growth rate. It is applied after the multiplicative growth factors, and it depends on the prior year's state, which is why Q-CRAFT computes this recursively year by year.

Predict, observe, explain

Predict. Run your country with the rule off first. Predict whether turning the rule on will change the 2099 debt ratio a lot or a little, and say why in terms of where the off-rule path ends up relative to the target.
Observe. Run the same country with the fiscal rule on and off. Watch both the debt path and the primary expenditure path.
Explain. With the rule off and unfavourable debt dynamics, meaning the interest rate exceeds growth, debt-to-GDP can rise without bound. Turn the rule on and debt converges toward your target, with the primary expenditure chart showing the spending cuts required to get there. The gap between the two runs is the fiscal effort your target implies. If the off-rule path already sits near the target, turning the rule on does almost nothing, and that is itself a finding worth reporting.

! Defend your choice: rule on or off

Which run belongs in a Fiscal Risk Statement: rule on or rule off?

DRAFT FOR TEAL: a defensible answer on rule on or off

Both belong, because they answer different questions. Rule **off** is the risk statement: this is where we end up if policy does not respond, which is the honest baseline for a document whose job is to disclose risk. Rule **on** is the policy statement: this is the effort required to hold the anchor. Publishing only the rule-on run understates the risk, because it assumes the response you are trying to justify. Publishing only the rule-off run invites the objection that no government would sit still for seventy years. Report the pair and say which is which.

Document it

Record: **rule on or off for the headline run**, **the paired run**, and **one sentence on which question each answers**.

4.6.5 Expenditure rigidity (0.0 - 1.0)

What it is. Expenditure rigidity measures how far government spending resists downward adjustment when climate shocks reduce GDP. It affects climate scenarios only, never the baseline.

Why it matters. This is the parameter that decides how much of the climate cost becomes debt. When climate change reduces GDP, government revenue falls with it, because revenue tracks nominal GDP. What happens to spending then determines how much of the shortfall shows up as additional borrowing.

- **1.0 (fully rigid, worst case):** spending stays at its baseline level in local currency terms even as GDP falls. Revenue declines and expenditure does not adjust, so the entire climate impact passes through to the primary balance and onto the debt ratio. This is a government that cannot or will not cut spending when growth disappoints, because of a large civil service wage bill, entitlement programmes, or political constraints on adjustment.
- **0.0 (fully flexible):** spending falls in proportion with GDP, holding the expenditure-to-GDP ratio at its baseline level. The debt impact of climate change is smaller, because the government absorbs the shock by reducing real spending per person.
- **Values between 0 and 1** represent partial adjustment. Most countries sit somewhere in between, because some spending categories are rigid (wages, pensions, debt service) while others can adjust (capital investment, discretionary programmes).

How to set it. Start from the composition of the budget:

- Countries with large public sector wage bills, universal social programmes, or rigid entitlement commitments: lean toward 0.7 to 1.0.
- Countries with more fiscal flexibility, smaller governments, or well-established expenditure review processes: 0.3 to 0.6.
- The default of 1.0 is deliberately conservative, showing the worst case where all climate damage becomes additional borrowing.

For detailed methodology, see the IMF User Guide, pp. 20 and 35-36 on the expenditure rigidity parameter and fiscal effects of climate change.

What the source shows. Rigidity is a claim about how spending has behaved, and the WEO record carries evidence on it.

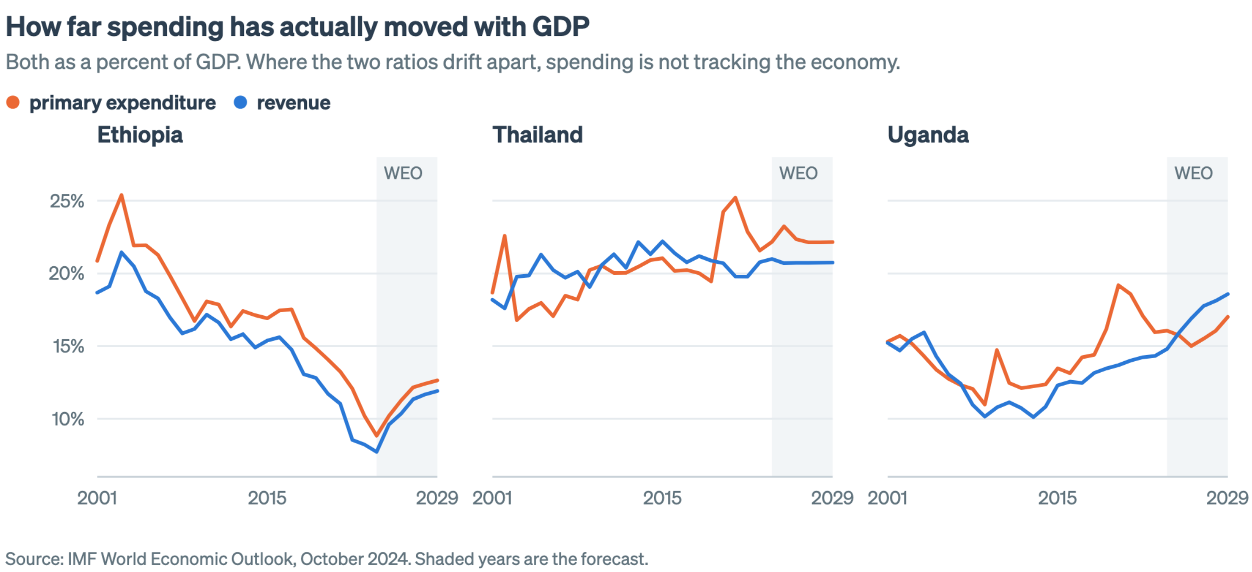


Figure 4.7: Rigidity is a claim about how these two lines move together. Where expenditure holds its share of GDP while revenue falls, the budget has behaved as though rigidity is high.

Read the two lines against each other rather than separately. Revenue holds a roughly constant share of GDP by construction in the projection, but in the historical record it moves, and the question is what expenditure did when it moved. A budget whose expenditure ratio climbs while revenue falls has behaved as though rigidity is high. That is evidence you can put in the packet, and it is stronger than a view about political will.

> **WIDGET-TODO: expenditure rigidity**
>
> Embed the expenditure rigidity intuition widget here (lane 2, run 3: `apps/qcraft-web/widgets/*`). Iframe under Quarto for now, native React once the site moves to Astro. The integration pass happens after this restructure and run 3 both land, and this callout is the anchor point. The prose above must carry the full idea on its own, because the default state of any interactive has to work for readers who never touch it.

> **Predict, observe, explain**
>
> **Predict.** On the Analysis tab you are about to compare the climate fan at rigidity 1.0 against rigidity 0.0. Write down which one produces the wider fan, and by roughly what factor.
> **Observe.** Set rigidity to 1.0 and read the 2099 gap between the baseline and Hot Unadapted. Set it to 0.0 and read the same gap.
> **Explain.** At 1.0 the fan spreads wide, because the whole climate revenue loss accumulates as debt, so the distance between Paris-aligned and Hot Unadapted is large. At 0.0 the scenarios compress toward the baseline, because the government absorbs the shock through spending. The distance between those two fans is your country's fiscal exposure to expenditure rigidity, and it is usually the largest single lever in the tool. If you predicted that rigidity matters less than the choice of climate scenario, check it, because for most countries it does not.

> **SCREENSHOT-TODO**
>
> Side-by-side Analysis tab for Uganda, the worked case, at rigidity 1.0 and rigidity 0.0, with the same axis limits on both, so the compression is visible rather than described. Same-axis is the point, because two auto-scaled charts will hide the effect.

> **Defend your choice: rigidity**
>
> This is the judgment call you will actually be challenged on. A colleague says 1.0 is alarmist: "no government holds spending flat in local currency for seventy years while the economy shrinks." Another says anything below 1.0 assumes a fiscal discipline your country has not demonstrated. What do you do?
>
> > **DRAFT FOR TEAL: defending a rigidity choice**
> > Both colleagues are making empirical claims, and the parameter is where you settle them with evidence rather than adjectives.
> > **The evidence to reach for** is the composition of the budget, not a view about political will. What share of primary expenditure is wages, pensions, statutory transfers and other commitments that require legislation to change? That share is close to a floor on rigidity. In most low-income countries it is high, and the flexible

remainder is dominated by domestically financed development spending, which is exactly the spending a government protects last.

The defensible position is to run 1.0 as the headline and a lower value as the sensitivity, and to say why in one line: the default is conservative by design, and a fiscal risk statement is the document where conservatism belongs. Then give the second colleague their number, which is what the sensitivity run is for: at rigidity 0.6, the 2099 gap narrows from X to Y percentage points.

What not to do is pick a middle value because it feels balanced. An unexplained 0.5 is weaker than 1.0 with a stated rationale, because 1.0 at least has the tool's own conservative-default argument attached. A middle number with no reasoning behind it is the assumption that gets quietly reused for five years by people who never knew it was a guess.

The sentence for the packet: "Rigidity set at 1.0, the tool default, because roughly N percent of primary expenditure is wages, pensions and statutory commitments. The sensitivity run at 0.6 changes the 2099 climate-baseline debt gap from X to Y percentage points."

Document it

Record: **rigidity used**, **the budget-composition evidence behind it**, and **the sensitivity run and its effect on the headline gap**. This is the entry the capstone rubric weights most heavily.

4.7 Wrapper: what you can now do

- You can set all five controls for a country and give a reason for each that cites evidence rather than preference.
- You can predict each parameter's direction of effect before moving it, which is what tells you when the tool is behaving oddly.
- You have five **Document it** entries. Together they are the assumption-rationale section of your export packet.

Common errors on this material: inverting the rigidity scale (see Chapter 1, question 2), moving the debt target while the fiscal rule is off and wondering why nothing happened, and choosing a middle value to avoid an argument.

On your desk this week: find one number in a projection your department published and ask who chose it and why. If the answer takes more than a minute to find, you have made the case for the export packet.

Chapter 5

A worked example, end to end

In this module

You will run one country the whole way through, from parameter choices to publishable prose. The worked case is Uganda, because it is the tool's golden-master verification country and every figure in it can be checked against two published documents. The scaffolding then fades onto other countries: a completion problem on Ethiopia where you finish the interpretation, and an independent problem on a third country that you do alone. The target is not a chart but a write-up, in the format of the IMF's C-PIMA presentation of Q-CRAFT results and Section III of Uganda's Fiscal Risk Statement.

Fast path

There is no fast path through this module, because it is the module the capstone is built from. If you are short of time, do Section 5.6 and Section 5.8 and skip the middle.

5.1 Warm-up

From Chapter 4 and Chapter 2:

1. Which two parameters change the climate result but not the baseline?
2. In which year do climate scenarios begin to diverge from the baseline?

(Answers: expenditure rigidity, and the climate scenario itself. Everything else moves both. 2030.)

5.2 Two documents already show what the output looks like

You are not inventing a format. Two published documents show what Q-CRAFT output looks like when it reaches a policymaker. Both are about Uganda, because Uganda is where the tool has been through a full cycle from workshop to published statement. The genre is not Ugandan, and your ministry's version of it will look much the same.

The IMF's C-PIMA write-up. *Uganda: PFM Climate Assessment: Public Investment and Fiscal Risks Management* (IMF, 2024) reports the September 2023 workshop with staff of Uganda's

Ministry of Finance, Planning and Economic Development (MoFPED, the finance ministry) in a single paragraph. Climate change looks milder in Uganda than in other sub-Saharan African countries, it says, and it still matters for growth and fiscal sustainability, because by end of century GDP loss could pass 4 percentage points with debt at 66 percent of GDP against 47.5 percent in the baseline. One paragraph. Three numbers. A comparative judgment.

Uganda's Fiscal Risk Statement FY 2024/25. Published by the Ministry of Finance, Planning and Economic Development, this is the fuller treatment: scenario definitions, a baseline fiscal path table, GDP deviation charts, fiscal balance and debt charts, and a closing policy sentence. Its climate section runs pages 13 to 17 of a 31-page document, and three source lines under its exhibits read "QCRAFT (2023)."

Your capstone is the shorter of the two. This module builds it.

> **ℹ The published section, page by page**
>
> Every line below is from the document itself, so you can open it and check.
> **Page 4, the introduction.** The statement groups fiscal risks into three categories and says the climate analysis "expanded its scope on fiscal risks associated with climate change by using the Quantitative Climate Risk Assessment Fiscal Tool (Q-CRAFT) for long term fiscal sustainability analysis." That is the tool named in the ministry's own words, in the framing section.
> **Page 13, the section opens.** The body heading reads "III. CLIMATE CHANGE AND NATURAL DISASTER FISCAL RISKS". The table of contents lists the same section as "CLIMATE CHANGE FISCAL RISKS". The page is national hazard context rather than Q-CRAFT output: floods and epidemics at about 75 percent of recorded events between 1985 and 2021, footnoted to the World Bank's Climate Risk Country Profile 2021, and annual economic losses from natural disasters estimated at US$87 million. Table 4 on that page is sourced to the World Bank Group (2022).
> **Pages 14 to 15, the set-up.** "Using the Q-craft tool, an assessment of the impact of climate change on Uganda's economic growth and fiscal sustainability was undertaken." The tool quantifies effects "under five different climate scenarios, against a baseline", and the five are named and defined on the page: Paris, Moderate, High, Hot, Vulnerable. A footnote cites Kahn et al. (2021) for the temperature-to-growth empirics. Page 15 carries Figure 8, the baseline decomposition of nominal GDP growth, and Table 5, the baseline fiscal path reproduced below, under the line "Source (Fig 8 & Tab 5: QCRAFT (2023)".
> **Page 16, the macroeconomic effects.** Figure 9, deviations from the baseline in level GDP and in labour productivity growth, "Source: QCRAFT (2023)". The text reports "a 4 percent reduction in nominal GDP by the end of the century under the Hot scenario", with Paris and Moderate described as benign for Uganda.
> **Page 17, the fiscal effects and the finding.** Figures 10 and 11, balances and the debt ratio by scenario, "Source: QCRAFT (2023)". Then the two sentences the whole section exists for: the primary deficit under Hot ends "0.7 percent worse than the baseline, thereby raising public debt by over 18 percent of GDP", and "In the High, Hot, and Vulnerable scenarios, public debt surpasses the 50 percent of GDP fiscal rule ceiling, taking on an unsustainable upward trajectory." The section closes on a policy sentence about meeting Paris commitments. Five pages. One table, four figures, one threshold finding. That is the size of the thing you are learning to write.

5.3 By the end of this module you can

- **Execute** a complete Q-CRAFT run for a country, from parameter selection to CSV export
- **Apply** the Baseline Sanity Check before interpreting any climate result
- **Interpret** the three chart families, saying what each one licenses you to claim
- **Draft** a two-paragraph fiscal risk write-up in the register of a published Fiscal Risk Statement
- **Assemble** the export packet: inputs, rationale, output

5.4 Where you are in the course

The map below lights the far end of the chain, which is where this module lives.

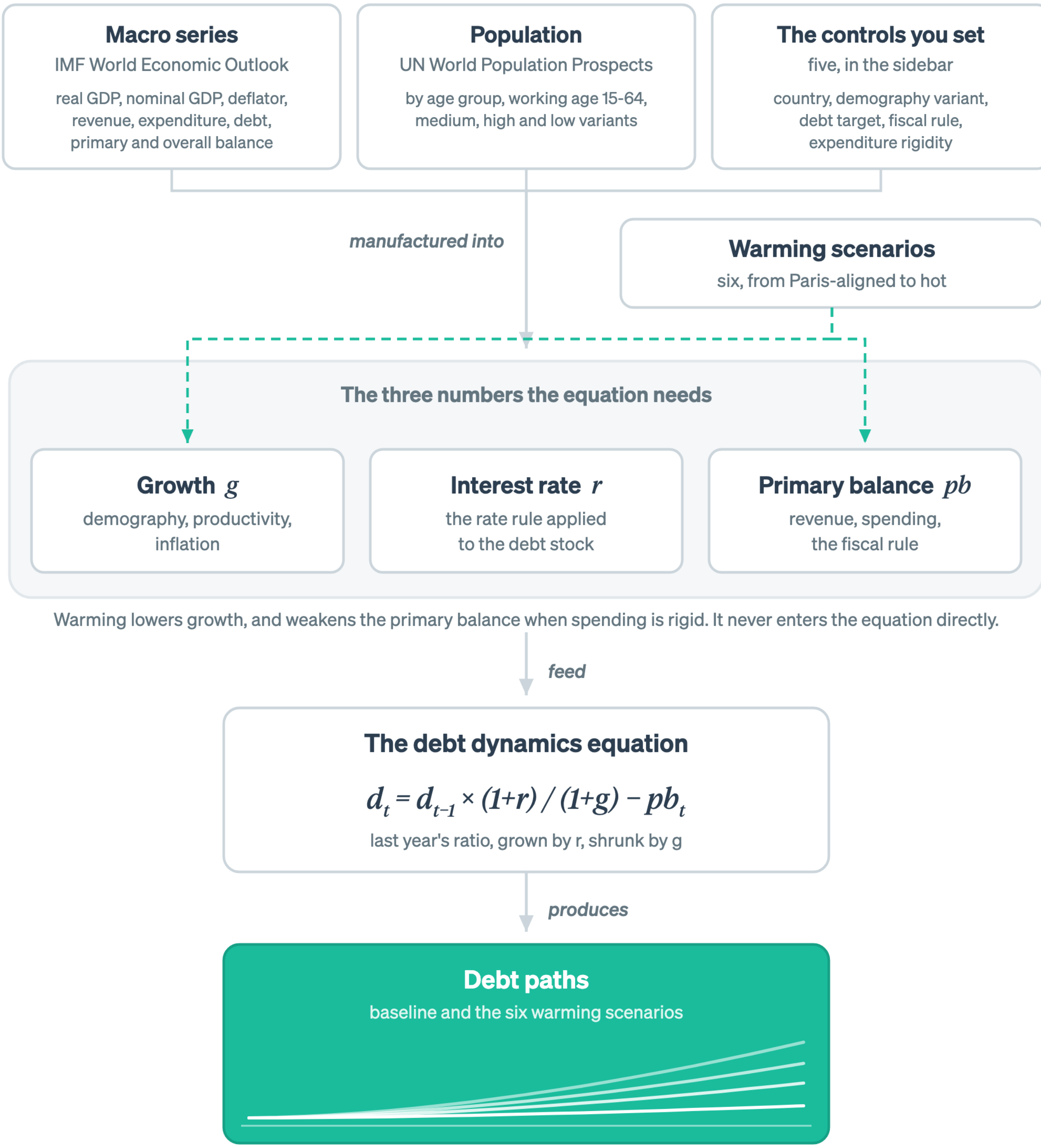


Figure 5.1: This module is the output end: the debt paths, and the write-up that comes out of them.

The last node is lit: the debt paths, and the write-up you build from them.

5.5 The target format

The Q-CRAFT part of that section is organised in three moves, numbered in the document itself. Copy that structure, because it has been published and it survived contact with its readers.

Move	What it contains	Where your material comes from
1. Set up the analysis	What the tool does, the scenarios used, and the baseline fiscal path	Chapter 4 parameters, Baseline tab
2. Macroeconomic effects	GDP deviation from baseline by scenario, and the productivity channel behind it	Climate tab
3. Fiscal effects	Primary and overall balance by scenario, debt-to-GDP by scenario, read against the fiscal anchor	Analysis tab

The published baseline table is worth copying too. Uganda's Table 5 reports four years, not seventy:

Summary fiscal path	2023	2050	2075	2099
Primary expenditure (% of GDP)	19.9	19.6	18.8	19.4
Interest expenditure (% of GDP)	3.3	1.3	1.8	2.4
Interest rate (%)	7.7	3.9	5.5	5.5
Primary balance (% of GDP)	-6.3	-1.7	-0.8	-1.4
Overall balance (% of GDP)	-9.6	-3.0	-2.7	-3.8
Debt-to-GDP ratio (%)	47.1	36.2	35.8	47.5

Source: Uganda Fiscal Risk Statement FY 2024/25, Table 5, sourced to QCRAFT (2023).

Four columns is a deliberate editorial choice, because a seventy-row table is not a policy document. Note what that baseline says before any climate damage is applied. Debt falls to the mid-30s by mid-century and comes back to 47.5 by 2099. Read the interest rate row against the growth path and you can see why.

5.6 The worked case: Uganda, start to finish

Uganda is the worked case because its numbers are published and checkable, not because it is the country you work on. What transfers to your own country is the method, which is the seven steps below.

Seven steps follow, and you should follow along in the app. Every step names what you are doing and what it licenses you to say.

The figure below lays the seven out, and it puts step 3 across the middle rather than beside the others, because that is where it belongs.

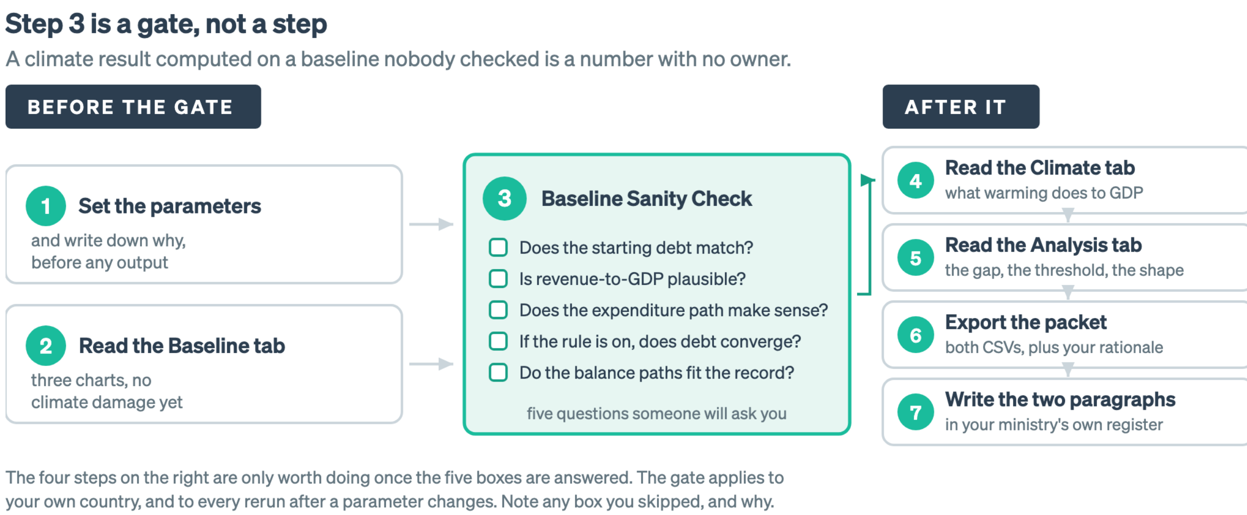


Figure 5.2: Steps 1 and 2 produce a baseline. Steps 4 to 7 spend it. The five boxes in the middle are the only thing standing between the two, and each one is a question a reviewer will ask you anyway.

Everything to the right of that gate is only as good as the baseline underneath it, which is why the sanity check comes before any climate reading rather than after.

5.6.1 Step 1: set the parameters, and write down why

Open Q-CRAFT Explorer and select Uganda. Use these settings for the worked case:

Parameter	Setting	Rationale for the packet
Country	Uganda	Analysis subject
Demography variant	Medium	Reference variant, with Low tested as a sensitivity
Debt target	50% of GDP	The ceiling Uganda's own Fiscal Risk Statement reads results against
Fiscal rule	Off for the headline run	Discloses the risk without assuming the policy response, with the rule-on run reported alongside
Expenditure rigidity	1.0	Tool default, conservative, with budget composition evidence per Chapter 4 and 0.6 tested as a sensitivity

That table is the first page of your export packet. Write it before you look at any output, so the settings are choices rather than post-hoc justifications.

5.6.2 Step 2: read the Baseline tab

The Baseline tab shows the no-climate-change scenario: the country's fiscal trajectory under current trends and the assumptions you have set. Three charts share the tab.

Debt-to-GDP trajectory (top) is the headline chart. Follow the trajectory from the historical period, shaded through 2029, into the projection period. Is debt rising, stable, or falling? If the fiscal rule is on, debt should converge toward your target. If it is off, the trajectory reveals the underlying fiscal dynamics: favourable when growth exceeds interest rates, unfavourable when interest rates exceed growth.

Revenue and Expenditure (% of GDP) (bottom left) shows two ratios that can drift apart. Revenue-to-GDP stays constant by assumption, because it grows with nominal GDP. Primary expenditure-to-GDP may diverge, because expenditure grows with total population and productivity while revenue grows with working-age population. In countries where the working-age population grows faster than total population, which is the demographic dividend, expenditure-to-GDP falls and fiscal space opens. In ageing countries it rises, which is a structural fiscal pressure independent of any policy choice.

Fiscal Balances (bottom right) carries two lines: the primary balance, meaning revenue minus non-interest expenditure, and the overall balance, which includes interest payments. The zero line is the reference: a primary surplus means the government is generating enough non-interest revenue to cover non-interest spending. The gap between the two lines is the interest burden on existing debt.

⚠ SCREENSHOT-TODO

Baseline tab for Uganda, all three charts, with callouts on: (a) the end of the shaded WEO period at 2029, (b) the 2099 debt ratio, (c) the primary-to-overall balance gap. This is the image the whole module refers back to.

5.6.3 Step 3: run the sanity check before you interpret anything

Baseline Sanity Check

Before interpreting climate results, verify the baseline makes sense:

- ☐ Does the initial debt level match your country's actual current debt-to-GDP ratio?
- ☐ Is the revenue-to-GDP ratio plausible given WEO forecasts?
- ☐ Does the expenditure path reflect realistic spending trends?
- ☐ If the fiscal rule is on, does debt converge toward your target?
- ☐ Do the primary and overall balance paths make economic sense given the country's fiscal history?

! DRAFT FOR TEAL: the sanity check applied to Uganda

Run the five boxes against what is on your screen and against the published figures.
Initial debt level. The published baseline puts 2023 at 47.1 percent of GDP, and Uganda's Fiscal Risk Statement reports 46.9 percent at June 2023 on the ministry's own basis. Those agree. If your Explorer run shows something materially different, the likely cause is the WEO

vintage, because the Explorer ships October 2024 and the published table came from the 2023 run. Note the difference rather than fixing it by overriding data.
Revenue-to-GDP. Revenue is held at a constant share of GDP by construction, so the check is not whether it moves. It is whether the level it is held at is the right one. For a country in the middle of a revenue mobilisation strategy, a flat revenue ratio is a conservative assumption and worth one sentence in the write-up.
Expenditure path. The published table shows primary expenditure drifting from 19.9 to 19.4 percent of GDP over seventy-six years, dipping to 18.8 around 2075. That is the demographic dividend showing up: total population growth drives spending, working-age population growth drives GDP, and for a while the second outruns the first. If your expenditure path rises instead of drifting down, check the demography variant.
Fiscal rule convergence. With the rule off, skip this box, and say in your packet that you skipped it and why.
Balance paths. The published baseline has the primary deficit narrowing from 6.3 percent of GDP in 2023 to 1.4 percent by 2099, with the overall deficit at 3.8. That is a large near-term consolidation built into the WEO forecast period rather than something the model invented. Whether it is credible is a question about the WEO, and it belongs in your write-up as a caveat rather than a correction.
The rule this step encodes: a climate result computed on a baseline you have not checked is a number with no owner. Every one of the five boxes is a question someone will ask you.

5.6.4 Step 4: read the Climate tab

The Climate tab shows how climate change affects real GDP under six warming scenarios, using empirical estimates from the FADCP Climate Dataset (Centorrino, Massetti, and Tagklis, 2024), building on Kahn et al. (2021).

What to look for. Two charts show absolute GDP levels and a GDP index set to 100 in 2029. The six scenarios run from Paris-aligned, below 2°C warming by 2100, to Hot Unadapted, high warming with slow adaptation. Impacts begin in 2030, and before that year all scenarios match the baseline. The gap between scenarios is the range of GDP outcomes across different climate futures.

Countries closer to the equator generally show larger GDP losses. The empirical estimates reflect historical relationships between temperature deviations and economic output, and countries already in warm climates suffer more from additional warming.

A country showing a large gap between Paris and Hot Unadapted faces high climate-fiscal vulnerability. A small gap suggests lower direct climate exposure in the model, though it does not account for the indirect effects that Q-CRAFT's conservative methodology leaves out: natural disasters, sea-level rise, and trade disruptions (User Guide, p. 5).

! DRAFT FOR TEAL: what the Uganda GDP chart licenses you to say

The published result is a level GDP loss of around 4 percent by end of century under the Hot scenario, and the C-PIMA summary describes Uganda's impact as milder than other sub-Saharan African countries.
Four percent needs a perspective sentence, because on its own it reads as small. Two are available and they point in opposite directions, so pick deliberately and say which you picked.

The deflating one: 4 percent of GDP spread over seventy-five years is a few hundredths of a percentage point off annual growth, well inside the noise of any single year's outturn.
The alarming one: it is a permanent level loss that compounds into the debt ratio every year thereafter, and 4 percent of GDP is a recurring annual amount comparable to a large sector budget.

> **TODO: verify the perspective figure**
> The second framing needs a checked comparator before it ships: 4 percent of Uganda's GDP expressed against a named line of the approved budget (health, or the road sector), sourced to the Approved Estimates for the relevant year. Do not ship an approximate comparison in a document aimed at the ministry that publishes the actual number.

Both framings are true. The second is the one a fiscal risk statement is for, and the first is the one you should expect back from a sceptical reader. Write the paragraph so it survives both.
What the chart does not license: any statement about what Uganda's GDP will be. This is a deviation between two runs of the same model, not a growth forecast.

⚠ SCREENSHOT-TODO

Climate tab for Uganda, GDP index chart, with the six scenario lines labelled on the data rather than in a legend, and the 2030 divergence point marked.

5.6.5 Step 5: read the Analysis tab

The Analysis tab is the comparison view, overlaying baseline and all climate scenario debt trajectories on a single chart.

The climate-fiscal risk premium. The spread between the baseline debt trajectory and the climate scenario trajectories is the country's climate-fiscal risk premium: the additional debt burden attributable to climate change. Three factors set that spread.

1. **Climate exposure:** how much GDP the country loses under each scenario, from the Climate tab.
2. **Expenditure rigidity:** how much of the GDP loss passes through to the primary balance and debt. Higher rigidity means a wider spread.
3. **Starting fiscal position:** countries with unfavourable baseline debt dynamics, where interest exceeds growth, see climate shocks amplified through the debt dynamics equation.

Reading the chart. A baseline showing stable or declining debt alongside a Hot Unadapted scenario that rises rapidly means climate change could destabilize an otherwise sustainable fiscal position. That is the core finding Q-CRAFT is designed to surface, and it answers one question: even if our fiscal policy is sound today, could climate change make it unsustainable?

The figure below is a real fan chart with the three readings marked on it, so you can see which part of the picture each one is looking at.

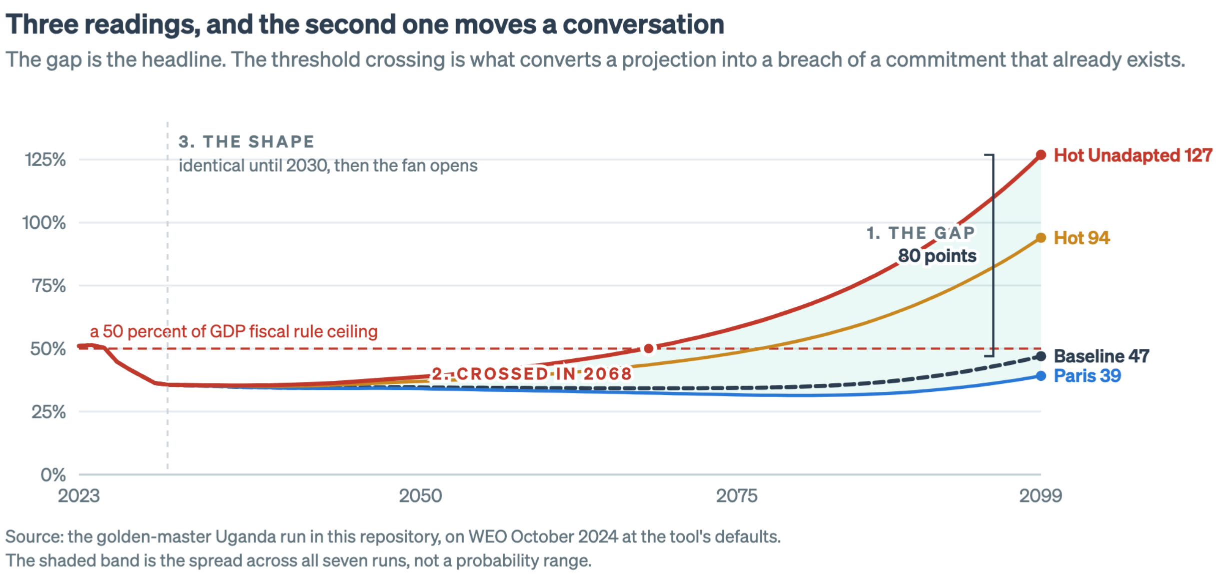


Figure 5.3: These are the Explorer's own numbers, on its current bundled data at its default settings. They run well above the published 2023 workshop figures quoted in this module, and the whole of that difference is vintage and parameter choice.

The gap is the number people quote, the crossing year is the number that changes a meeting, and the flat stretch to 2030 is the reminder that none of this is a forecast of the next decade.

⚠ TODO: two sets of Uganda numbers now coexist in this module

The figure above is the Explorer's own run on WEO October 2024 at the tool's defaults, and it ends around 127 percent of GDP under Hot Unadapted. The prose in this module quotes the published 2023 workshop figures, which used an earlier vintage and its own parameter choices, and which end at 66 percent under Hot. Both are correct for what they are, and the difference is exactly the vintage-and-parameters point Chapter 2 makes. Whether to keep both in one module, and which one leads, is an editorial call for review.

! DRAFT FOR TEAL: reading the Uganda fan chart

Three readings, in the order a reviewer will want them.
The gap. Baseline at 47.5 percent of GDP in 2099, Hot at 66. Eighteen and a half percentage points is the number that goes in the first sentence. Uganda's own Fiscal Risk Statement says the same thing from the other end: the primary deficit in the Hot scenario ends 0.7 percentage points worse than baseline, "thereby raising public debt by over 18 percent of GDP." A 0.7 point annual deficit gap producing an 18 point debt gap is the compounding traced in Step 3 of Chapter 3, and it is worth saying out loud, because it is the part non-specialists miss.
The threshold. The gap only becomes a policy fact when it crosses something. Uganda's Statement reports that in the High, Hot and Vulnerable scenarios debt passes the 50 percent of GDP fiscal rule ceiling. That sentence does more work than the 18 points, because it converts a projection into a breach of an existing commitment.
The shape. Look at when the lines separate, as well as where they end. Scenarios are identical

until 2030, then fan slowly. A path that crosses the ceiling in 2085 and one that crosses in 2060 imply very different urgency, and the fan chart shows the difference at a glance where a 2099 table does not.

What to do when the baseline itself is rising. If your run has debt rising in the baseline, the climate gap is no longer the interesting number, because both paths are unsustainable. Say so, and report the year each path crosses the anchor instead.

5.6.6 Step 6: export the packet

The Data tab holds an interactive data grid and two CSV downloads.

- **Download Baseline CSV:** the baseline fiscal projection, with no climate scenarios. Useful for folding Q-CRAFT projections into your own fiscal framework or DSA.
- **Download All Scenarios CSV:** baseline plus all six climate scenarios, stacked with a scenario identifier column. Use this for custom analysis, cross-country comparison, or integration with other tools.

All values are in billions of local currency units, except ratios, which are percentages of GDP. The data covers the full projection period from 2009 through 2099.

Download both. Add them to the parameter table from Step 1 and the Document it entries from Chapter 4. That is the export packet: inputs, rationale, output, in one place, reproducible by someone who has never met you.

This is the fourth of the four reasons in Chapter 2, arriving as files. The assumptions, your written remarks on them and the presentation-ready output come out of the same run, so the half hour that would have gone on copying numbers into a document goes on the paragraph in Step 7 instead, which is the part anyone will read.

5.6.7 Step 7: write the two paragraphs

! DRAFT FOR TEAL: model two-paragraph write-up

Built from the published Uganda figures so every number in it is checkable. Learners replace them with their own run.

> Climate change presents a measurable long-term risk to Uganda's fiscal position. Using the IMF's Quantitative Climate Risk Assessment Fiscal Tool, the baseline projection, which assumes no climate damage, has public debt at 47.5 percent of GDP in 2099 after falling to around 36 percent by mid-century as the demographic dividend widens the gap between growth and the effective interest rate. Under the Hot scenario, in which emissions follow SSP3-7.0 and temperatures track the 90th percentile of model projections, output is around 4 percent below baseline by the end of the century and public debt reaches 66 percent of GDP. The mechanism is indirect: higher temperatures slow labour productivity growth, slower growth reduces revenue, and with primary expenditure held at its baseline level in shilling terms the shortfall accumulates as debt. An annual primary deficit only 0.7 percentage points wider than baseline compounds into a debt ratio more than 18 percentage points higher over seventy years.

The policy significance lies in the threshold rather than the magnitude. In the High, Hot and Vulnerable scenarios, public debt passes the 50 percent of GDP fiscal rule ceiling and takes on a rising trajectory, which means climate damage alone is sufficient to move Uganda from compliance to breach without any other adverse shock. These estimates are conservative. They capture the slow productivity effects of temperature change and exclude natural disasters, sea-level rise, non-market damages and the public expenditure required to adapt, so the true fiscal exposure is larger than reported here. The results are also sensitive to the expenditure rigidity assumption: at full rigidity, the tool's default, the entire revenue loss becomes borrowing, while at 0.6 the end-century gap narrows materially. Meeting Uganda's Paris commitments and building expenditure flexibility both reduce this exposure, and the two are complements rather than alternatives.

Six things make this a Fiscal Risk Statement paragraph rather than a report of a model run:

1. It names the tool and the scenario in one clause each, then moves on.
2. Every number is a comparison between two runs, never a forecast.
3. It gives the mechanism in one sentence, so a reader who does not know the debt dynamics equation can still follow the causal chain.
4. It states the threshold, because a breach of an existing commitment is what makes a projection actionable.
5. It states the conservatism caveat before anyone else can, which is what makes the rest credible.
6. It ends on something the ministry can do.

One editorial query for Teal: the closing sentence of the second paragraph does policy advocacy inside a risk document. Whether that register is right is a call I should not make on your behalf.

! Now do your own country

The seven steps are the part that transfers. Uganda supplied numbers you could check them against. Nothing in the sequence depended on the country: set the parameters and write down why, check the baseline before you interpret anything, read the three tabs in order, export the packet, write two paragraphs.

The rest of this module runs that method twice more, on countries that behave differently from Uganda and from each other, with less help each time. The capstone in Chapter 7 runs it on yours.

5.7 Completion problem: Ethiopia under the High scenario

New country, same method, most of the setup fixed for you. Ethiopia earns its place here because its demography runs the other way from most countries in a fiscal risk conversation: the working-age population is still climbing, which pushes the baseline in one direction while climate damage pushes the other.

Done for you:

- The method: the same seven steps, in the same order. You are not inventing a procedure.

- The scenario: **High** is SSP3-7.0 at the average of model temperature projections, rather than the 90th percentile.
- The parameter template from Step 1, with one open cell. The debt target is Ethiopia's own fiscal anchor if you can establish one, and 50 percent of GDP if you cannot. Record which you used and why.

Your turn, six tasks:

1. Run the Baseline Sanity Check on Ethiopia. This is a different baseline from the worked case, so the box is now yours and it comes before any climate reading.
2. Read the 2099 debt ratio under High from the Analysis tab. Record it. ______
3. Compute the gap to baseline. ______ percentage points.
4. Read the year the High path crosses your target, if it does. ______
5. Write one sentence explaining why High sits below Hot when both use the same emissions pathway.
6. Rewrite the first sentence of the model paragraph above using your Ethiopia numbers.

> **i** DRAFT FOR TEAL: checking yourself on task 5
>
> The distinction is the temperature distribution rather than the emissions. High takes the average temperature outcome across climate models running SSP3-7.0, and Hot takes the 90th percentile of that same set. Same policy world, different draw from the uncertainty. Which one belongs in a fiscal risk statement is a real question: risk documents conventionally report adverse tail outcomes, which argues for Hot, while a central-case fiscal framework argues for High. Report both and say why you led with the one you led with.

5.8 Independent problem: adaptation speed, on a third country

No scaffolding this time, a checklist only, and a country whose fiscal position looks like neither of the first two.

The task. Compare **Hot + Adapted** against **Hot + Unadapted** for **Zambia**, or for another country in the dropdown whose interest-growth differential runs against it. Temperatures are the same in both scenarios, and the difference is how quickly the economy adjusts.

Deliver five things:

- ☐ The 2099 debt ratio under each, and the gap between them
- ☐ One sentence on what that gap represents in policy terms
- ☐ One sentence on what it does **not** represent, which is the trap in this comparison
- ☐ The same comparison at expenditure rigidity 0.6, with a note on whether the adaptation gap or the rigidity gap is larger for your country
- ☐ A two-paragraph write-up in the format of Step 7

> **i** DRAFT FOR TEAL: the trap, if you want to check before you write
>
> Faster adaptation in Q-CRAFT is a faster economic adjustment to a given temperature path. It is not adaptation spending. The model does not cost the investment that would produce that adjustment, and the User Guide says so directly (p. 6). So the Adapted-to-Unadapted

gap is the benefit side of an adaptation programme with the cost side missing. Reporting it as the value of adaptation without that sentence attached is the likeliest way to misuse this tool in a budget discussion.

5.9 Wrapper: what you can now do

- You have run three countries, one with full support and one alone, and produced an export packet with rationale attached.
- You can apply the Baseline Sanity Check and say what each box protects you from.
- You can write two paragraphs that a Fiscal Risk Statement editor would recognise.

Common errors on this material: interpreting climate results before checking the baseline, quoting a 2099 level as a forecast, and reporting the adaptation gap as the value of adaptation.

On your desk this week: take the two paragraphs you drafted to whoever owns your ministry's fiscal risk chapter and ask what they would cut. The answer tells you more about the format than this module can.

Chapter 6

What the tool can and cannot tell you

In this module
You will start with what the tool does well and why the IMF built it that way, because a caveat is only useful to someone who knows what it qualifies. Then you will learn what Q-CRAFT leaves out and which direction the omissions run, so you can state the limit before a reviewer states it for you. You will learn to read a fan chart whose baseline is clipped at zero when the climate scenarios are not. Last, you will practise telling a Q-CRAFT question from a debt sustainability question, in both directions.

Fast path

No fast path. This is the shortest module and it is the one that keeps you out of trouble. If you only read one section, read Section 6.7.

6.1 Warm-up

From Chapter 5:

1. What does the gap between Hot Adapted and Hot Unadapted represent, and what does it leave out?
2. Which check comes before any climate interpretation?

(Answers: the benefit of faster adjustment, with the cost of achieving it not modelled. The Baseline Sanity Check.)

6.2 The question that comes after the presentation

You have presented the fan chart. The senior official who will have to defend it looks at the Hot Unadapted line and asks: “So this is what climate change will cost us?”

Your whole credibility rests on the next thirty seconds. This module is that answer, in its several parts.

6.3 By the end of this module you can

- **Name** four things Q-CRAFT does well, and **say** which design choice buys each one
- **List** the specific damage channels Q-CRAFT excludes, and **state** the direction of the resulting bias
- **Explain** why baseline and climate scenarios treat negative debt differently, and **adjust** your reading of a chart accordingly
- **Compare** two countries with similar debt paths and **distinguish** the different drivers behind them
- **Decide** whether a given question calls for Q-CRAFT, the LIC-DSF, both, or neither, and **justify** the choice

6.4 Where you are in the course

The map below lights the two boxes this module is about.

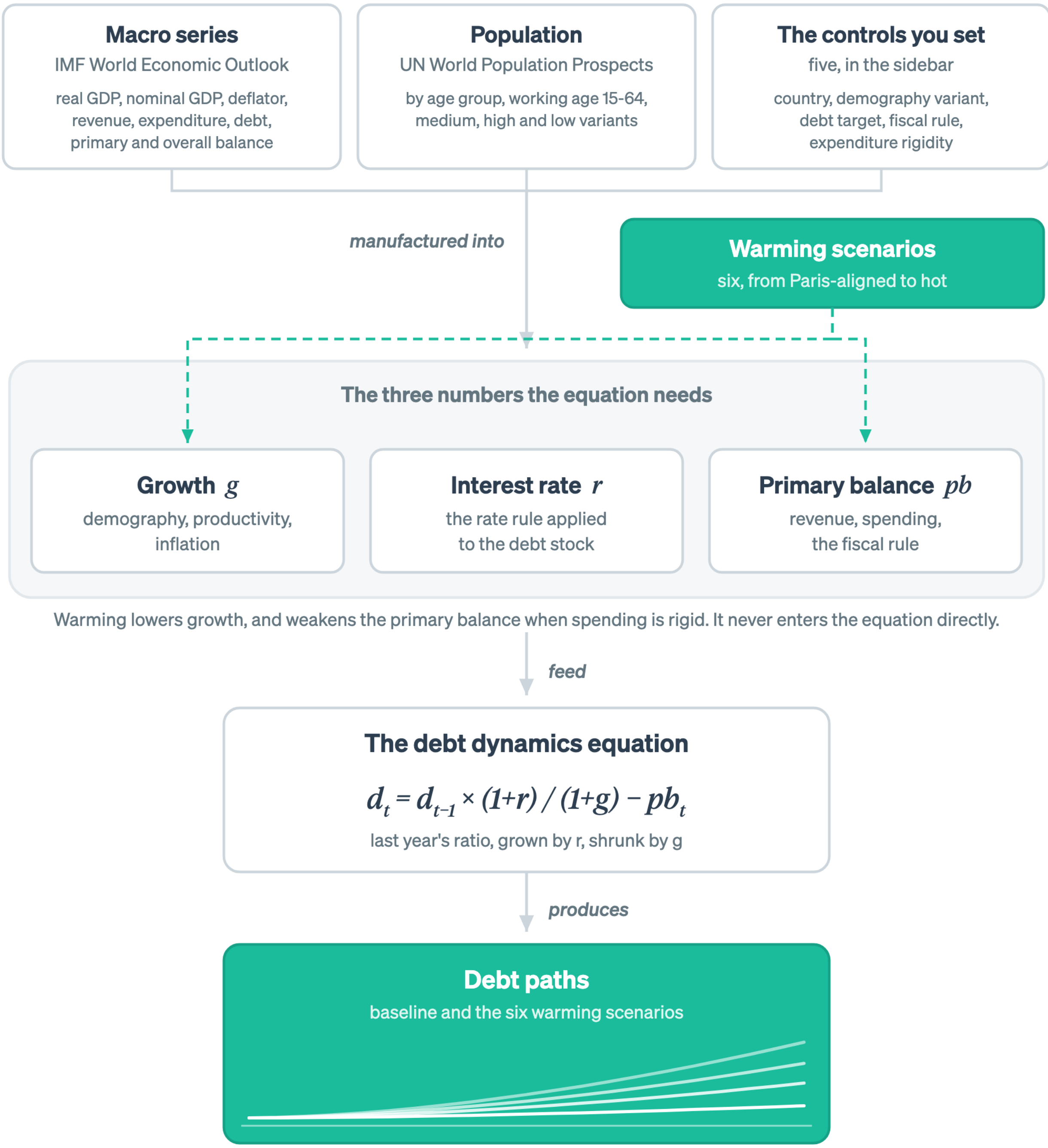


Figure 6.1: This module is about what the warming scenarios contain, and what the debt paths therefore license you to write.

Two nodes are lit, and they are the two ends of the climate channel: what the warming scenarios contain, and what the debt paths let you write.

6.5 What the tool is good at, and why it was built that way

Here is the whole module in one breath, before the detail arrives. **The strengths:** Q-CRAFT runs for most of the world, on data that is already published, with no bespoke country dataset needed, and its arithmetic is verified against the IMF's original workbook. **The limitations:** the scenarios capture the slow productivity channel and exclude tipping points, nonlinearities, natural disasters, sea-level rise, adaptation costs and country-specific channels, so the results read conservative, a lower bound on fiscal impact rather than a central estimate of the total (User Guide, pp. 5-6).

Those two halves are one design choice seen from two sides. Q-CRAFT does four things well, each one carries a cost, and the strengths come first because a caveat is only useful to someone who knows what it qualifies.

It isolates one channel and models it from data. Temperature affects growth, growth affects revenue, and revenue affects debt. Q-CRAFT takes the temperature-to-growth step from cross-country empirical estimates using the Kahn et al. (2021) methodology, updated in the FADCP Climate Dataset (Centorrino, Massetti, and Tagklis, 2024), then carries it through a standard debt dynamics equation. One channel, estimated from data, pushed through arithmetic a reviewer can check line by line. That is a narrower claim than an integrated assessment model makes, and it is a claim you can defend in a room.

It runs on data every country already has. The User Guide is explicit that the tool can be deployed "without any additional data, even in low-capacity settings", building its scenarios from IMF, UN and World Bank sources (p. 6). No new collection, no survey, no waiting for a mission. A team can have a first pass by the end of the week. The same page is equally explicit that country expertise is still needed to make the baseline match the country's own fiscal framework, which is what Chapter 4 is for.

It is comparable across countries. The same production function, the same debt dynamics equation and the same six scenarios run for every economy in the tool. Two countries' results can therefore be set beside each other, which is what makes the diagnosis exercise later in this module work at all.

It produces the shape a fiscal risk chapter needs. The output is a comparison between scenarios rather than a forecast, and a risk chapter is a comparison between scenarios. The User Guide positions Q-CRAFT as "a starting point for climate change fiscal risk analysis" and as a foundation for deeper work with general equilibrium models where capacity allows (pp. 5-6).

Those four follow from one brief. The Fiscal Affairs Department needed something a country team could run in a week, on data that already exists, producing a defensible comparison rather than a false forecast. Every limitation in the rest of this module follows from the same brief, which is why they read as costs of a choice rather than as oversights.

6.6 What the numbers mean, and what they do not

Q-CRAFT Explorer produces stylized long-term projections. They are useful for understanding the direction and magnitude of climate-fiscal risks, comparing scenarios, and identifying the fiscal parameters that matter most for a given country.

The results are not forecasts. Projections to 2099 depend on assumptions about productivity, inflation, interest rates, and climate that involve deep uncertainty. The value sits in the comparison across scenarios and the sensitivity to assumptions, never in any single projected number.

Q-CRAFT is designed to complement existing fiscal analysis rather than replace it. The IMF User Guide positions it as "a starting point for climate change fiscal risk analysis" that supplements country macroeconomic models (User Guide, pp. 5-6). Use Q-CRAFT to identify which climate-fiscal channels matter most for your country, then investigate those channels with more detailed models and country-specific data.

6.7 Every exclusion is a cost left out, so the estimate is a floor

The scenarios capture one channel: the slow effect of temperature change on productivity, and through it on growth. Six things are excluded, the User Guide names all six, and every one of them runs in the same direction.

The figure below puts the modelled channel and the six exclusions on one page, with the direction they share drawn along the bottom.

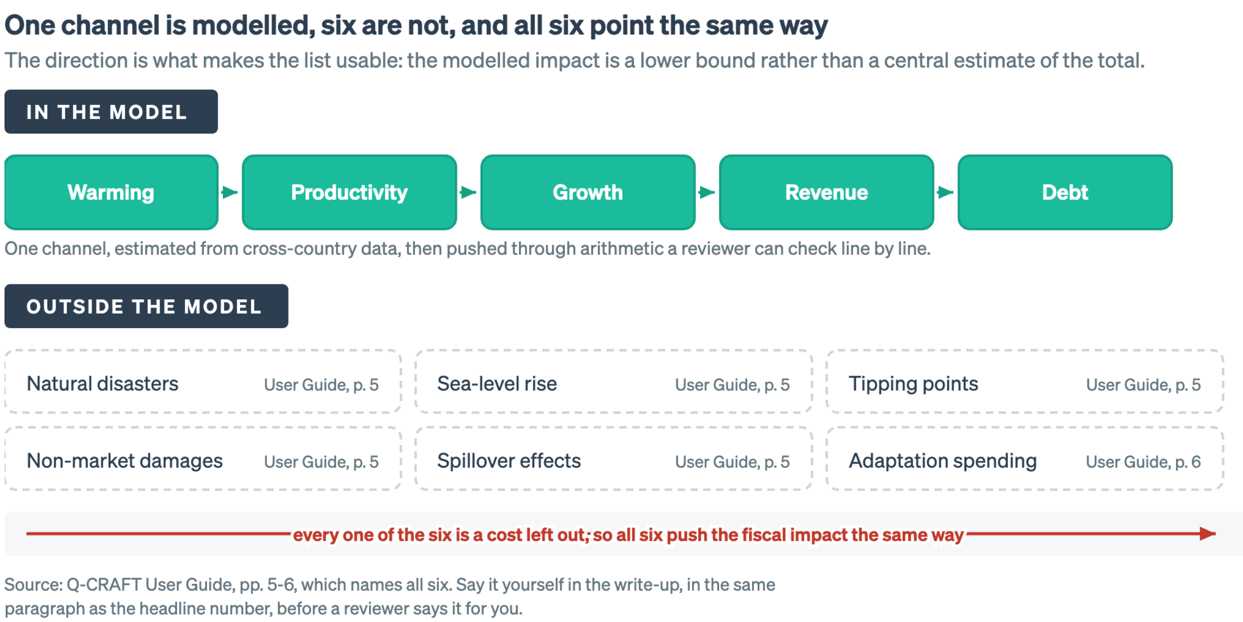


Figure 6.2: Each exclusion carries the User Guide page that documents it, so the list is quotable in a footnote without further research. What runs across the top is everything that is left.

Nothing on that page is a cost the model overstates, which is why the estimate is a floor rather than a central case. The table repeats the six with the reason each one matters for a risk statement.

Excluded	Documented at	Why it matters for a fiscal risk statement
Climate-induced natural disasters	User Guide, p. 5	Often the channel a national risk statement leads with. Uganda's puts floods and epidemics at about 75 percent of recorded events, 1985 to 2021
Sea-level rise	User Guide, p. 5	Absent for a landlocked economy, decisive for Bangladesh or a small island state
Tipping points	User Guide, p. 5	The estimates are fitted to historical temperature variation and cannot extrapolate to discontinuities
Non-market damages	User Guide, p. 5	Morbidity, mortality, conflict, food insecurity: real costs, absent from GDP
Spillover effects	User Guide, p. 5	Trade partners, remittance sources and commodity markets are hit too
Adaptation spending	User Guide, p. 6	Even the Hot Adapted scenario models faster adjustment, not the public expenditure that buys it

The direction is what makes this list usable. Every exclusion is a cost left out, so the modelled fiscal impact is a lower bound on the modelled channels rather than a central estimate of total impact. The User Guide reaches the same conclusion in its own voice: the results "do not account for the potential impacts of climate change induced natural disasters, sea-level rise risks and other environmental risks, rendering the outcomes conservative" (p. 5). One sentence follows from it: *actual fiscal impacts are likely larger than these results suggest.*

Say it yourself, in the write-up, before a reviewer says it for you. A conservative number with the conservatism disclosed is stronger than a larger number without it.

One more limitation sits inside the estimates themselves. The six above are things left outside the model. This one is a property of what is inside it. The damage estimates are fitted to historical variation in temperature, so they carry information about the range the world has already lived through and none about the range it has not. Nonlinearities beyond that range are therefore outside the estimate rather than judged to be small. Whether that understates the damage from warming well outside the historical record is an open empirical question. Expect to be asked it, and answer it the way this module keeps answering things: report the number, name the channel it covers, and name the channels it does not.

Partial equilibrium is a limit of a different kind. The User Guide describes the set-up as "essentially a partial-equilibrium" one (p. 5) and says plainly that Q-CRAFT "is not a forecasting model nor a general equilibrium model of the economy" (p. 6). Fiscal policy does not feed back into growth or interest rates, so cutting spending inside the model never slows the economy down. That limit has no obvious sign, unlike the six above: it flatters a consolidation scenario, and it also leaves out whatever growth an adaptation programme would buy. What it means in practice is that Q-CRAFT cannot cost a policy. It can show you what happens to the debt path under assumptions about policy that you supplied, and that is a different thing.

! DRAFT FOR TEAL: what conservatism does not buy you

Conservatism protects the credibility of a number. It does not make the number safe to quote out of context, and it creates one specific hazard worth naming.
A conservative estimate that gets picked up as *the* estimate becomes a ceiling in someone else's argument. "The analysis shows climate costs us 18 points of debt" is a sentence that starts life as your careful lower bound and ends life as a budget office's reason to fund adaptation at 18 points' worth. The defence is structural rather than rhetorical: put the exclusion list in the same paragraph as the headline number, not in a footnote, and not in an annex.

6.8 The baseline floors debt at zero and the climate scenarios do not

i The rule

The baseline scenario applies a floor of zero to the debt-to-GDP ratio: if the equation produces a negative value, debt is set to zero. Climate scenarios do not apply this floor, and debt is allowed to go negative, representing net asset positions. The asymmetry is intentional, because it avoids masking the full range of climate-scenario outcomes.

The two panels below are the same projection, drawn as the chart shows it and then as it actually is.

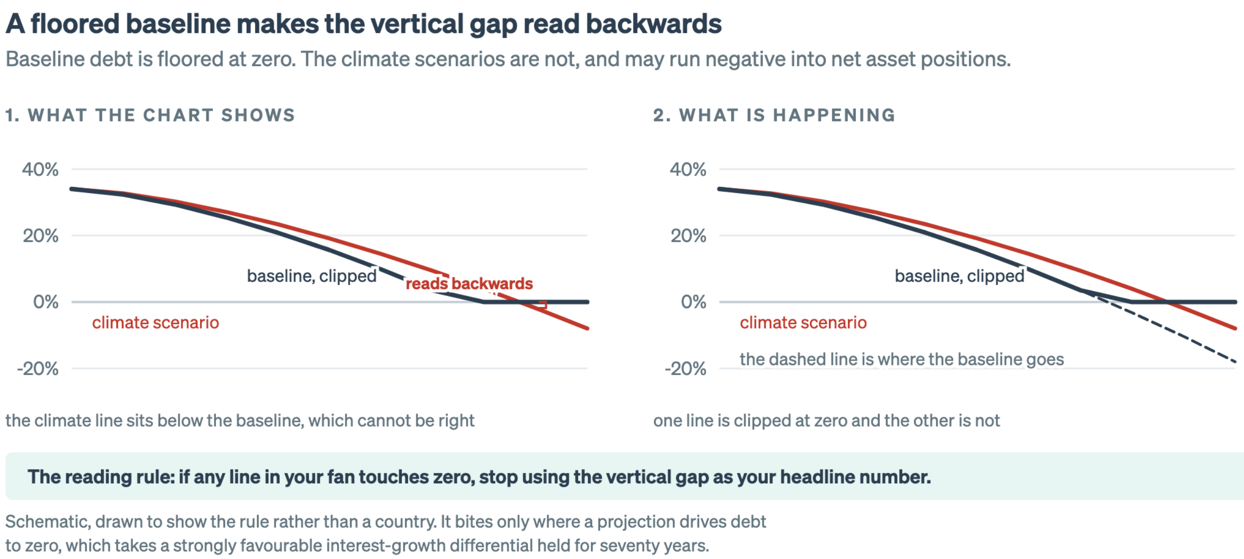


Figure 6.3: The floor is a deliberate choice rather than a bug. Flooring the climate scenarios too would compress exactly the range the tool exists to show, so the asymmetry stays and the interpretive burden lands on you.

In the left panel the climate line sits below the baseline, which cannot be true, and the right panel shows the reason: the baseline stops at zero and the climate scenario does not.

! DRAFT FOR TEAL: how the asymmetry changes what you are looking at

For a narrow set of countries the asymmetry matters a lot.

When it bites. Only when a projection drives debt to zero. That happens for countries with strongly favourable interest-growth differentials and sustained primary surpluses over a seventy-year horizon. It is not a problem in the worked case, whose published baseline lands at 47.5 percent of GDP.

What it does to the picture. In such a country the baseline flattens along zero while the climate scenarios continue downward into negative territory. Read naively, the chart shows climate scenarios with *lower* debt than baseline, which is nonsense. What you are actually seeing is one line clipped and the others not.

The reading rule. If any line in your fan chart touches zero, stop using the vertical gap as your headline number. Check the underlying series in the Data tab, then either report the comparison over a window before the floor binds or report the primary balance gap instead, which is not clipped.

Why the asymmetry is a design choice rather than a bug. Flooring the climate scenarios too would compress exactly the range the tool exists to show. The choice preserves the spread and pushes the interpretive burden onto you. This module is where that burden gets paid.

6.9 Two countries, same chart, different problem

Transfer is the thing this course is actually for: recognising which situation you are in when the labels come off. Structured comparison is how it gets taught.

Comparison exercise: same path, different drivers

Find the pair. In the Explorer, work through countries in the dropdown until you find two whose baseline 2099 debt ratios are within about 5 percentage points of each other, but whose baseline-to-Hot-Unadapted gaps differ by 10 percentage points or more.
Then diagnose. For each country, decide which of the three drivers from Chapter 5 explains the difference:

1. Climate exposure, meaning how much GDP is lost, from the Climate tab
2. Expenditure rigidity, meaning how much of that loss becomes debt
3. Starting fiscal position, meaning whether the interest-growth differential amplifies or damps it

Write one sentence per country saying which driver dominates, and one sentence saying what policy conclusion follows for each. The conclusions should differ. If they do not, you have picked a pair that is too similar.
Then reverse it. Find two countries with similar climate exposure on the Climate tab whose debt outcomes diverge. The driver is now somewhere in the fiscal position, and finding it is the exercise.

TODO: seed the comparison with verified pairs

This exercise currently sends learners hunting. Two or three verified country pairs, run and checked against the current data vintage, would make it usable in a 20-minute workshop breakout. Needs a real run rather than an assumption about which pairs work.

! DRAFT FOR TEAL: why this exercise and not a worked comparison

The instinct is to hand over the pair and the answer. Doing the search is the exercise, because the skill being built is diagnosis from surface features that mislead, and a country pair you were handed has already had the diagnosis done for you.
The general principle is the one to carry out of this module: a debt path is an outcome, not a diagnosis. Two countries arriving at 60 percent of GDP in 2099, one because it is highly exposed to warming and one because it borrows at rates above its growth, need different policies and different chapters in different documents. Q-CRAFT will show you the outcome. Deciding which country you are looking at is your job, and it is the part that cannot be automated.

6.10 Q-CRAFT and the LIC-DSF answer different questions

Q-CRAFT and the IMF-World Bank Low-Income Country Debt Sustainability Framework both produce debt projections for low-income countries. They are not substitutes, and using one where the other belongs is a common and expensive error.

	Q-CRAFT	LIC-DSF
Question it answers	How much does climate damage change the long-run fiscal path?	Is this country's debt sustainable, and at what risk rating?
Horizon	To 2099	Medium term, with longer-run tails
Output	Scenario comparison, no rating	A risk rating that carries operational consequences
Climate	Central: six scenarios	Not the organising principle
Debt detail	One aggregate stock	External and public splits, currency, creditor composition
Used for	Fiscal risk disclosure, C-PIMA annexes, long-term planning	Lending decisions, concessionality terms, programme design

Self-check: five questions, which tool?

For each, name Q-CRAFT, LIC-DSF, both, or neither, and say why in one line.

1. "The Fiscal Risk Statement needs a section on climate. What do we run?"
2. "Are we still at moderate risk of debt distress?"
3. "If we take this Eurobond instead of an IDA credit, what happens to our risk rating?"
4. "What is the fiscal cost of the National Adaptation Plan?"
5. "Does climate risk change our debt sustainability assessment?"

DRAFT FOR TEAL: self-check answers

The test in every case is which question the tool was built to answer, never which tool you happen to have open.

1. **Q-CRAFT.** This is its designed use, and it is the use Uganda already made of it in the FY2024/25 statement.
2. **LIC-DSF.** Q-CRAFT produces no risk rating and has no thresholds to breach. Answering this question from a Q-CRAFT chart is the error this section exists to prevent.
3. **LIC-DSF.** The question is about creditor composition and terms, which Q-CRAFT does not represent, because it carries one aggregate debt stock and one effective interest rate.
4. **Neither.** Q-CRAFT does not cost adaptation, by design (User Guide, p. 6), and the LIC-DSF does not either. This question needs a costed adaptation plan, and the honest answer is to say so rather than to produce a number from a tool that cannot see it.
5. **Both, in sequence.** This is the interesting one. Q-CRAFT establishes whether the climate channel is large enough to matter for the fiscal path. If it is, the finding belongs alongside the DSF assessment as a qualitative risk consideration rather than as an input to it, because the two run on different horizons and different definitions of debt. Anyone who tells you they have mechanically fed Q-CRAFT output into a DSF has done something that needs explaining.

The reverse direction is the part usually skipped. A question that arrives framed for the DSF may be a Q-CRAFT question. “Our risk rating is fine, so climate is not a debt issue” is a DSF answer to a long-horizon question the DSF does not ask. Uganda’s own result makes the point: a baseline that is comfortable on any medium-term view still crosses the fiscal rule ceiling by end of century once climate damage is applied.

6.11 Wrapper: what you can now do

- You can say what the tool does well and which design choice buys each strength.
- You can list what the scenarios exclude and state the direction of the bias without looking it up.
- You can spot a clipped baseline in a fan chart and say why the vertical gap has stopped meaning what it usually means.
- You can take a question and say which tool it belongs to, in both directions.

Common errors on this material: quoting a conservative estimate as a total, reading a floored baseline as a favourable climate result, and answering a sustainability-rating question from a Q-CRAFT chart.

The thirty-second answer, for reference: “No. It is what climate change does to the debt ratio through one channel, growth, under stated assumptions about how spending responds. It excludes disasters, sea-level rise and the cost of adapting, so it is a floor rather than a total. What it tells us is that the channel is large enough to breach the ceiling on its own.”

On your desk this week: find a projection in circulation in your ministry and write down one question it cannot answer. Circulate that sentence with it.

Chapter 7

The capstone

In this module

You will produce the capstone: an export packet plus a two-paragraph fiscal risk draft, for a country you choose. It is marked on whether you can defend your assumptions, never on whether your numbers match anyone else's. The module also closes the loop on Chapter 1, so you can see what changed.

Fast path

Short on time? Read the brief and the rubric, then do the work. The retake at the end takes ten minutes and is the cheapest way to find out whether the economics landed or only the clicking did.

7.1 Warm-up

From across the course, without looking:

1. Which two of the three numbers does climate damage move?
2. What does expenditure rigidity 1.0 mean?
3. Name one thing the climate scenarios exclude.

(Answers: g and pb. Spending is sticky, so the revenue loss becomes borrowing. Natural disasters, sea-level rise, tipping points, non-market damages, spillovers, or adaptation spending.)

7.2 By the end of this module you can

- **Produce** a complete export packet for a country of your choosing
- **Defend** each parameter choice against a specific challenge
- **Assess** your own draft against the rubric before anyone else sees it
- **Compare** your current capability against your Chapter 1 self-assessment

7.3 Where you are in the course

The map below is lit end to end for the first and last time.

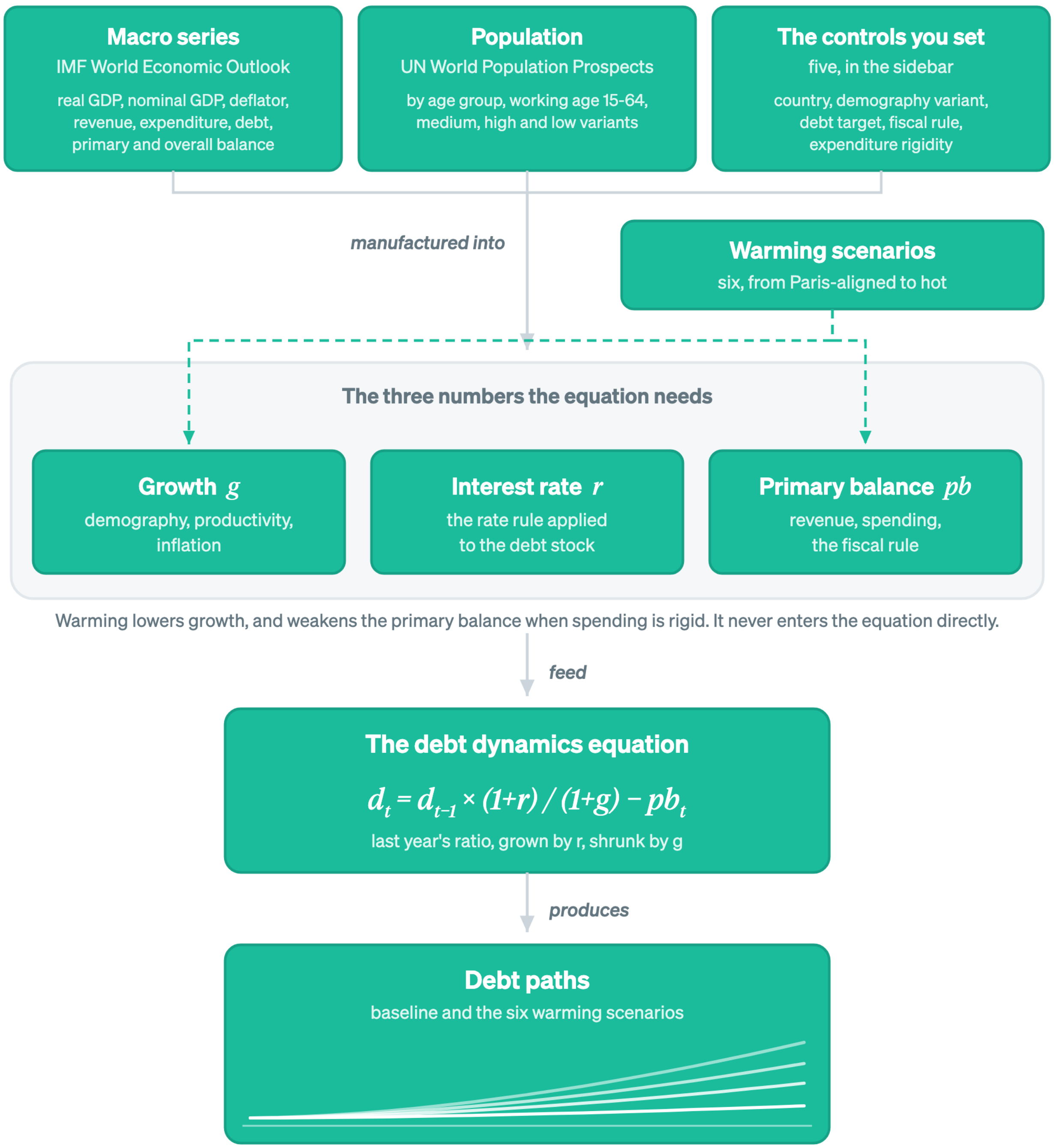


Figure 7.1: The capstone runs the whole chain, from the sources the tool loads to the paths you hand over.

Every node is lit, because the capstone runs the chain end to end.

7.4 The capstone brief

The scenario. You have been asked for a climate-fiscal risk section for your ministry's next fiscal-risk documentation, and you will defend it to the senior officials who sign it off. You have the Explorer, the WEO vintage it ships with, and whatever you know about the country. You have one week and half a page.

What to hand in, in three parts.

1. **The export packet.**
   - The parameter table: every setting, with a one-line rationale each (the five **Document it** entries from Chapter 4)
   - The data vintage, stated
   - Both CSV exports, baseline and all scenarios
   - At least two sensitivity runs, with what changed and by how much
2. **The two-paragraph draft**, in the format of Chapter 5, Step 7.
3. **Three challenge answers.** Written responses of two to three sentences each to:
   - "Is this a forecast?"
   - "Why did you choose that rigidity value?"
   - "So this is what climate change will cost us?"

The packet is the fourth reason from Chapter 2 in its finished form. Your assumptions, your remarks on them and presentation-ready output travel together as one handover, which is what leaves the week free for the policy question rather than for assembling the evidence by hand.

Country choice. Pick the country you work on, because the capstone is a rehearsal for the document you will actually have to write. Pick Uganda instead if you want a run you can compare against the published section Chapter 5 works through. Avoid Ethiopia and Zambia unless you want to extend Chapter 5, since you have already worked those.

What is not required. Matching anyone else's published figures. You are on a later WEO vintage and possibly different parameters, and reproducing someone else's numbers is not the skill being assessed.

7.5 The rubric

! DRAFT FOR TEAL: capstone rubric

Four criteria, and the weights are a proposal rather than a decision.

Criterion	Weight	Meets the standard	Falls short
Assumptions defended	40%	Every parameter has a rationale citing country evidence. Rigidity in particular has budget-composition reasoning behind it, and a sensitivity run.	Parameters listed without reasons, or reasons that restate the setting ("rigidity 1.0 because spending is rigid").

Baseline checked	20%	The five sanity-check boxes are answered against the actual run, including any that were skipped and why. Differences from the ministry's own figures are explained rather than reconciled away.	Climate results interpreted with no evidence the baseline was examined.
Interpretation	25%	Numbers presented as comparisons between runs. The mechanism stated in one sentence. A threshold identified. The conservatism caveat in the same paragraph as the headline number.	A 2099 level quoted as a prediction, a debt gap described as a cost, or the caveat relegated to a footnote.
Written for the reader	15%	Reads like your ministry's own risk chapter. A non-specialist can follow the causal chain. Two paragraphs, not five.	Reads like a description of a model run. Jargon undefined. Over length.

One error disqualifies on its own, and it recurs often enough to name separately: reporting the Adapted-to-Unadapted gap as the value of adaptation. It is the benefit side with the cost side missing (Chapter 5, independent problem).

Note for the live workshop. The rubric doubles as the peer-review instrument, run in three passes. First pass, each reviewer marks assumptions only. Second pass, interpretation only. Third pass, the writing. Splitting the passes stops reviewers spending all their attention on prose, which is what happens when they are asked to review everything at once.

The figure below puts the three parts of the hand-in beside the weights they are marked against.

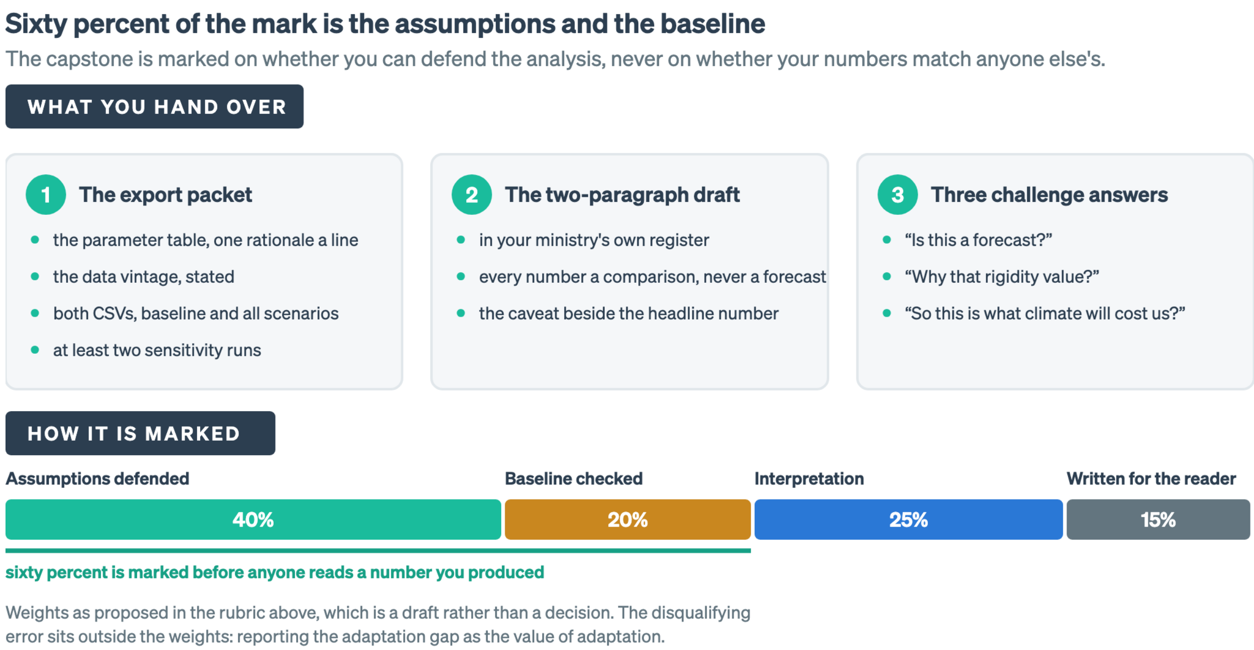


Figure 7.2: Nobody is marking whether your 2099 number matches a published one. The one thing that fails a capstone outright sits outside the weights entirely: reporting the adaptation gap as the value of adaptation.

Sixty percent of the mark is settled by your assumptions and your baseline check, both of which happen before anyone reads a number you produced.

7.6 Common errors to watch for

> ⚠ TODO: replace with pilot-observed errors
>
> The list below is built from the misconceptions this course was designed against, not from watching people make them. The redesign plan treats a pilot with two or three target-profile testers as non-negotiable, and the error log from that pilot should replace this list. Until then, treat it as a hypothesis.

- Reading rigidity backwards, so the headline risk number comes out inverted
- Interpreting climate results before checking the baseline
- Quoting a 2099 level as a forecast
- Adjusting the debt target while the fiscal rule is off, then concluding the target does nothing
- Describing a debt-ratio gap as a cost in local currency
- Reporting the adaptation gap as the value of adaptation
- Answering a debt-sustainability-rating question from a Q-CRAFT chart

7.7 Where you started, where you are

Go back to your three answers from Chapter 1. Answer them again now, before reading on.

On long-term fiscal projection: A1 (not worked with a DSA) / A2 (can follow one) / A3 (could build one this week)

On climate-fiscal analysis: B1 (climate side only) / B2 (can describe the use) / B3 (have run or interpreted one)

On the tool itself: C1 (never opened it) / C2 (clicked around) / C3 (ran it and explained the output)

The A and B rows are the ones worth looking at. If you have completed the capstone, C3 is true by construction, because you have run the tool and written the explanation. Movement on A and B is the course working. No movement on A and B with C3 achieved means you can drive the tool while the economics underneath it did not land, and that is worth a conversation rather than a shrug.

Retake the three questions from Module 0

Same three items, unchanged. Answer from memory before opening anything.

1. Zero primary balance, $r = 9$ percent, $g = 6$ percent. Debt ratio next year: falls, flat, or rises?
2. Expenditure rigidity 1.0 means what about spending under a climate scenario?
3. Debt at 66 percent under Hot against 47.5 percent baseline. What is the defensible statement of what that means?

   DRAFT FOR TEAL: the answers, and what a wrong one means now
   A wrong answer here costs more than it did in Chapter 1, because now it is attached to a document you are about to circulate.

   1. **Rises**, to about 51.4 from 50. If this one is still wrong, Step 1 of Chapter 3 did not do its job and nothing in your capstone's interpretation section can be trusted, because every climate result in the tool is this arithmetic with a lower g.
   2. **Spending stays at its baseline level in local currency terms**, so the whole revenue loss becomes borrowing. Getting this wrong at the end of the course means your headline number may have the wrong sign on its sensitivity, which is the error most likely to reach print.
   3. **Under stated assumptions, climate damage of this magnitude adds roughly 19 percentage points of GDP to the debt ratio by 2099 relative to the same projection without it.** A comparison between two runs, with the assumptions attached. If you reached for the forecast framing or the cost framing, reread Chapter 6 before you circulate anything.

7.8 What to use on your desk this week

- The debt-stabilizing primary balance for your country: three numbers, one division, and it opens most credible fiscal risk write-ups.
- The **Document it** habit: every assumption gets a line saying why. It costs a minute per parameter and it is the difference between an analysis and a set of numbers nobody can defend a year later.
- The which-tool question. When a request arrives, ask what question it is asking before opening anything.

7.9 Spaced follow-up

Two things arrive after the workshop, as part of the course rather than as an afterthought.

- **Day 3:** one retrieval question by email. No preparation, no marking.
- **Week 2:** an application check. What did you run, what did you find, what got used.

Spacing beats massing, and the IMF's own evaluation of capacity development found that training with hands-on follow-up sticks while the standalone workshop decays. The follow-up is the part that makes the rest count.

> ⚠ TODO: workshop materials
>
> The live sessions need artefacts this guide does not yet contain: breakout task cards, the anonymous poll items, and the peer-review sheet derived from the rubric above. Tracked separately from the book.

7.10 Wrapper: the whole course in six lines

- One equation decides the debt path, and the rest of the tool builds its three inputs. (Chapter 2)
- Two things move the ratio: the interest-growth gap, and the primary balance. (Step 1, Chapter 3)
- Every parameter is a judgment call, and the record of the call is the deliverable. (Chapter 4)
- The output is a comparison between two runs, written up in your ministry's own format. (Chapter 5)
- The estimate is a floor, the baseline is floored at zero and the scenarios are not, and some questions belong to a different tool. (Chapter 6)
- What you hand over is an analysis you can defend. (this module)

Appendix A

Appendix: from Q-CRAFT to the LIC-DSF

This appendix is not part of the course

The modules teach you to run the tool and defend the analysis. This appendix is about the project behind the tool: why it was built, what it is a proof of concept for, and how to shape the next version. Nothing in the capstone depends on it. Read it if you want to influence what V2 does.

This is not an IMF product

Q-CRAFT Explorer, this course and everything proposed below are an independent project by Teal Insights and NatureFinance. None of it is an official IMF product and none of it carries IMF endorsement. The work is meant to be complementary to the IMF's own training materials and to the Q-CRAFT User Guide (Tim and Rahman, 2024), which remain the authoritative references. Nothing in this appendix should be read as a criticism of a tool the Fiscal Affairs Department built for its own technical assistance and made available to everyone.

What you need to know:

- Q-CRAFT is a proof of concept for a larger goal: making the LIC-DSF more usable through human-centered design
- The biggest barrier to effective fiscal projection tools is ergonomics, not economics
- We are looking for co-design partners, not endorsements

A.1 The real barrier: ergonomics, not economics

The biggest barrier to effective use of fiscal projection tools is not the economics. It is the ergonomics. Smart people struggle not because the methodology is too complex, but because the tools do not guide them through the decisions they need to make.

This observation has emerged from years of conversations with MoF economists, central bank staff, and IFI economists. It has inspired a more organized human-centered design approach: structured design sprints and organized discussions with practitioners who use these tools in their daily work.

A.2 What we built and what we proved

Q-CRAFT Explorer reimplements the IMF's Q-CRAFT Excel workbook as an open-source Python web application. The calculation engine is a standalone package with automated parity tests against the original Excel tool.

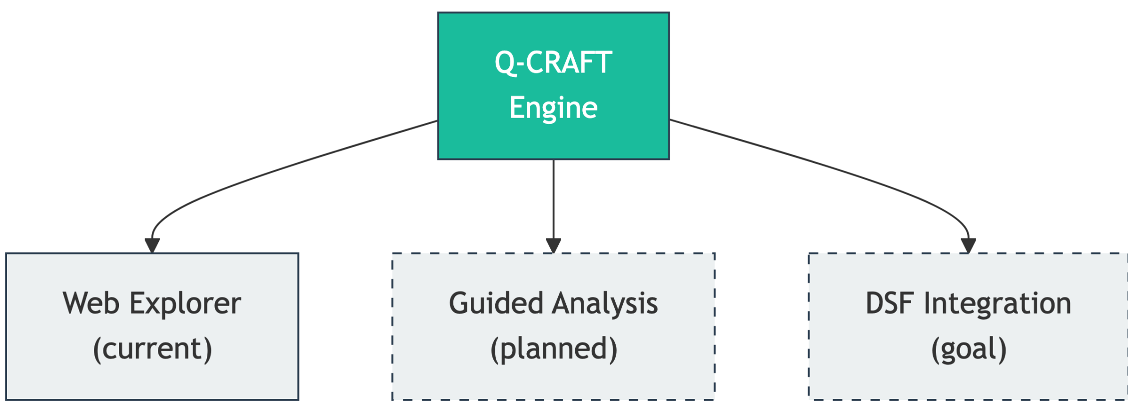
The testing pipeline is public and automated. Baseline parity is exact for 147 of 147 tested countries: zero percentage-point deviation across debt-to-GDP, revenue, primary balance and primary expenditure as shares of GDP, with a further 25 parameter sensitivity combinations passing. Climate-scenario parity is confirmed for ratio metrics, meaning debt-to-GDP, revenue and expenditure as percent of GDP. This kind of transparent, automated verification is essential for tools where the stakes are higher. Chapter 2 has the fuller statement of what the claim covers.

A.3 Q-CRAFT is the proof of concept

Q-CRAFT is deliberately small: five user-facing parameters, one core equation. We chose it because it demonstrates the approach in miniature.

The real goal is the LIC-DSF, the Low-Income Country Debt Sustainability Framework. The LIC-DSF modeling chain (Q-CRAFT, DIGNAD, LIC-DSF) is far more complex, far more consequential, and used for every low-income country. We are already working with end-users of the LIC-DSF, and we want to work with its designers too.

A.4 A modular engine, multiple interfaces



One engine, multiple interfaces

The same engine powers different front-ends for different needs:

1. **Web Explorer (current).** Interactive charts for quick country analysis. Select a country, adjust parameters, compare scenarios.
2. **Guided Analysis Tool (planned).** AI-assisted reasoning through input assumptions. Always available, cheaper than a technical assistance mission. Helps users understand what to

enter and why.
3. **DSF Integration (the goal).** Q-CRAFT and DIGNAD feeding into LIC-DSF assumptions, with the same ergonomic layer applied to a far more consequential tool.

The current process makes it hard to do things the right way. Documentation is separated from implementation. If you do not know what to enter for a parameter, you search the user guide and hope it is there (and often it is not). The goal is design that makes it easy to do the right thing and hard to do the wrong thing.

A.5 The co-design invitation

We built V1. Here are specific areas where practitioner input shapes V2.

1. **Parameter guidance.** How should we help users choose productivity growth, expenditure rigidity, and debt target values? What heuristics do experienced practitioners use?
2. **Default assumptions.** What are sensible defaults by country type? Should defaults vary by region?
3. **Capacity development integration.** How would this tool fit into an actual CD workshop or IMF mission?
4. **Validation priorities.** Which countries and scenarios should we prioritize for verification?
5. **Missing features.** What is the first thing you would want that is not here?

What we are asking for: a few hours from a few people for design feedback, continued communication so we build credibly, and feedback on whether we are making something useful.

What we are NOT asking for: official endorsement (we understand institutional constraints), access to confidential information (we work with publicly available materials), or money (this is grant-funded, MIT-licensed open source).

A.6 Why start here

Two reasons to start with Q-CRAFT:

1. **Immediate value.** A co-designed Q-CRAFT tool could be in use within weeks. It costs nothing and requires minimal commitment to evaluate. It complements, not replaces, the Excel workbook.
2. **Pioneer the approach.** Success with Q-CRAFT builds trust and methodology for the LIC-DSF. We prove the design process works on a small tool before applying it to the one that matters most.

A.7 The SovTech vision

Central banks have been applying technology to supervisory functions for years. The BIS documented "SupTech generations" as a framework for technology-driven supervisory and regulatory solutions. The question is no longer whether such tools belong in public financial institutions. The question is when sovereign debt management will catch up.

We call this **SovTech**: the application of SupTech principles to sovereign debt analysis. Three principles guide this work:

Modularity that respects how people work. Same engine, different interfaces. Components like building blocks: swap one module without rebuilding everything.

Open source as institutional infrastructure. Transparency means anyone can examine how analysis is done. MIT license, no barriers to adoption. The calculation engine and the interface are separate concerns, developed and tested independently.

AI as enabler, with a trust layer. The cost of building analytical software has dropped dramatically. But sovereign debt analysis serves policy decisions worth billions of dollars. Verification pipelines, automated testing, and human-in-the-loop governance are requirements, not optional features. This is why we built the parity verification pipeline described in Chapter 2.

Q-CRAFT Explorer is our attempt to show what this looks like. The source code is on GitHub. We welcome your input.

Glossary

Climate scenarios Six warming pathways used in Q-CRAFT, based on IPCC Shared Socioeconomic Pathways (SSPs). The coolest is Paris-aligned (below 2°C warming by 2100, SSP1-2.6). The hottest is Hot Unadapted (high warming with slow adaptation, SSP3-7.0 at the 90th percentile). Climate impacts on GDP are derived from the FADCP Climate Dataset (Centorrino, Massetti, and Tagklis, 2024), building on the empirical estimates of Kahn et al. (2021). See the IMF User Guide, Section IV.B for detailed methodology.

C-PIMA (Climate Public Investment Management Assessment) An IMF framework for evaluating how well a country's public investment management institutions account for climate change. C-PIMA assessments include an annex that uses Q-CRAFT to model the fiscal impacts of climate scenarios for the assessed country. See the C-PIMA Handbook (IMF, 2025) for the full framework.

DSA (Debt Sustainability Analysis) A framework used by the IMF and World Bank to assess whether a country's debt is sustainable (that is, whether the government can meet its current and future debt obligations without requiring debt relief or accumulating arrears). Q-CRAFT projections can complement DSA analysis by providing long-term climate-adjusted fiscal trajectories.

Debt dynamics equation The core equation in Q-CRAFT: $d_t = d_{t-1} \times \frac{1+r_t}{1+g_t} - pb_t$. Next year's debt-to-GDP ratio equals the current ratio, adjusted for the interest-growth differential, minus the primary balance. When interest rates exceed growth ($r > g$), debt rises automatically: the government must run a primary surplus even to hold debt still. See the IMF User Guide, Section IV.A on debt dynamics for detailed derivation and corollaries.

Debt-to-GDP ratio Total government debt expressed as a percentage of nominal GDP. The standard measure of a country's debt burden, since it scales debt relative to the economy's ability to service it. Q-CRAFT projects this ratio through 2099 under baseline and climate scenarios.

Expenditure rigidity The degree to which government spending resists downward adjustment when climate shocks reduce GDP. Measured on a 0-1 scale. A value of 1.0 means spending is fully sticky: it stays at the baseline level in local currency terms even as GDP falls, representing the worst case for debt accumulation. A value of 0.0 means spending adjusts fully with GDP, maintaining the baseline expenditure-to-GDP ratio. This parameter only affects climate scenarios, not the baseline. See the IMF User Guide, pp. 20 and 35-36.

Fiscal rule A mechanism in Q-CRAFT that adjusts primary expenditure to steer debt toward a target ratio. When enabled, the model computes the debt-stabilizing primary balance and adjusts spending by the fiscal gap (the difference between the actual and debt-stabilizing primary balance). The adjustment is additive in levels (not a rate) and depends on the prior year's state, requiring recursive year-by-year computation. See the IMF User Guide, pp. 15-18.

DIGNAD (Debt, Investment, Growth, and Natural Disasters) An IMF dynamic general equilibrium model for analyzing the macroeconomic effects of natural disasters and public investment in developing countries. More complex than Q-CRAFT, with substantially more parameters. Discussed in the co-design appendix as an example of a tool that could benefit from the SovTech approach.

Golden master In software verification, a reference output used to detect unintended changes. Q-CRAFT's golden master tests compare the Python engine's outputs against values extracted from the IMF's Excel workbook for the same country and parameter inputs. Parity with the Excel tool (not any independent calculation) is the acceptance criterion.

LIC-DSF (Low-Income Country Debt Sustainability Framework) The IMF-World Bank framework for assessing debt sustainability in low-income countries. Uses country-specific debt burden thresholds based on a composite indicator of institutional quality and macroeconomic fundamentals. Compared against Q-CRAFT in Module 5, and discussed in the co-design appendix as a more complex tool that could benefit from the SovTech approach.

Interest-growth differential The difference between the nominal interest rate on government debt (r) and nominal GDP growth (g), expressed as $(r-g)/(1+g)$. When positive ($r > g$), debt dynamics are unfavorable: the debt ratio rises automatically. When negative ($g > r$), debt dynamics are favorable. This single variable determines whether debt is self-stabilizing or self-reinforcing.

Primary balance Government revenue minus non-interest expenditure (primary expenditure), expressed as a share of GDP. The primary balance is the fiscal variable that a government can control in the near to medium term through fiscal policy. A primary surplus means the government generates enough revenue to cover its non-interest spending. A primary deficit means it does not.

WEO (World Economic Outlook) The IMF's flagship publication providing analysis and projections of the global economy. Q-CRAFT uses the WEO database (currently October 2024 vintage) as the source for macroeconomic and fiscal data through the medium-term forecast horizon (currently through 2029). Beyond this horizon, Q-CRAFT's own projection methodology takes over.

SovTech The application of SupTech (supervisory technology) principles (modularity, open source, human-centered design) to sovereign debt analysis and public financial management tools. Q-CRAFT Explorer is a proof of concept for this approach.

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